



General Familiarization LEAP-1B

Training Manual

V0 - 621 - Level 1



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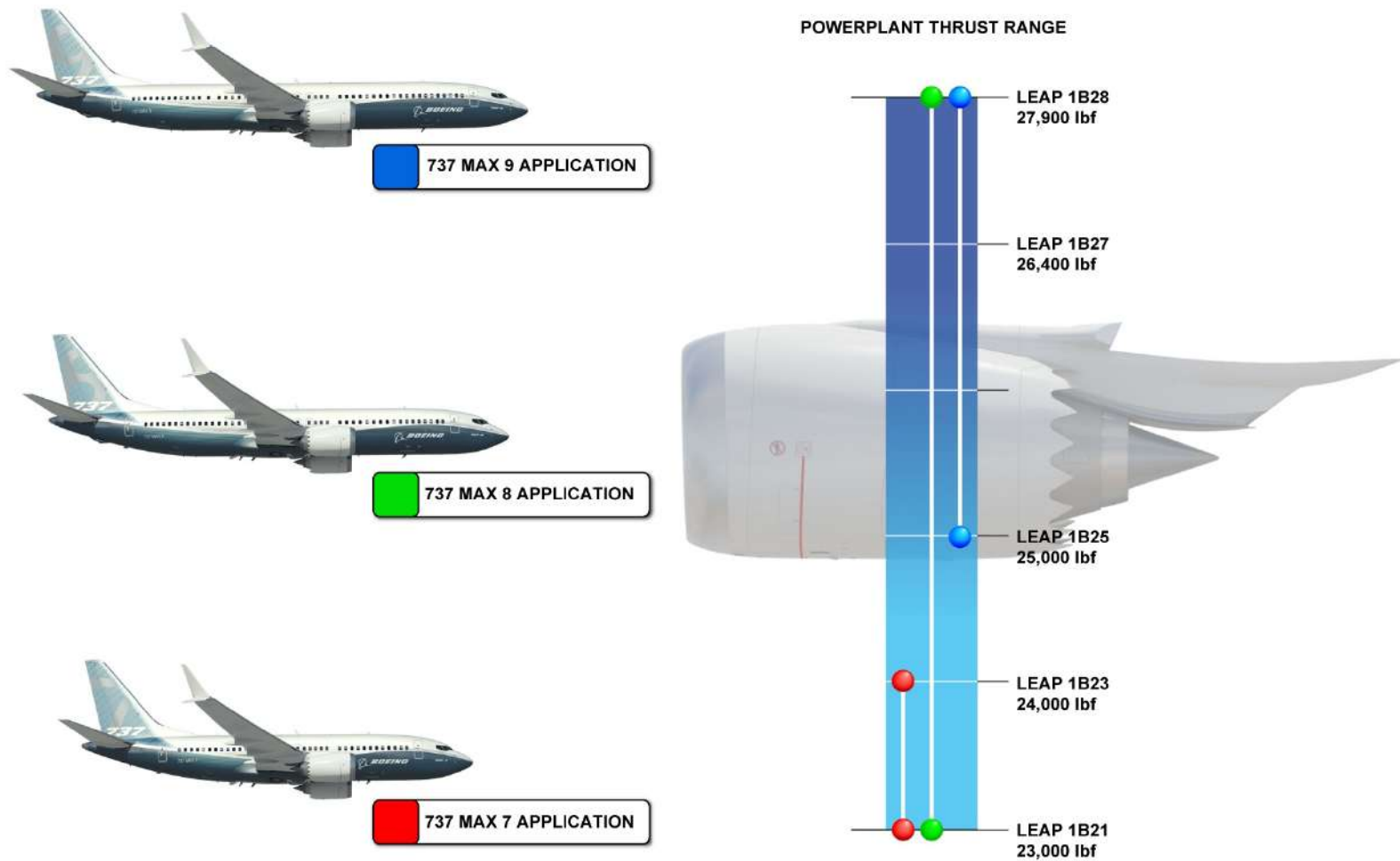
Introduction

LEAP-1B Family on 737 MAX

The LEAP-1B with its nacelle is the power source for the entire BOEING 737 MAX family:

- MAX 7,
- MAX 8,
- MAX 9.

The power range is 23,000 to 27,900 lbf at sea-level take-off thrust.



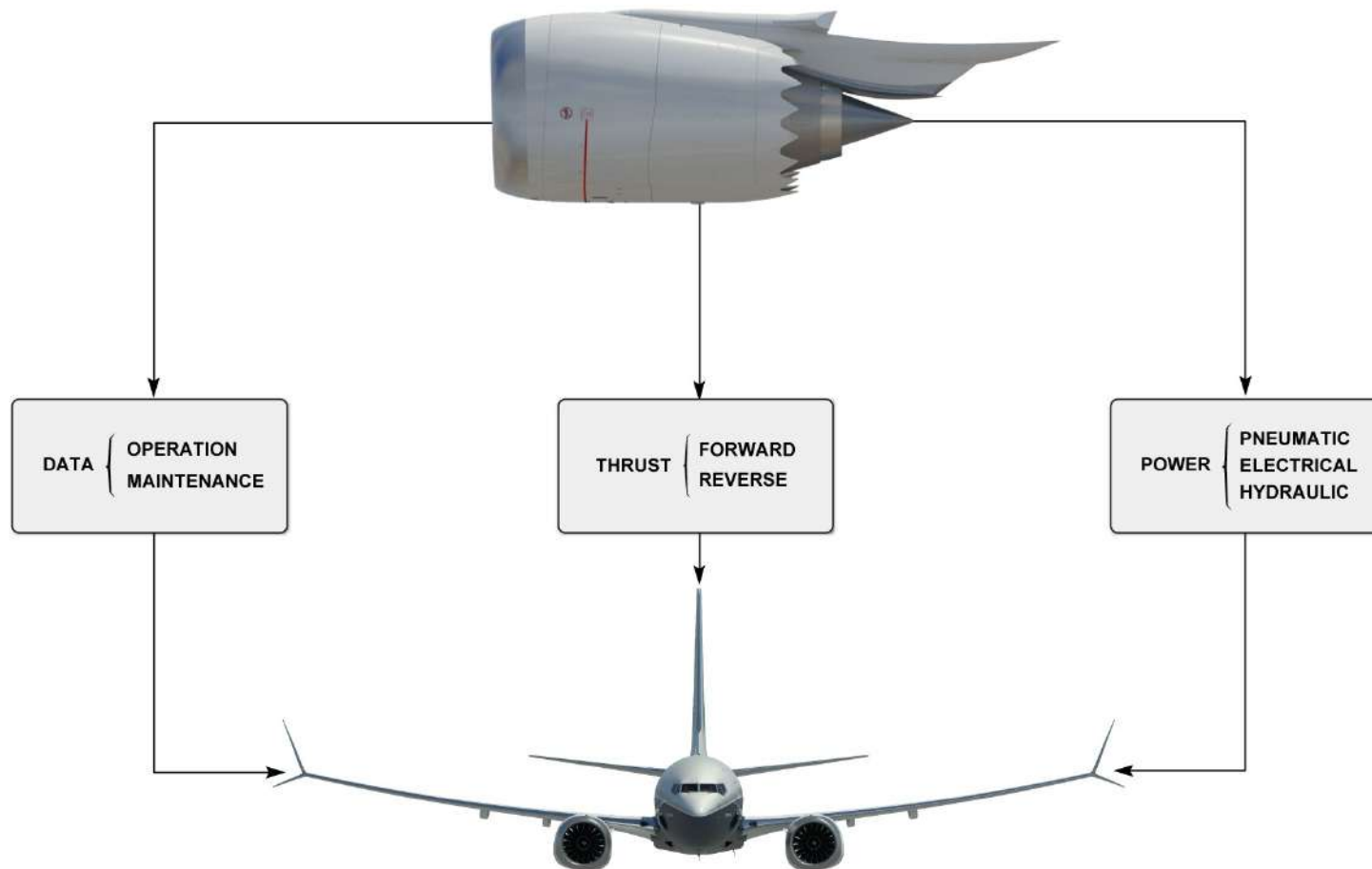
LEAP-1B Family on 737 MAX

Powerplant - Purpose

Purpose

The powerplant provides the aircraft with:

- forward thrust,
- reverse thrust only on the ground,
- several power sources: air, electricity and hydraulics,
- data for operation and maintenance.



Powerplant - Purpose

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Powerplant - Airflows

Identify and Locate

The LEAP-1B engine is a high bypass dual rotor axial flow turbofan engine. The two airflows are split behind the fan blades by the flow splitter:

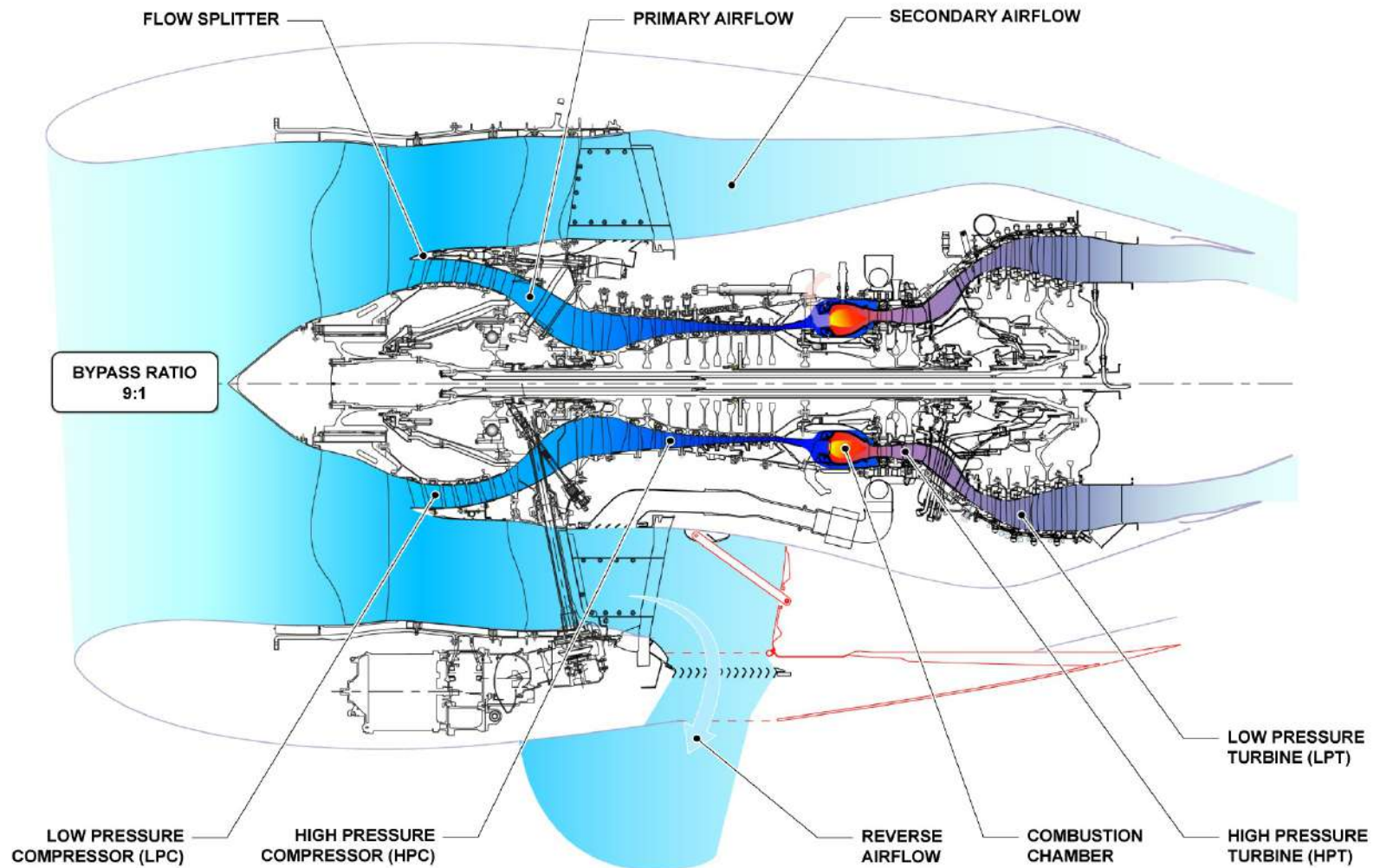
- The primary airflow passes through the inner portion of the fan blades and is directed to the booster. Then it enters the High Pressure Compressor (HPC) and goes to the combustion chamber. Mixed with fuel and ignited, the gas flow produced provides energy to the High Pressure Turbine (HPT) and to the Low Pressure Turbine (LPT). The HPT drives the HPC, and the LPT drives the Low Pressure Compressor (LPC).

- The secondary airflow passes through the outer portion of the fan blades and exits through the nacelle discharge duct.

On the ground at take-off, the secondary airflow produces most of the thrust.

The secondary airflow is also used by the thrust reverser system.

The engine bypass ratio is approximately 9:1.



Powerplant - Airflows

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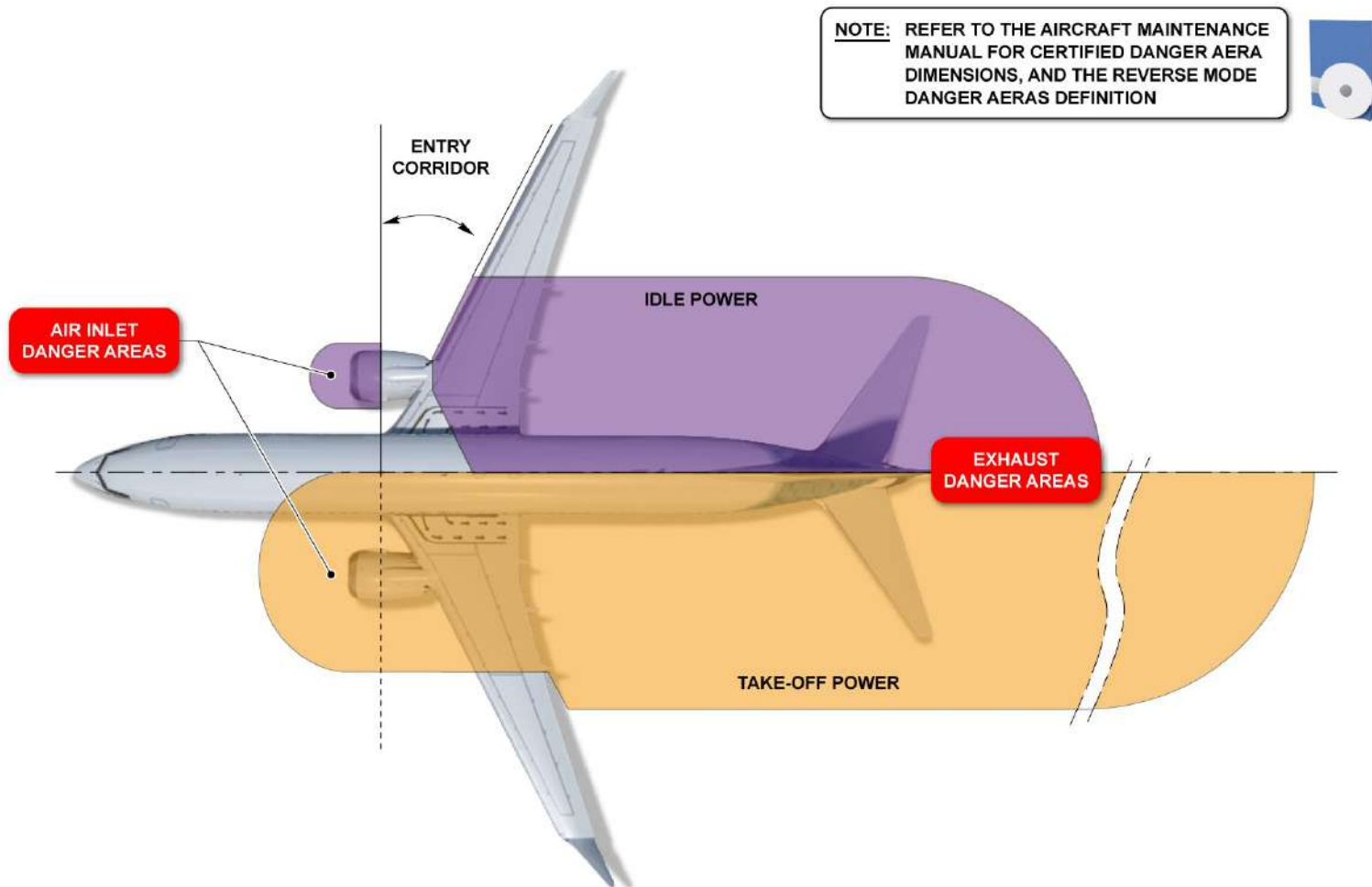
Hazard and Danger Areas

Identify and Locate

For safety purposes, hazard and danger areas have to be defined. Personnel must be aware of these areas when they perform engine run-ups for maintenance purposes.

The defined danger areas are the air intake suction area and the exhaust . An entry corridor is defined to access the engine in specific conditions.

Refer to the Aircraft Maintenance Manual (AMM) for certified danger area dimensions.



Hazard and Danger Areas

Acoustical Hazard Areas

Identify and Locate

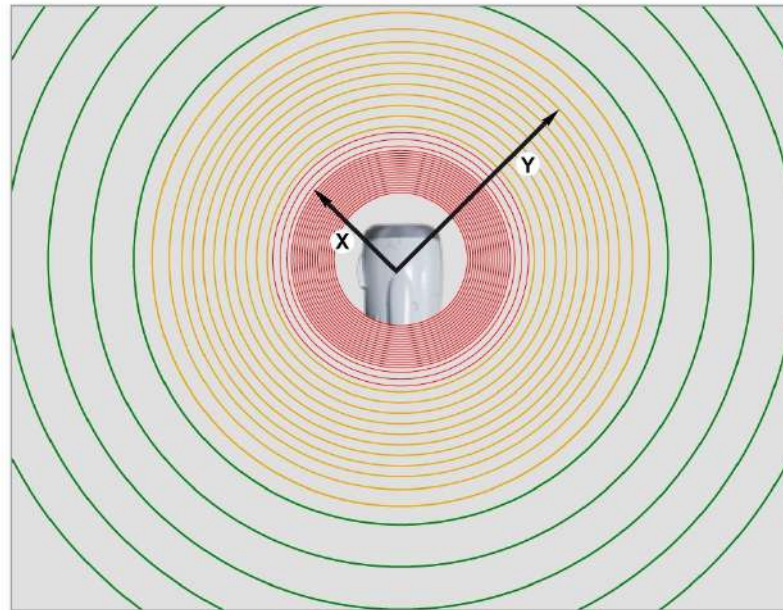
Engine noise can cause temporary and/or permanent loss of hearing.

The following charts provide information on the ear protection requirements, depending on the distance to the engine.

Notes:

- the charts provide information about ear damage even when wearing ear protection,
- refer to the Aircraft Maintenance Manual (AMM) for certified values.

NOTE: REFER TO THE AIRCRAFT MAINTENANCE MANUAL FOR CERTIFIED VALUES

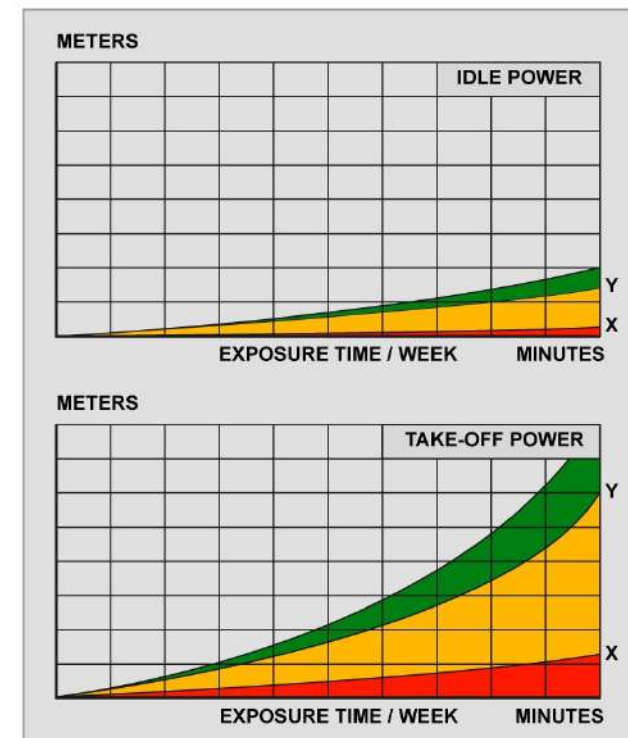


EAR DAMAGE EVEN WITH EAR PROTECTION

EAR PROTECTION REQUIRED

NO EAR PROTECTION REQUIRED

RADIAL DISTANCE FROM THE ENGINE IN METERS



Acoustical Hazard Areas

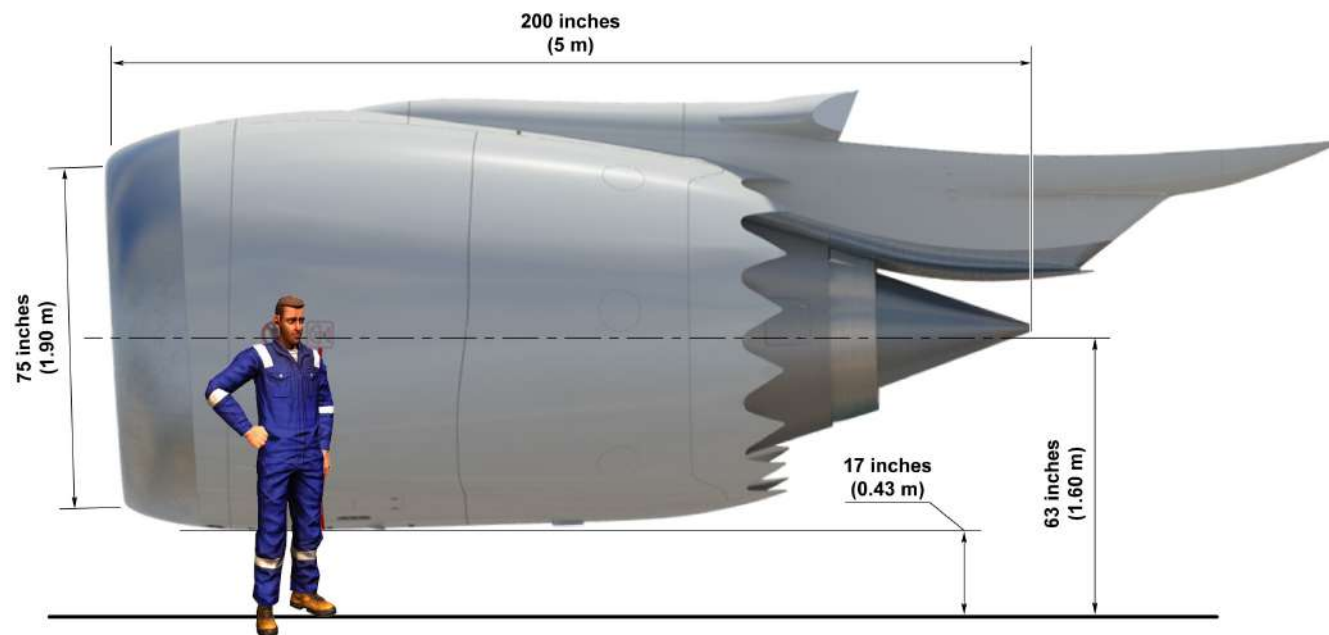
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Powerplant External Dimensions and Ground Clearances

For the entire powerplant, the external approximate dimensions are :

- length: 200 inches (5,08 m),
- outer diameter of the air inlet: 75,27 inches (1,91 m),
- ground clearance: 17 inches (0,43m).

NOTE: ALL VALUES ARE APPROXIMATE



Powerplant External Dimensions and Ground Clearances

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Nacelle Inlet Cowl

Inlet Cowl and Access Door

Identify and Locate

The inlet cowl is a fixed interchangeable aerodynamic fairing mounted on the forward face of the engine fan case.

On the outer barrel of the inlet cowl, there are a T12 sensor access door located at 12 o'clock and a vent opening for the anti-ice air exhaust at 6 o'clock.

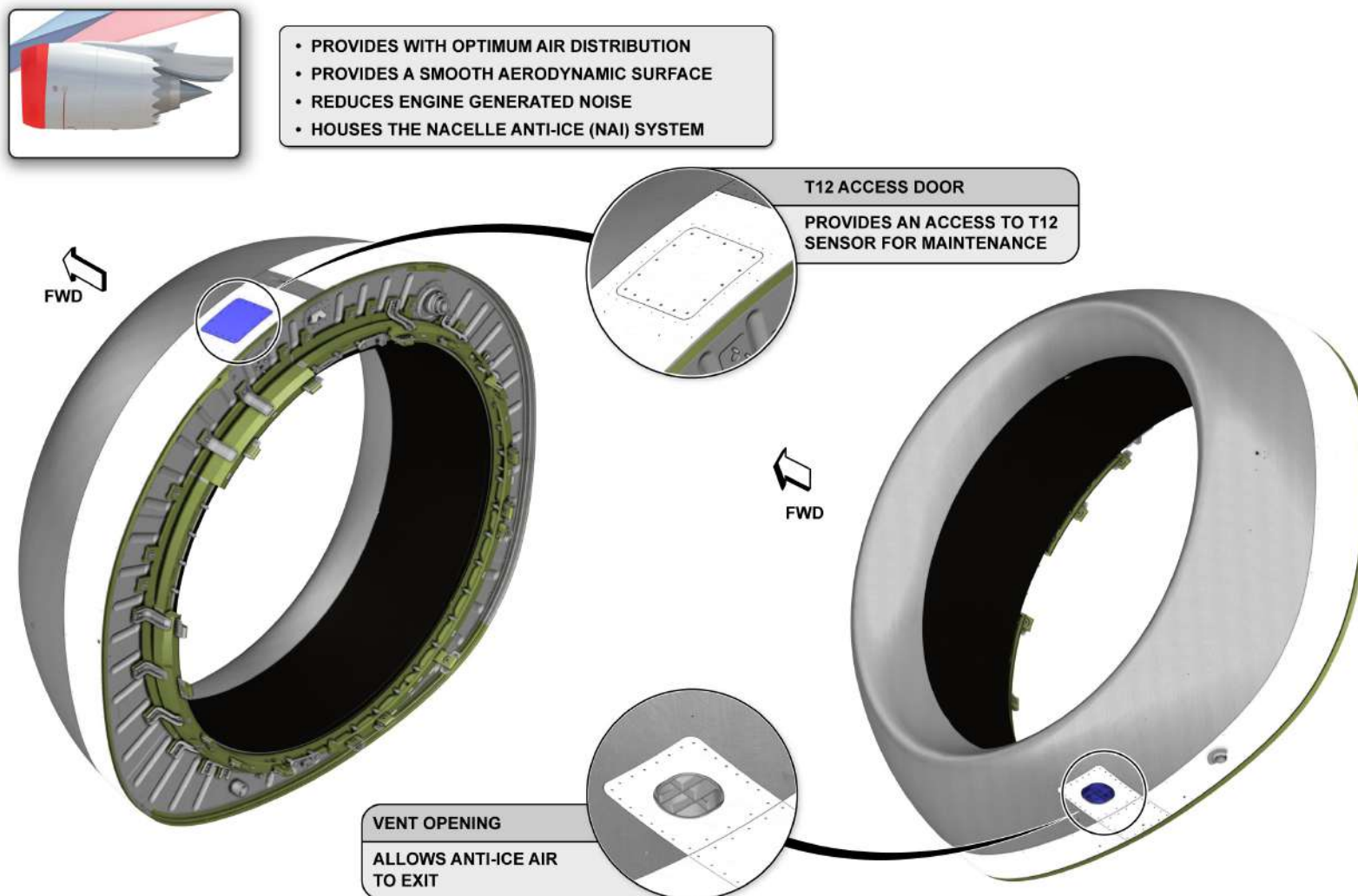
Purpose

The inlet cowl has 4 main functions:

- to provide the engine with optimum air distribution, whatever the external conditions,
- to provide a smooth aerodynamic surface to the airflow,
- to reduce engine generated noise,
- to house the Nacelle Anti-Ice (NAI) system.

The 12 sensor access door provides an access to the T12 sensor for maintenance actions.

The purpose of the vent opening is to allow the anti-ice air exit from the Nacelle Anti-Ice (NAI) duct.



Inlet Cowl and Access Door



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General Familiarization LEAP-1B



Nacelle Fan Cowl Panels

Left Hand Fan Cowl Panel

Identify and Locate

The left hand fan cowl panel is located between the inlet cowl and the thrust reverser and encloses the engine air inlet section.

The left hand fan cowl panel is fitted with:

- an Integrated Drive Generator (IDG) access door located at 8 o'clock,
- a Starter Air Valve (SAV) access located at 9 o'clock,
- three latch keepers located at 6 o'clock,
- a Spring Door Opening System (SDOS) located on the forward end of fan cowl,
- a Hold Open Rod (HOR) located on the aft end of fan cowl.

Purpose

The purposes of the fan cowls are:

- to ensure a continuous aerodynamic air flow around the nacelle from the inlet cowl to the thrust reverser,
- to protect the parts fitted on the fan case and the fan frame.
- to provide fire containment,
- to reduce lightning effect.

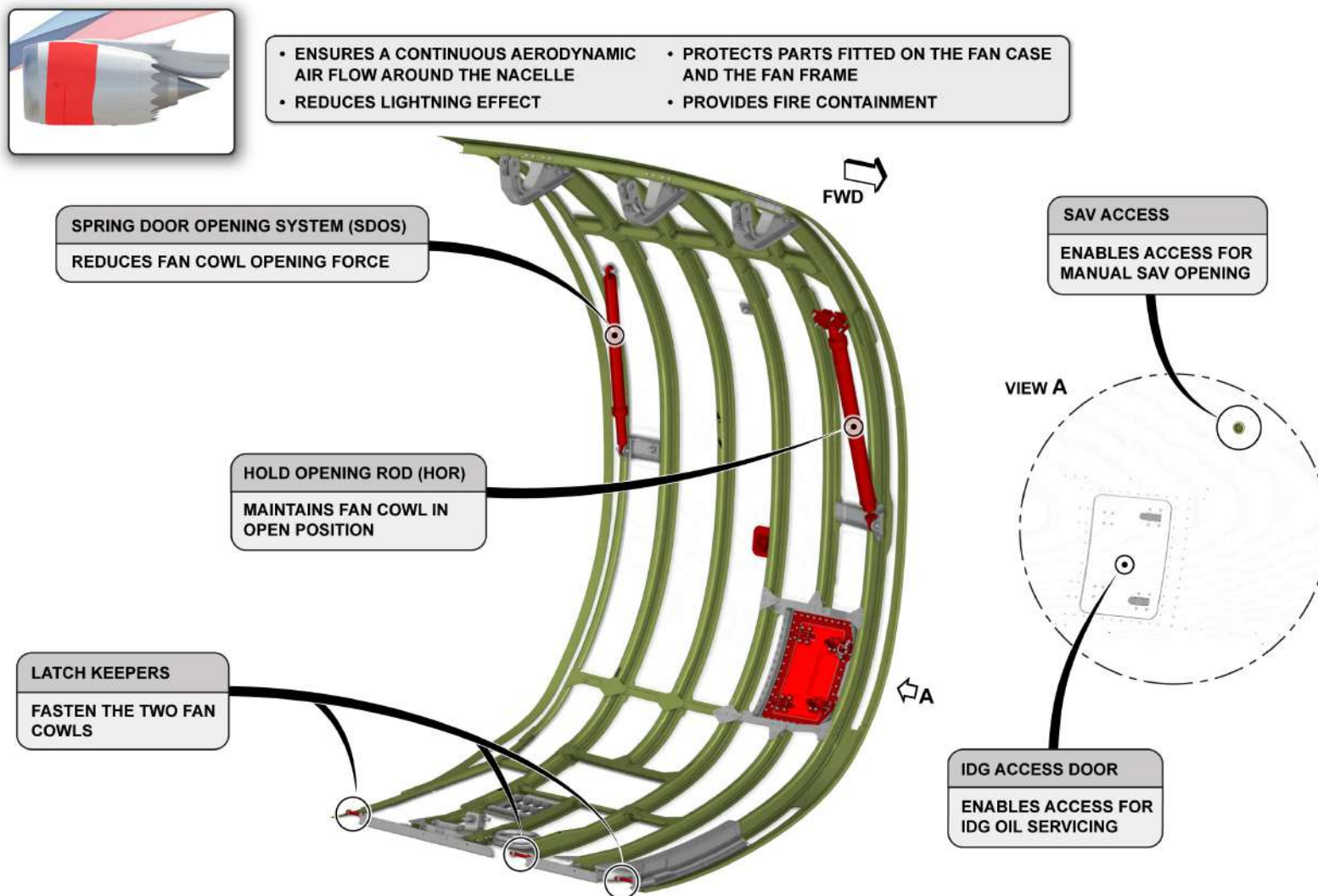
The IDG access door enables the IDG oil servicing.

The SAV access allows the SAV manual opening.

The purpose of the three latch keepers is to fasten the two fan cowl doors together during flight.

The SDOS contains a spring preload which reduces fan

cowl opening force to <35 lbs. The HOR help to maintain the fan cowl panels in the open position.



Left Hand Fan Cowl Panel

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Right Hand Fan Cowl Panel

Identify and Locate

The right hand fan cowl panel is located between the inlet cowl and the thrust reverser and encloses the engine air inlet section.

The right hand fan cowl panel is fitted with:

- an oil tank access door located at 4 o'clock,
- three fan cowl panel latch hooks located at 5 o'clock,
- a fan cowl chine located in the upper external surface of the fan cowl panel at 2 o'clock,
- a Spring Door Opening System (SDOS) on the forward end of fan cowl,
- a Hold Open Rod (HOR) on the aft end of fan cowl.

NOTE : on the engine 2, the chine is located on the left hand fan cowl panel at 10 o'clock.

Purpose

The purposes of the fan cowl are:

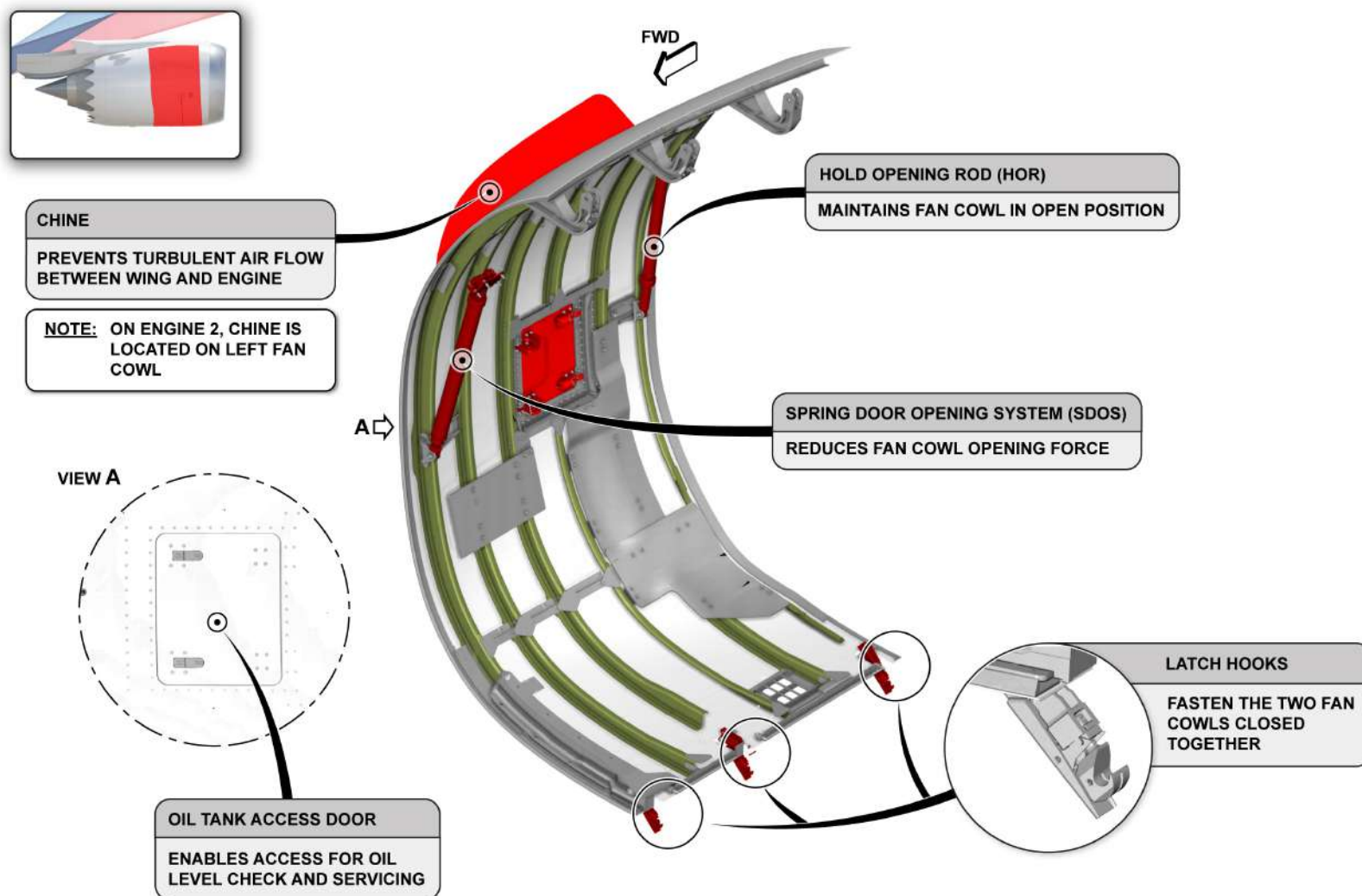
- to ensure a continuous aerodynamic air flow around the nacelle from the inlet cowl to the thrust reverser,
- to protect the parts fitted on the fan case and fan frame,
- to provide fire containment,
- to reduce lightning effect.

The oil tank access door enables an access to the oil tank for oil level check and servicing.

The purpose of the three latch hooks is to fasten the two fan cowl doors together during flight.

The chine prevents turbulent airflow at the engine / wing interface.

The SDOS contains a spring preload which reduces fan cowl opening force to <35 lbs. The HOR helps to maintain the fan cowl panels in the open position.



Right Hand Fan Cowl Panel

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Thrust Reverser

Identify and Locate

The LEAP-1B Thrust Reverser (T/R) is located from aft of the fan cowls and extends to aft of the primary exhaust attaché flange. It encloses the engine core and the Low Pressure Turbine (LPT).

The T/R consists of two halves hinged at the top of the pylon and latched together at the bottom.

The T/R halves are composed of:

- 2 translating sleeves located aft of the fan cowl,
- a fixed outer structure that features 16 graphite composite cascade vanes, 8 per half, located under the translating sleeves (in stowed position),
- 5 blocker doors per reverser half, mounted between the translating sleeve and the T/R inner wall assembly.

Purpose

In stowed position, the T/R ducts the secondary airflow up to the exit to produce reverse thrust.

In landing or Rejected Take Off (RTO) operation, the thrust reverser improves the aircraft braking capacity by redirecting forward, through cascade vanes, the secondary airflow.

In stowed position, the translating sleeves:

- are designed to optimize the fan air exhaust flow (pressure and velocity),
- protect the cascade vanes and the other core and LPT components.

In deployed position, the translating sleeves:

- expose the cascade vanes,
- move the blocker doors into the fan air exhaust flow to redirect the secondary airflow in the forward direction through the cascade vanes; that produces a reverse thrust.

The shape of the cascade vanes prevents as much as possible reingestion of debris from the ground during reverse operation at high power.



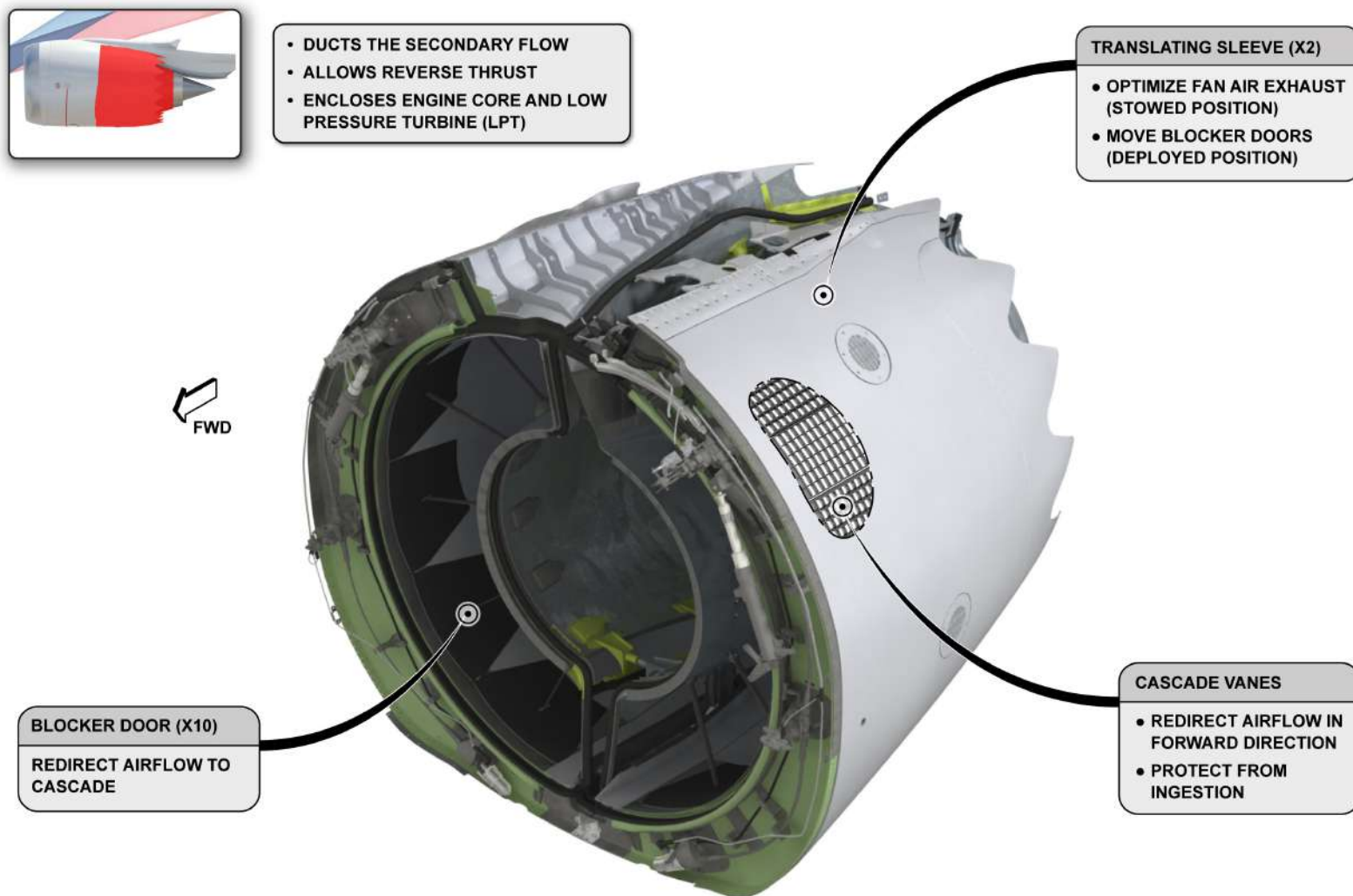
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Nacelle Thrust Reverser



Thrust Reverser

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Thrust Reverser Actuation System

Identify and Locate

Each translating sleeve of the thrust reverser can be moved backward (reverser in deploy position) or forward (reverser in stowed position) with the three linear hydraulic actuators located on forward face of thrust reverser:

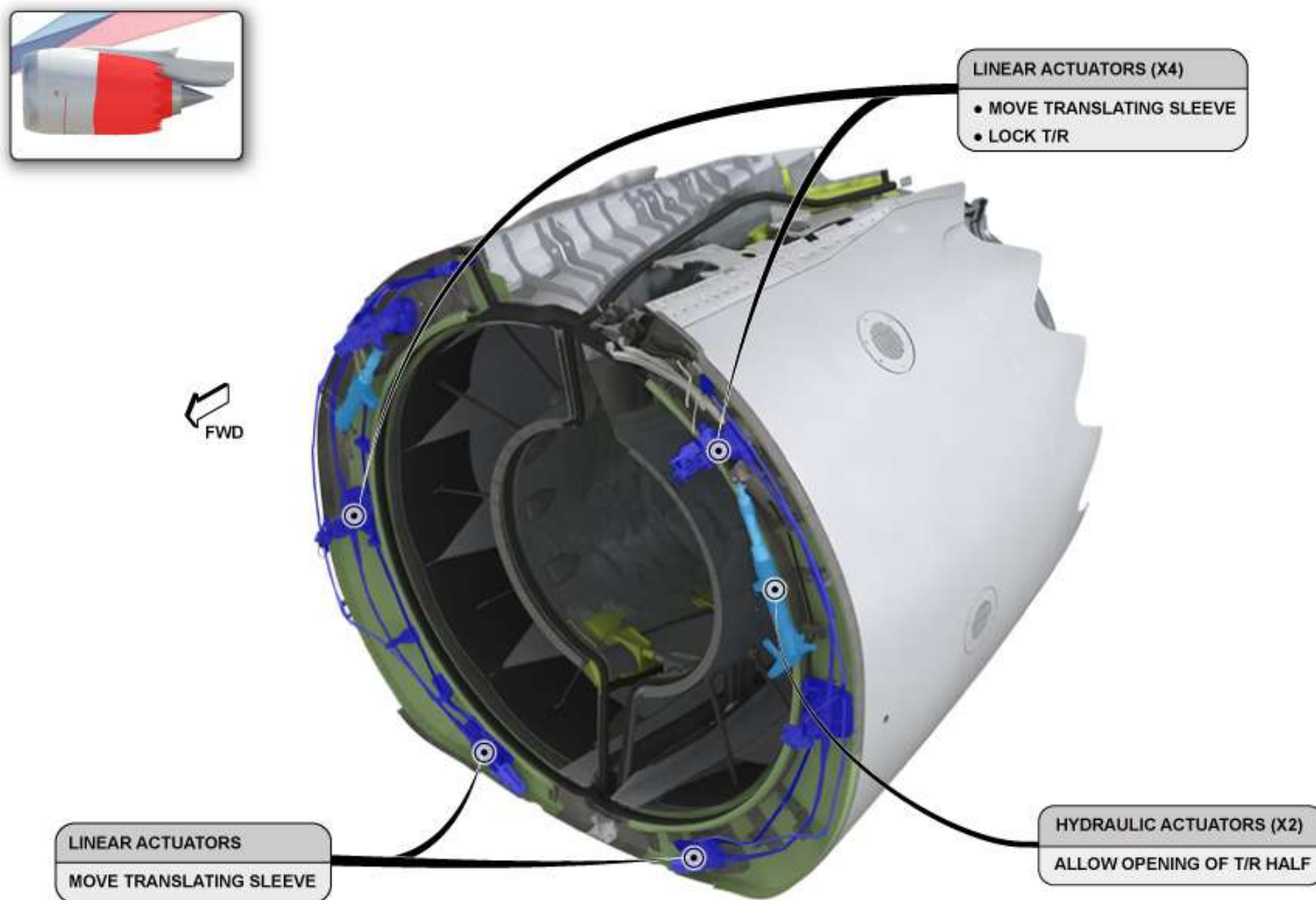
- the upper locking actuator located at 11 o'clock for left translating sleeve, at 1 o'clock for right translating sleeve,
- the middle locking actuator located at 9 o'clock for left translating sleeve, at 3 o'clock for right translating sleeve,
- the lower actuator located at 7:30 o'clock for left translating sleeve, at 4:30 o'clock for right translating sleeve.

In addition, a manually driven hydraulic actuator is located on each thrust reverser half on the forward face, between upper and middle locking actuator.

Purpose

The 6 linear actuators purpose is to deploy or stow the two thrust reverser halves. The 2 upper and the 2 middle actuators have a locking device to lock the thrust reverser in the fully deployed or stowed positions.

The 2 manually driven actuators allows the opening of each reverser halve for inspection and maintenance access to the engine, up to 37° for the inboard one and up to 55° for the outboard one.



Thrust Reverser Actuation System

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Nacelle Exhaust

Exhaust System

Identify and Locate

The turbine exhaust system is composed of a nozzle and a plug. They are attached to the rear flanges of the Turbine Rear Vane (TRV).

The Center Vent Tube Extension (CVTE) assembly is located in the exhaust plug.

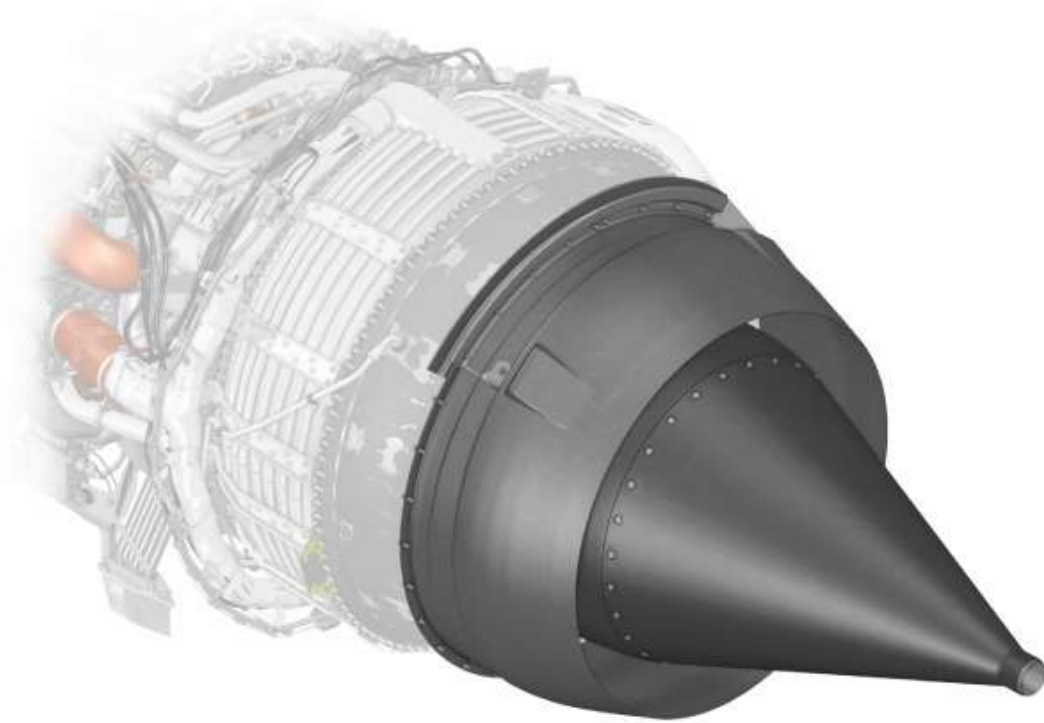
Purpose

The turbine exhaust system supplies an exit for the turbine exhaust gases. The system increases velocity of the gases to increase engine thrust.

The CVTE ensures the venting of the engine sump air.



- SUPPLIES AN EXIT FOR THE EXHAUST GASES AND INCREASES THEIR VELOCITY TO INCREASE THE THRUST
- ENSURES THE VENTING OF ENGINE SUMP AIR



Exhaust System

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General Familiarization LEAP-1B



Quick Engine Change

Quick Engine Change & Cowls

Identify and Locate

The main components of the LEAP-1B powerplant are:

- the Quick Engine Change (QEC),
- the fan cowl doors and thrust reverser cowl doors.

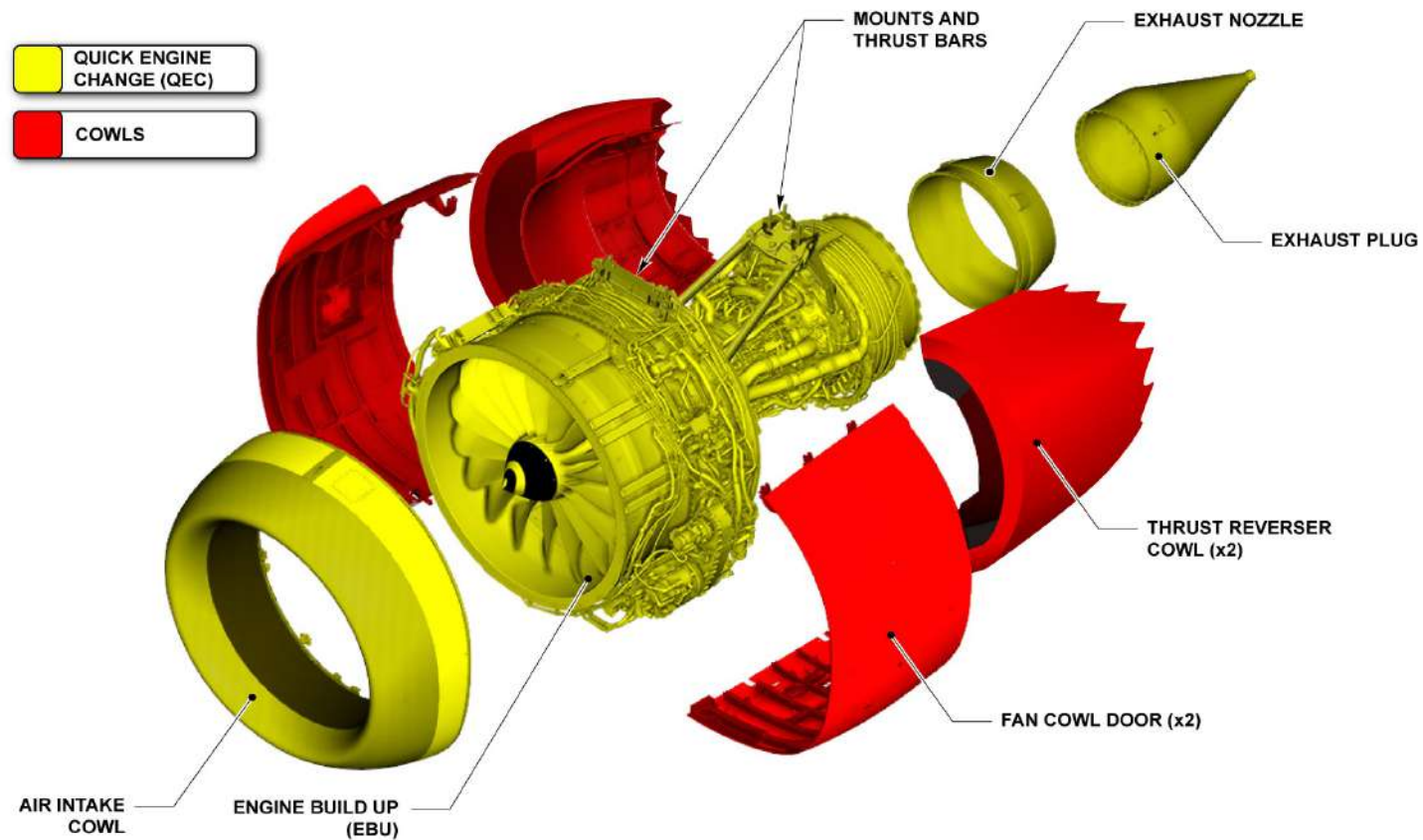
The QEC consists of the following components:

- the air inlet cowl,
- the Engine Build-Up (EBU),
- the exhaust nozzle and exhaust plug,
- the engine mounts and thrust links.

Purpose

In Quick Engine Change (QEC) configuration the engine can be removed/installed from/on the aircraft, using simple connections at the nacelle/aircraft pylon interface.

IN QEC CONFIGURATION THE ENGINE CAN BE REMOVED/INSTALLED USING SIMPLE CONNECTIONS AT THE PYLON INTERFACE



Quick Engine Change & Cowls

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Quick Engine Change Engine Mounts

Forward Engine Mount

Identify and Locate

The forward engine mount is linked to the fan frame shroud at 12 o'clock.

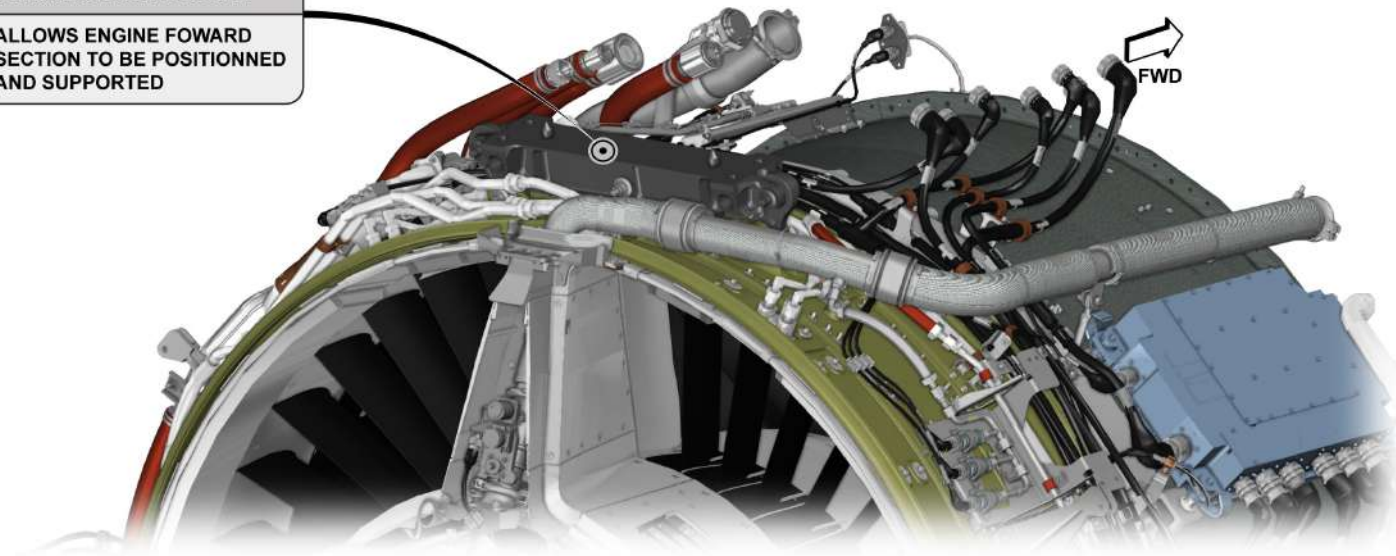
Purpose

The forward engine mount positions and supports the forward side of the engine to the pylon strut. With the aft engine mount, it ensures a demountable mechanical link of the engine to the aircraft.



FORWARD ENGINE MOUNT

ALLOWS ENGINE FOWARD
SECTION TO BE POSITIONNED
AND SUPPORTED



Forward Engine Mount

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Aft Engine Mount and Thrust Links

Identify and Locate

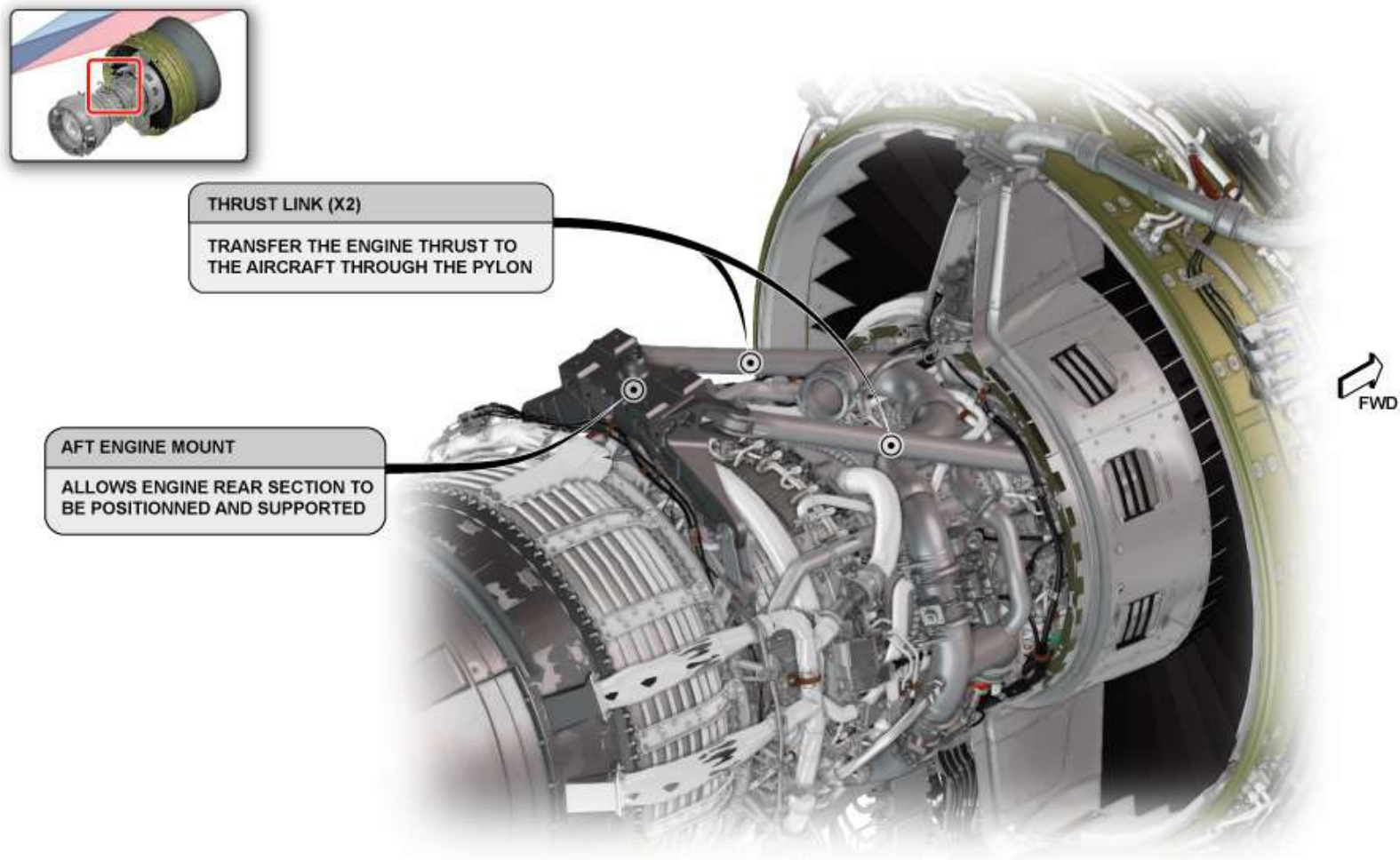
The aft engine mount is linked to the Turbine Center Frame (TCF) at 12 o'clock.

The two thrust links are connected on one end to the fan frame hub and on the other end to the aft engine mount.

Purpose

The aft engine mount positions and supports the aft of the engine to the pylon strut. With the forward engine mount, it ensures a demountable mechanical link of the engine to the aircraft.

The two thrust links transfer the engine thrust to the aircraft through the pylon.



Aft Engine Mount and Thrust Links

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Quick Engine Change Engine Build-up Units

Powerplant - Systems

Identify and Locate

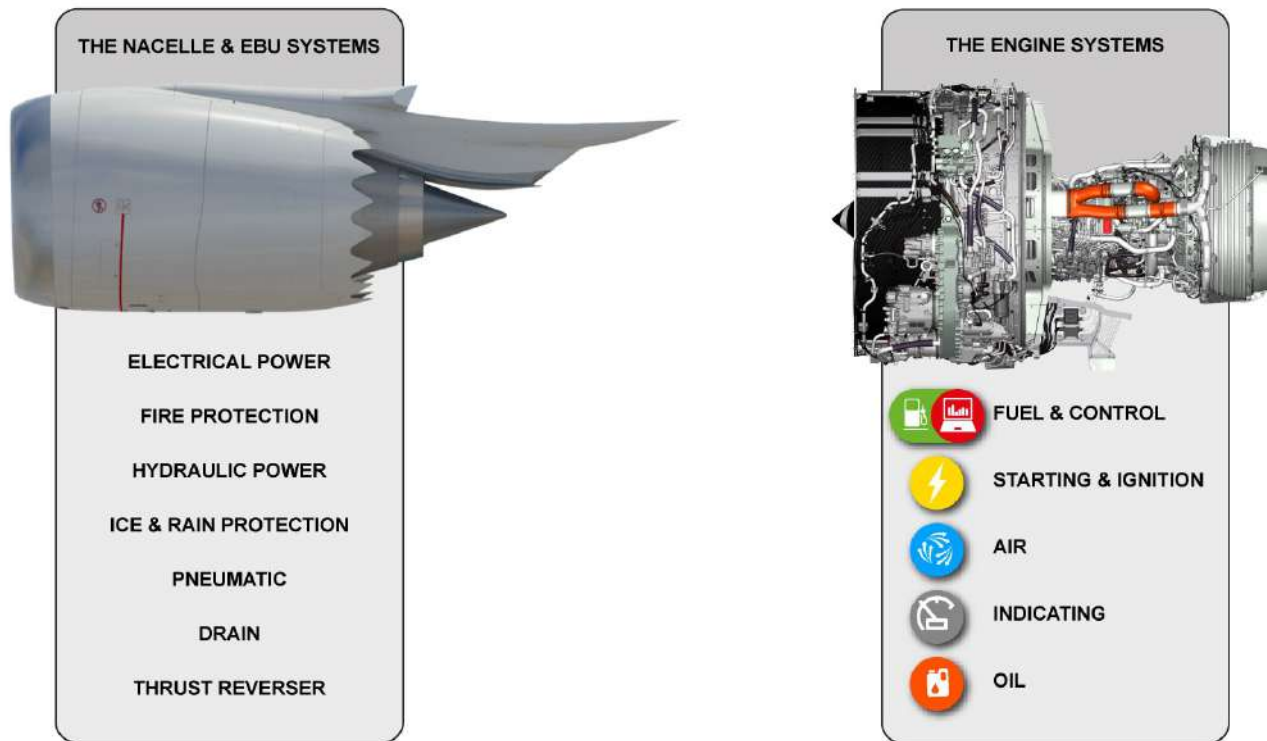
The powerplant includes two groups of systems:

- the engine systems:

- Air Transport Authority (ATA) chapter 73: engine fuel and control,
- ATA chapter 74: ignition,
- ATA chapter 75: air,
- ATA chapter 77: engine indicating,
- ATA chapter 79: oil,
- ATA chapter 80: starting.

- aircraft systems that are the aircraft manufacturer's responsibility but are partially fitted on the engine. The training course only describes the part of these aircraft ATA chapters that are fitted on the engine:

- ATA chapter 24: electrical power,
- ATA chapter 26: fire protection,
- ATA chapter 29: hydraulic power,
- ATA chapter 30: ice and rain protection,
- ATA chapter 36: pneumatic,
- ATA chapter 71: drain,
- ATA chapter 78: thrust reverser.



Powerplant - Systems

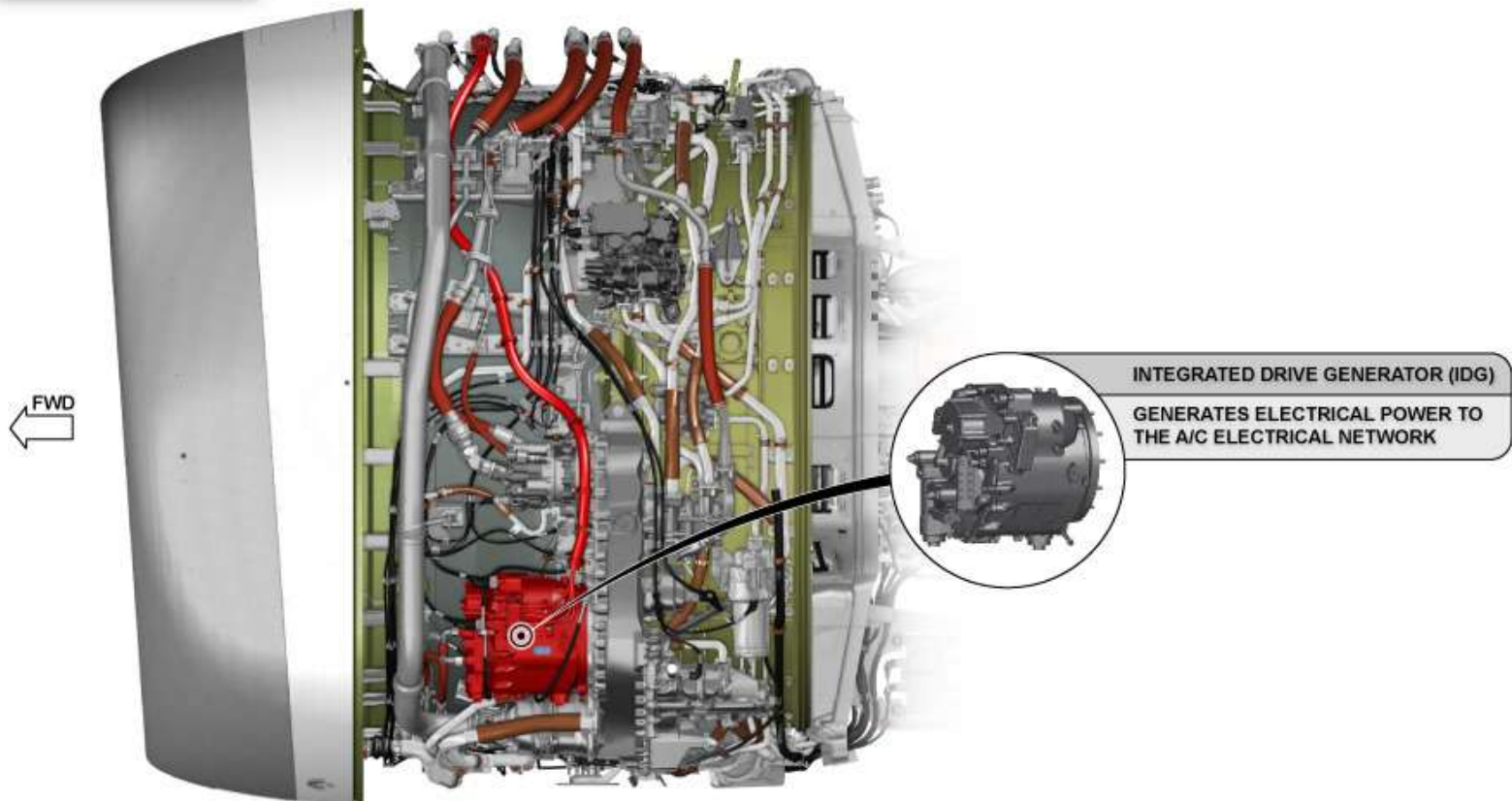
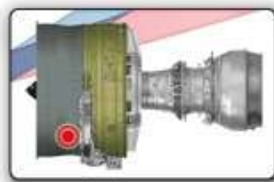
Integrated Drive Generator

Identify and Locate

The Integrated Drive Generator (IDG) is mounted on the forward face of the Accessory Gear Box (AGB) at 7 o'clock.

Purpose

The IDG is driven by the AGB and supplies three-phase 115/200VAC - 400Hz electrical power to the aircraft electrical network. In flight, it is the normal electrical power source of the aircraft.



Integrated Drive Generator

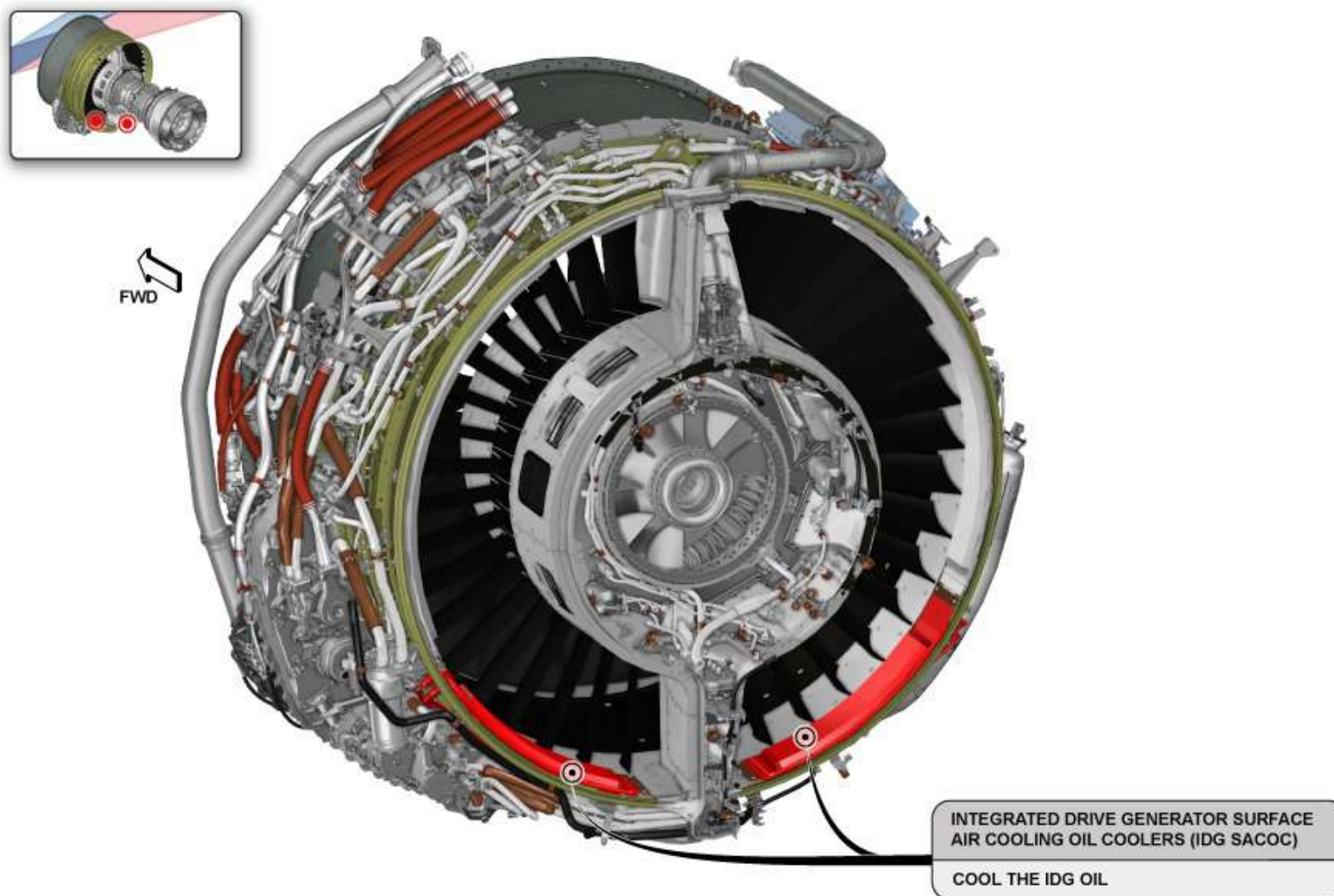
Integrated Drive Generator Surface Air Cooling Oil Cooler

Identify and Locate

The Integrated Drive Generator (IDG) Surface Air Cooling Oil Cooler (SACOC) is composed of two segments fixed to the inner surface of the fan frame, between 6 and 8 o' clock for the left segment and between 4 and 6 o' clock for the right segment of the IDG SACOC.

Purpose

The purpose of the right and left IDG SACOC is to reduce the temperature of the oil in the IDG oil circuit by thermal exchange with the secondary airflow.



Integrated Drive Generator Surface Air Cooling Oil Cooler

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Pneumatic

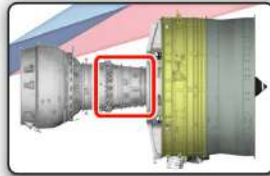
Identify and Locate

Some of the pneumatic components are located all around the core of the engine and up to the aircraft pylon.

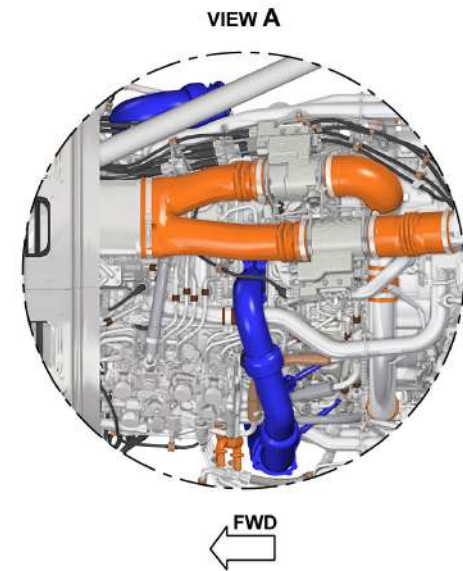
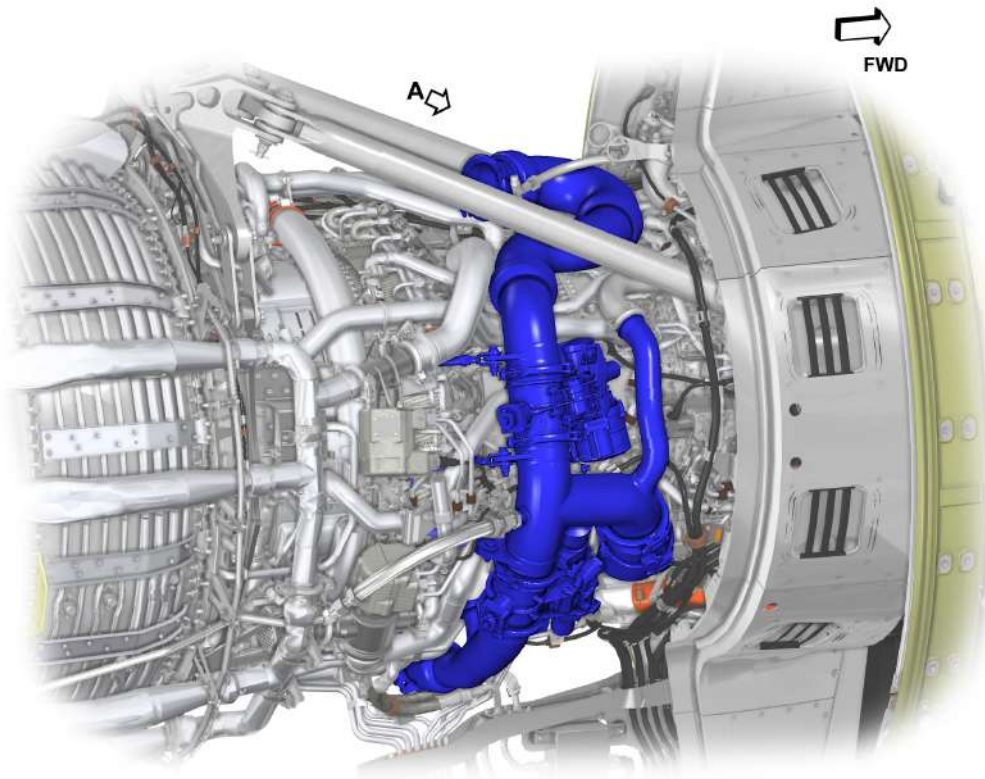
Purpose

The engine pneumatic system supplies pressurized air to the following systems and components:

- engine 1 and 2 starters,
- air conditioning packs,
- anti-ice systems (engine and wing),
- hydraulic reservoir and water tank for pressurization,
- fuel inerting.



- SUPPLIES PRESSURIZED AIR FOR:**
- AIR CONDITIONING
 - FUEL INERTING
 - ENGINE STARTING
 - HYDRAULIC AND WATER TANKS PRESSURIZATION
 - ANTI-ICE



Pneumatic

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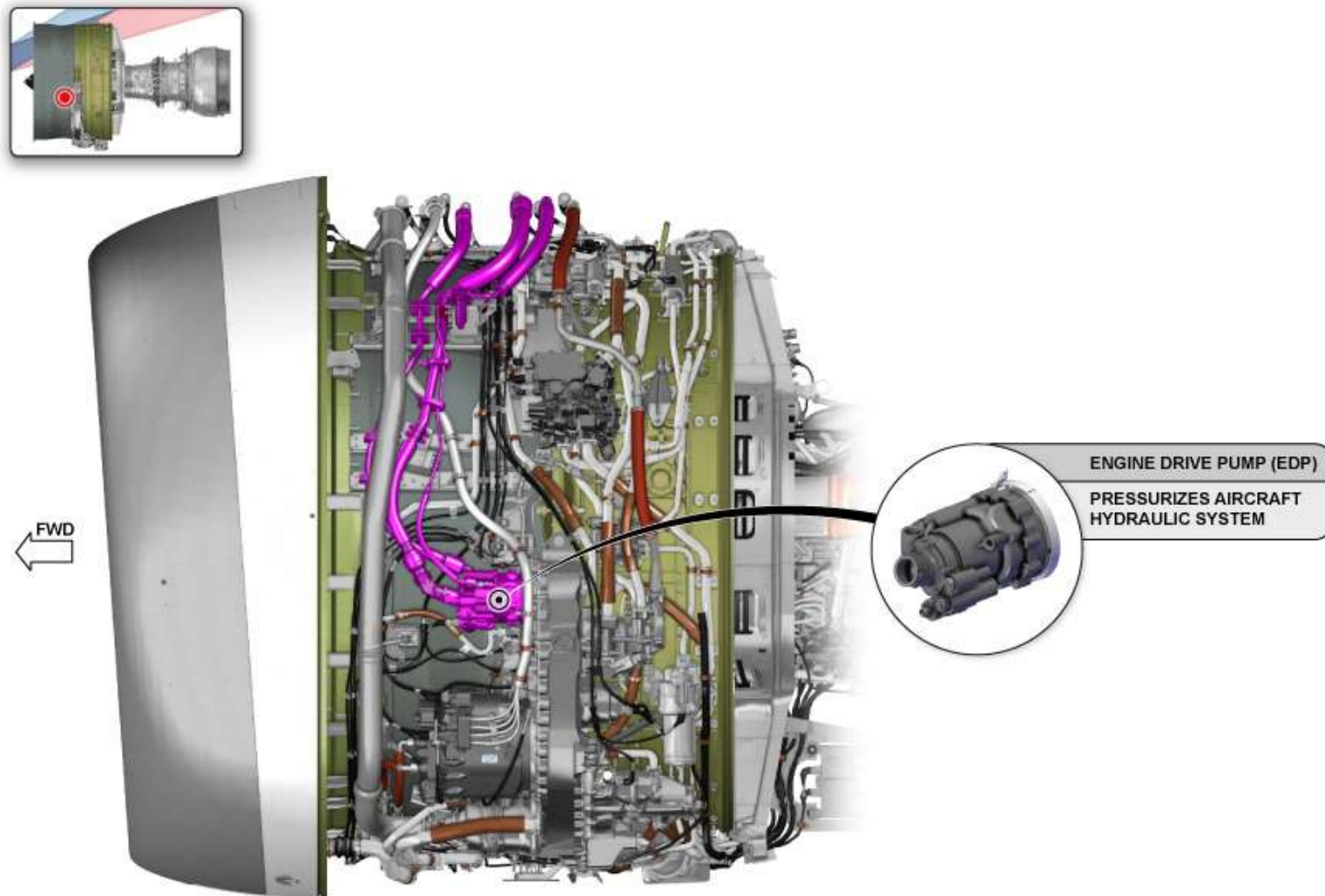
Engine Driven Pump

Identify and Locate

The Engine Driven Pump (EDP) is a hydraulic pump mounted on the forward face of the Accessory Gear Box (AGB) at 8 o'clock.

Purpose

EDP is driven by the AGB and pressurizes the hydraulic system A (engine 1) or B (engine 2) of the aircraft.



Engine Driven Pump

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Inlet Cowl Anti-Icing System

Identify and Locate

The engine inlet cowl anti-icing system is located on the engine core and the fan case:

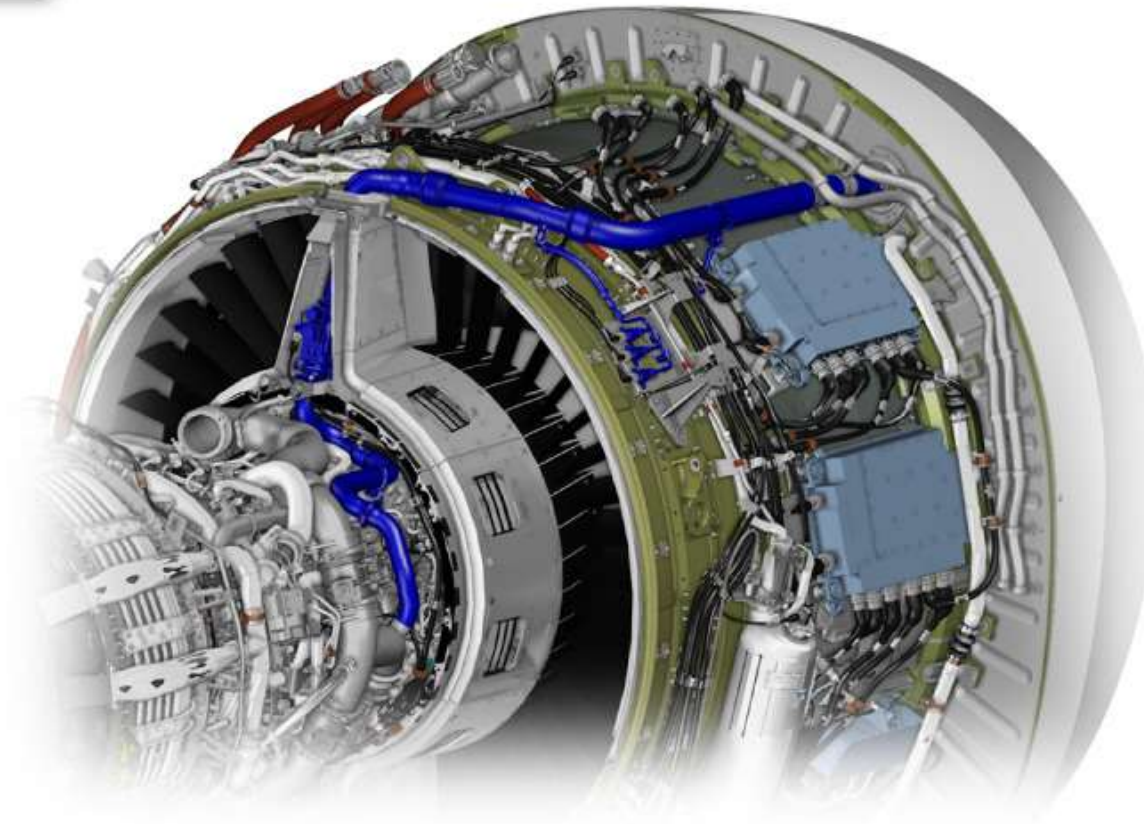
- Engine Anti-Ice (EAI) valve is located on front of engine core at 12 o'clock,
- EAI pressure switches are located on the fan case at 2 o'clock.

Purpose

The engine inlet cowl anti-ice system prevents ice formation on the inlet cowl surfaces in icing conditions during flight and ground operations.



PREVENTS ICE FORMATION ON THE INLET
COWL SURFACES



Inlet Cowl Anti-Icing System

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Thermal Anti Ice Valve and Pressure Sensors

Identify and Locate

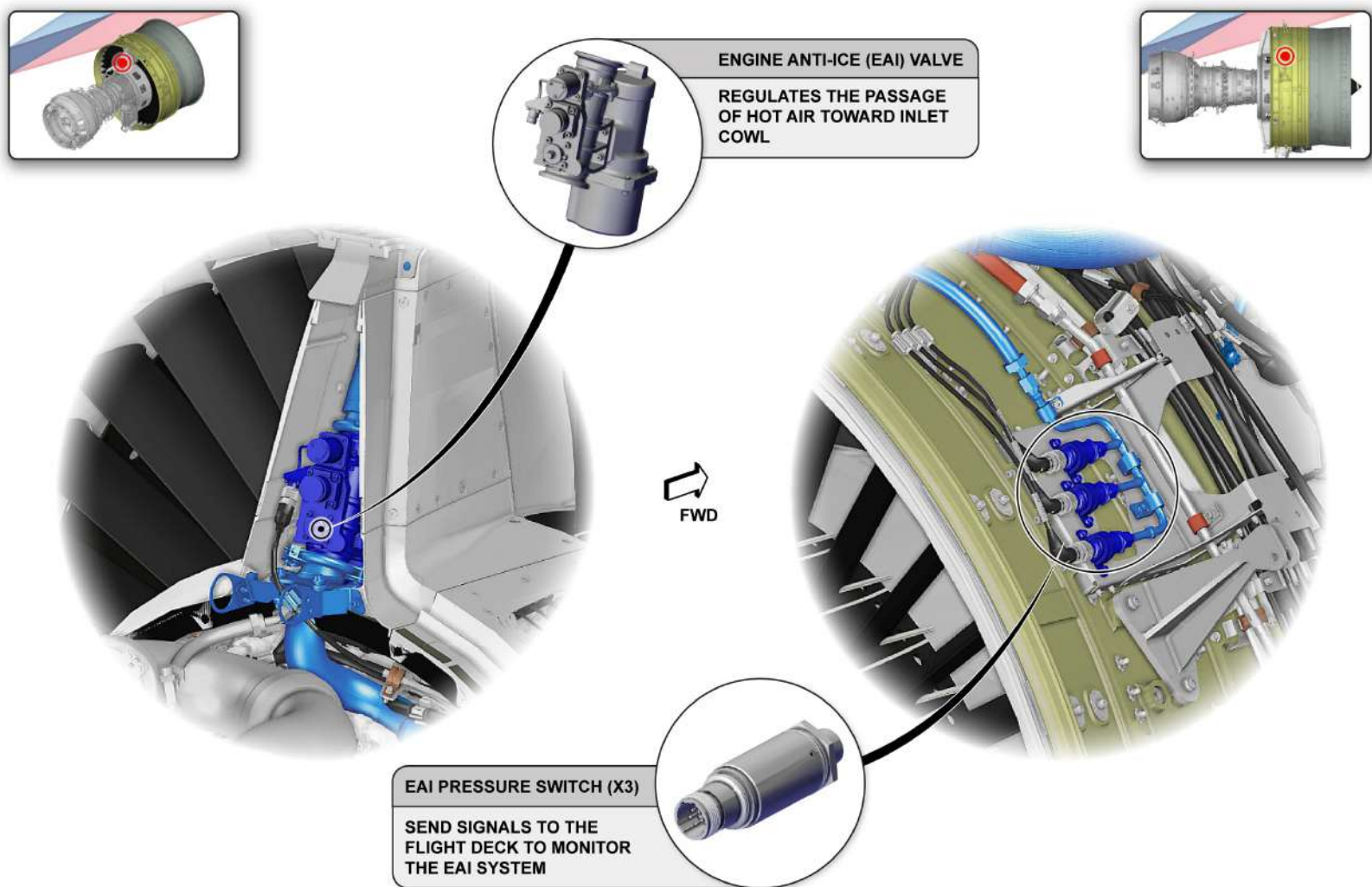
The Engine Anti Ice (EAI) valve is located downstream of the engine bleed manifold at 12 o'clock at the front of engine core.

3 pressure switches are located on the fan frame at 2 o'clock.

Purpose

The EAI valve permits and regulates the passage of hot air coming from the compressor through the duct towards the engine inlet cowl.

The 3 pressure switches are used for flight deck indication on EAI system.



Thermal Anti Ice Valve and Pressure Sensors

Engine Fan Section Fire Detection

Identify and Locate

On the engine fan section, the fire detection system uses fire detectors linked by 2 loops (A & B), which are located:

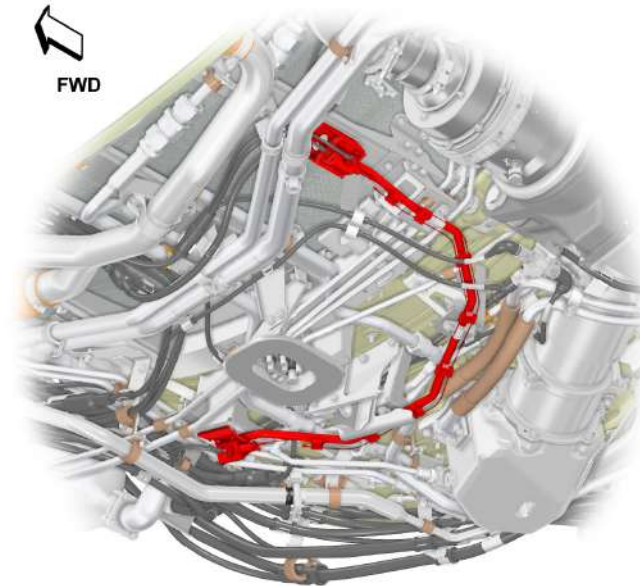
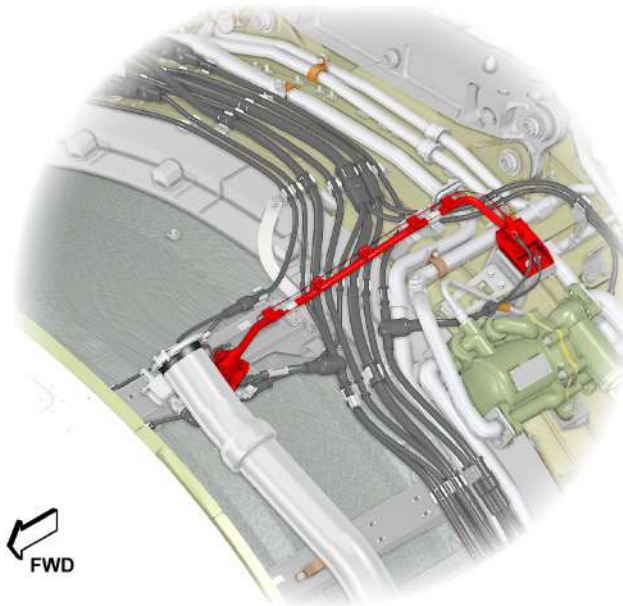
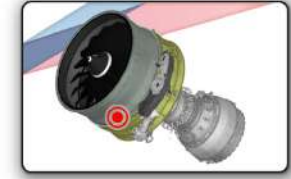
- one on the upper area,
- one on the lower area.

Purpose

The engine fire detectors provide the flight deck with adequate information on the engine condition: overheat or fire.



DETECTS FIRE OR OVERHEAT PRESENCE
AROUND THE FAN ENGINE SECTION



Engine Fan Section Fire Detection

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Engine Core Section Zone Fire Detection

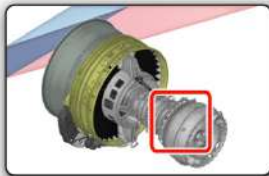
Identify and Locate

On the engine core section, the fire detection system uses fire detectors linked by 2 loops (A & B), which are located:

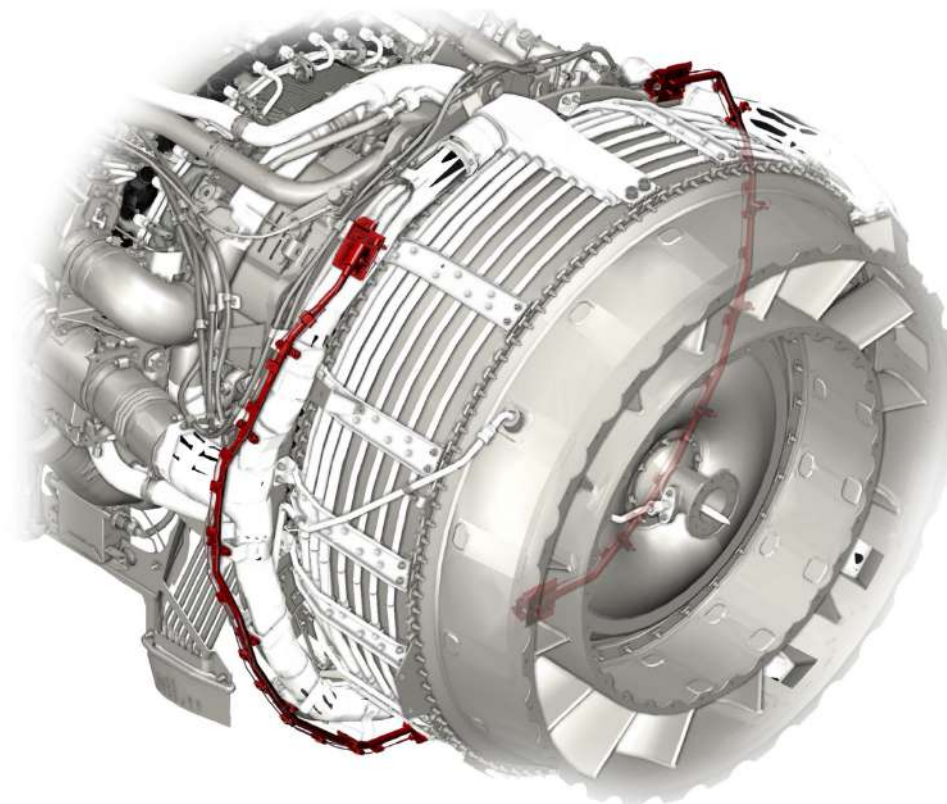
- one on the left area,
- one on the right area.

Purpose

The engine fire detectors provide the flight deck with adequate information on the engine condition: overheat or fire.



DETECTS FIRE OR OVERHEAT PRESENCE
AROUND THE ENGINE CORE SECTION



Engine Core Section Zone Fire Detection

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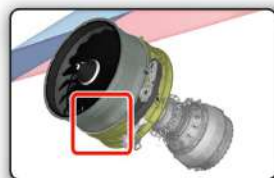
Engine Fan Section Drains

Identify and Locate

The engine fan section drain system consists of:
- the fan compartment drain lines located around the fan section,
the fan drain mast at 6 o'clock.

Purpose

The fan drain lines purpose is to drain all excess system fluids by gravity from the fan compartments and to provide a quick understanding of the leakage source.



- DRAINS OVERBOARD ALL EXCESS FLUIDS
- ALLOWS TO IDENTIFY THE LEAKAGE SOURCE



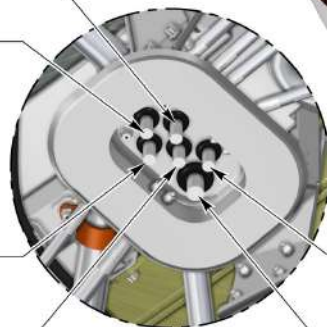
HAND CRANCKING
DRAIN LINE

INTEGRATED DRIVE
GENERATOR (IDG)
DRAIN LINE

OIL TANK
DRAIN LINE

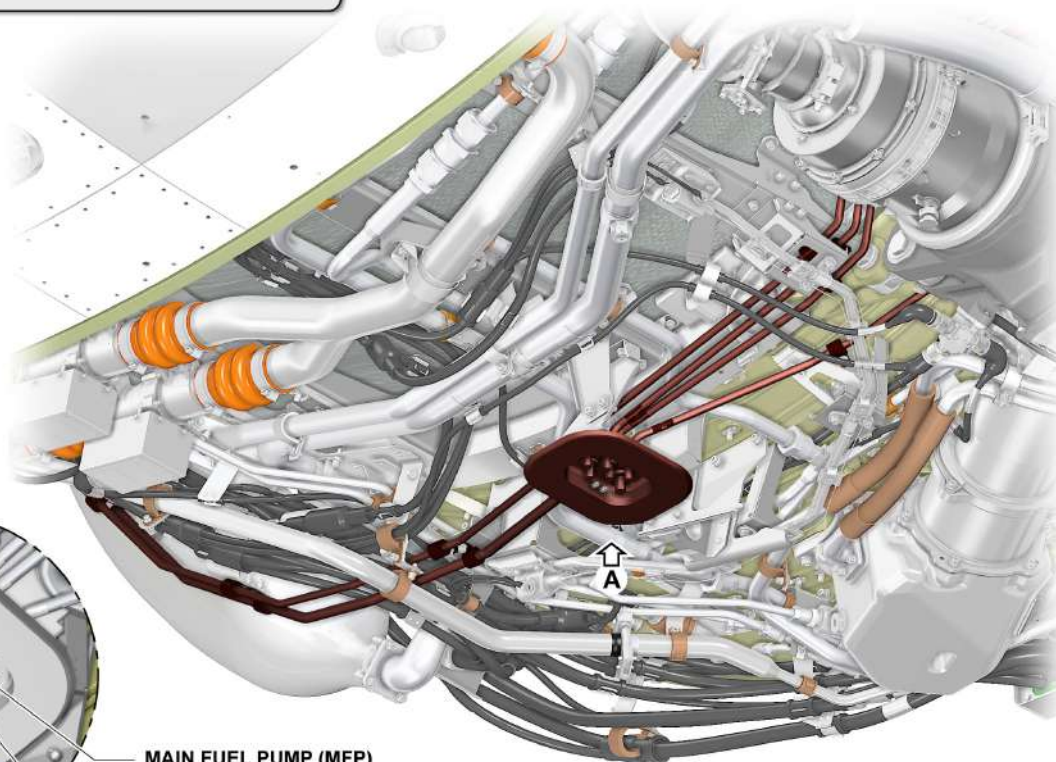
HYDRAULIC PUMP
DRAIN LINE

VIEW A



MAIN FUEL PUMP (MFP)
DRAIN LINE

PYLON DRAIN
LINE



Engine Fan Section Drains

Engine Core & Turbine Drains

Identify and Locate

The other engine drain system consists of:

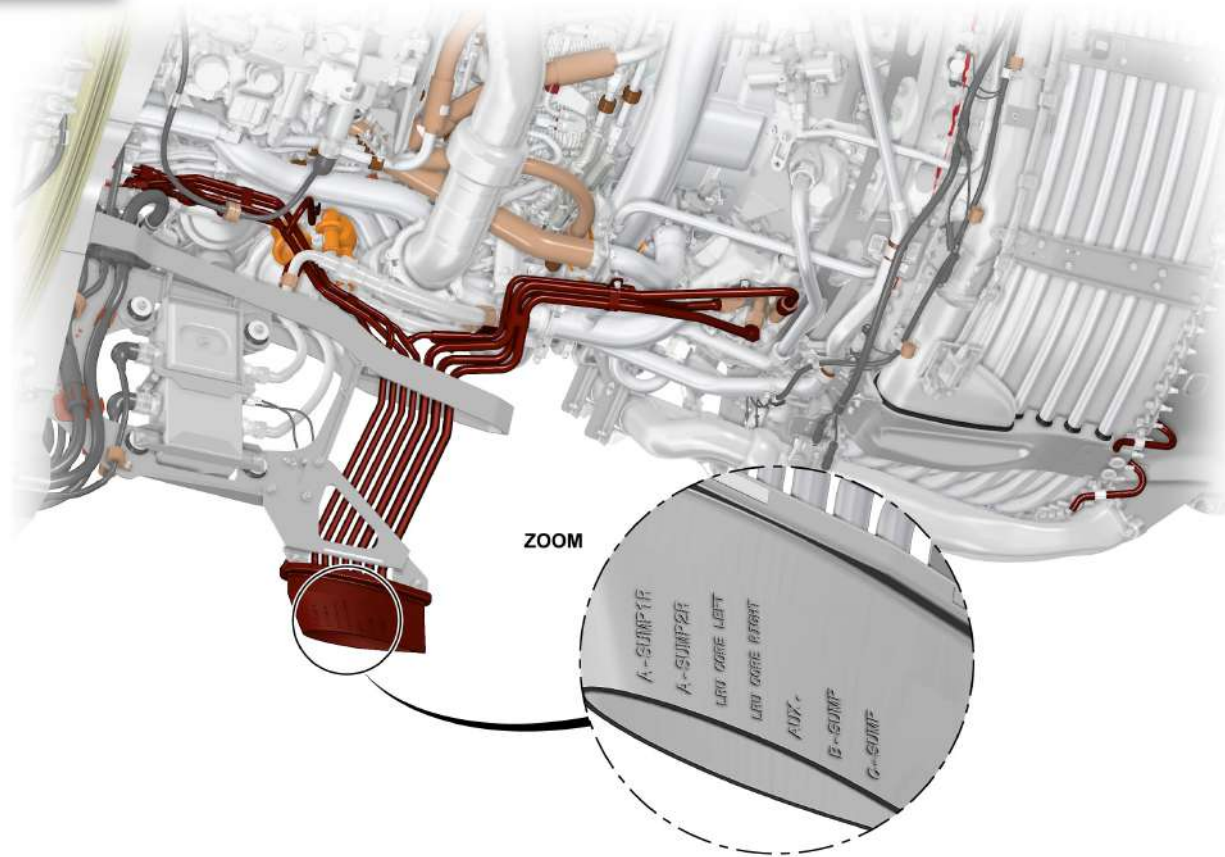
- the core compartment drain lines located around the core section and terminated at the core drain mast assembly at 6 o'clock,
- the Turbine Rear Vane (TRV) drain lines which are routed from the aft of combustion chamber at 6 o'clock to Thrust Reverser (T/R) mounted overboard drain.

Purpose

The core drain lines purpose is to drain all excess system fluids by gravity from the core compartments and to provide a quick understanding of the leakage source. The 2 TRV drain lines are used to drain fuel overboard in case of aborted engine start.



- DRAINS OVERBOARD ALL EXCESS FLUIDS
- ALLOWS TO IDENTIFY THE LEAKAGE SOURCE



Engine Core & Turbine Drains



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General Familiarization LEAP-1B



Basic Engine General

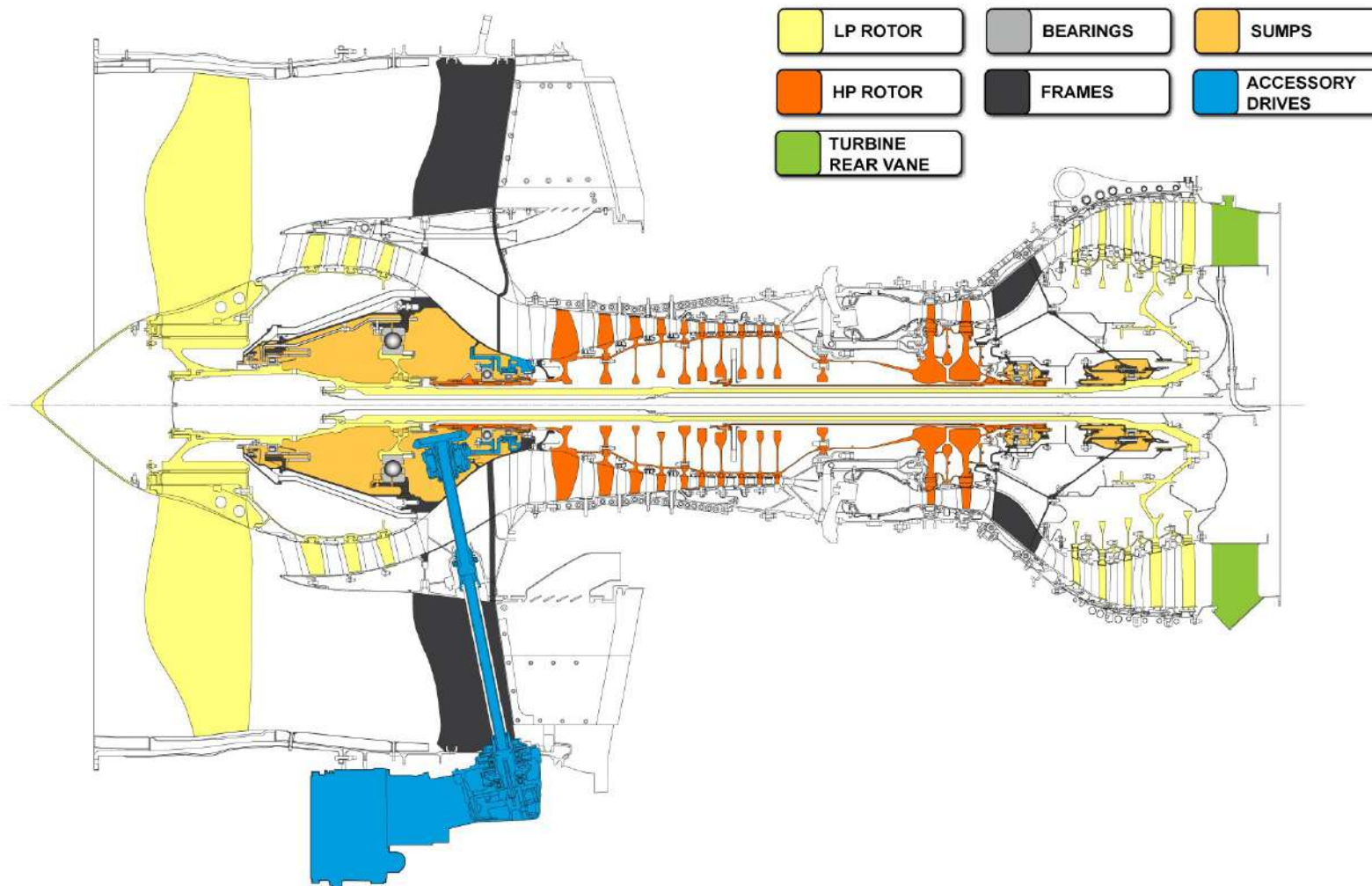
JAN 2016

Mechanical Arrangement

Identify and Locate

The LEAP-1B engine consists of:

- the Low Pressure (LP) rotating system (N1),
- the High Pressure (HP) rotating system (N2),
- 5 main bearings,
- 3 sumps,
- 2 structural frames,
- the Turbine Rear Vane (TRV),
- the accessory drives.



Mechanical Arrangement

V0

Main Shafts Bearings

Identify and Locate

The two engine rotors are supported by 5 bearings, identified from No.1 (at the front) to No.5 (at the rear).

No.1 roller bearing is inside sump A, at the front of the bearing No.1 and No.2 support assembly.

No.2 ball bearing is inside sump A, at the rear of the bearing No.1 and No.2 support assembly.

No.3 ball and roller bearings are inside sump A, in the No.3 bearing housing of the Inlet Gearbox (IGB).

No.4 roller bearing is inside sump B, between the hub of the Turbine Center Frame (TCF) and the rear shaft of the High Pressure Turbine (HPT).

No.5 roller bearing is inside sump C, between the No.5 bearing support and the shaft of the Low Pressure Turbine (LPT).

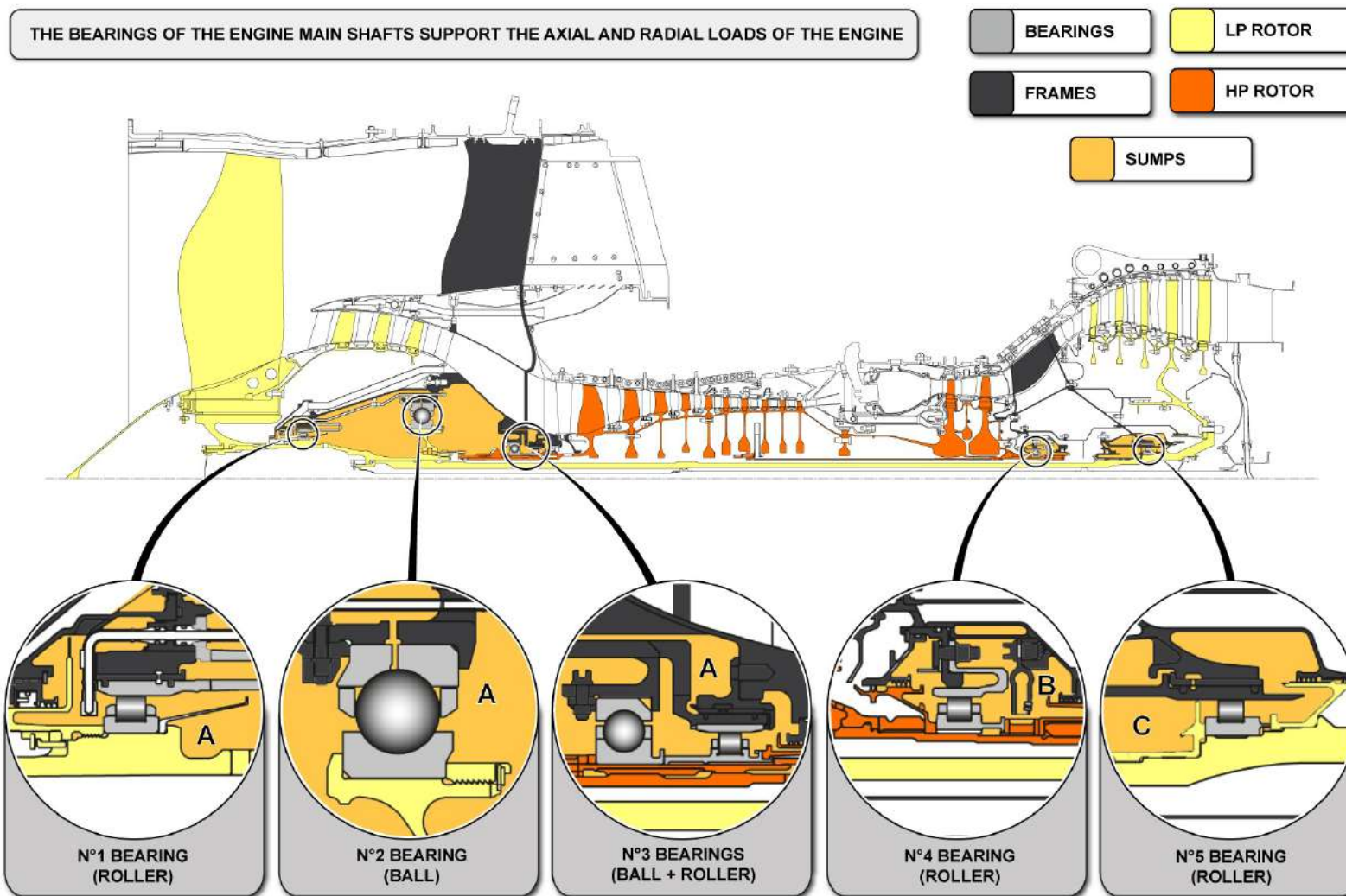
Purpose

There are two categories of bearings:

- the ball bearings which take up axial and radial loads,
- the roller bearings which take up only radial loads.

The Low Pressure Rotor (N1) is supported by two roller bearings (No.1 and No.5) and one ball bearing (No.2).

The High Pressure Rotor (N2) is supported by one roller bearing and one ball bearing (No.3) and one more roller bearing (No.4).



Main Shafts Bearings

Sump Seals

Identify and Locate

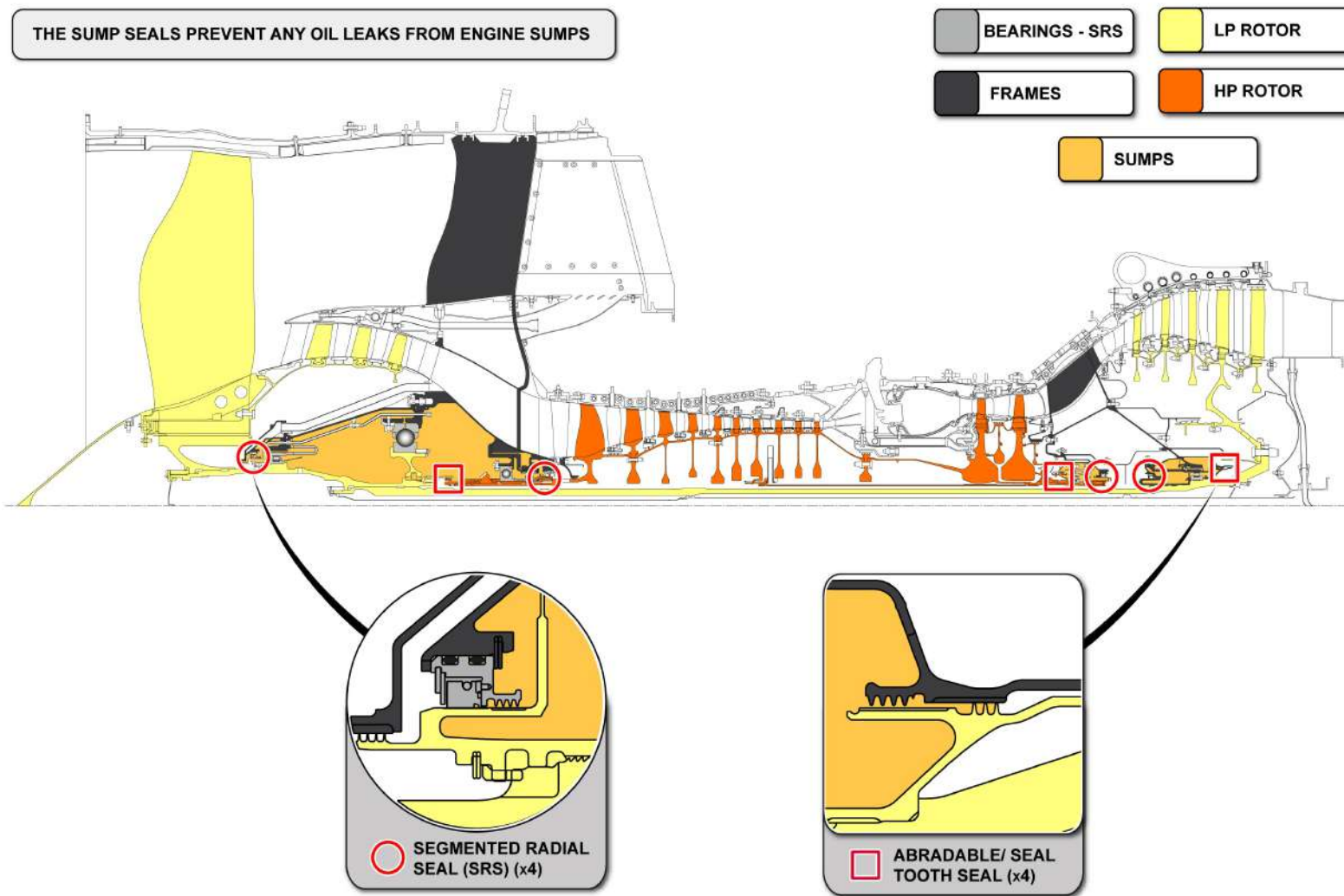
The vented A sump front sealing is realized with a Segmented Radial Seal(SRS) composed of a Carbon and a labyrinth seal, an intermediate classic abradable/seal tooth seal is located upstream the N°3 bearing and downstream a SRS ensures the aft sealing.

The non vented B sump sealing is done by an abradable/seal tooth seal upstream and a SRS downstream.

The non vented C sump sealing is done with a SRS upstream and an abradable/seal tooth seal downstream.

Purpose

The sump seals prevent any oil leaks from the engine sumps.



Sump Seals

V0

Performance and Limits

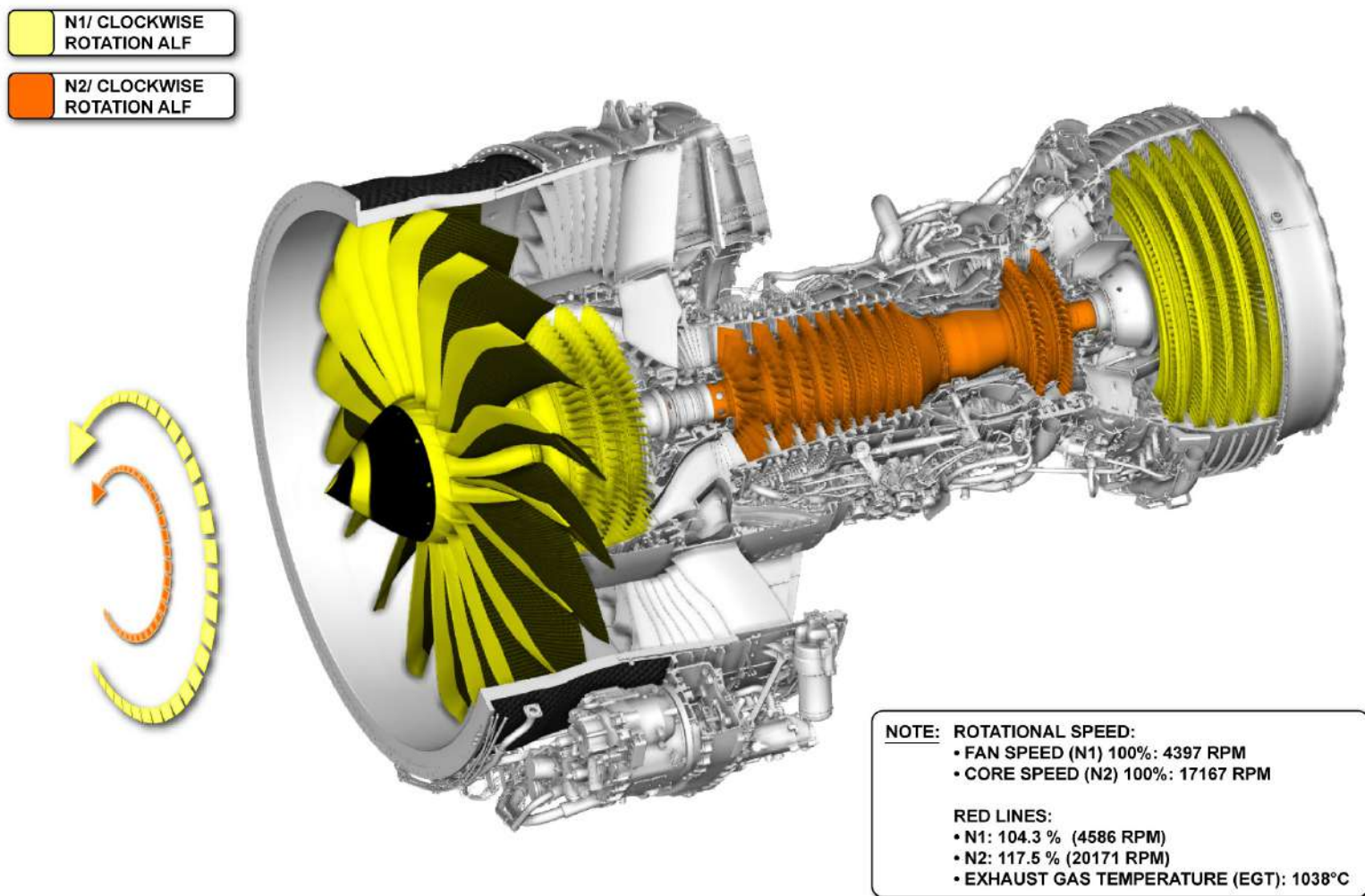
Identify and Locate

Rotational speed:

- fan speed (N1) 100%: 4397 RPM,
- core speed (N2) 100%: 17167 RPM.

Red lines:

- N1: 104.3% (4586 RPM)
- N2: 117.5% (20171 RPM)
- Exhaust Gas Temperature (EGT): 1038°C



Performance and Limits

V0

On-Condition Maintenance Philosophy

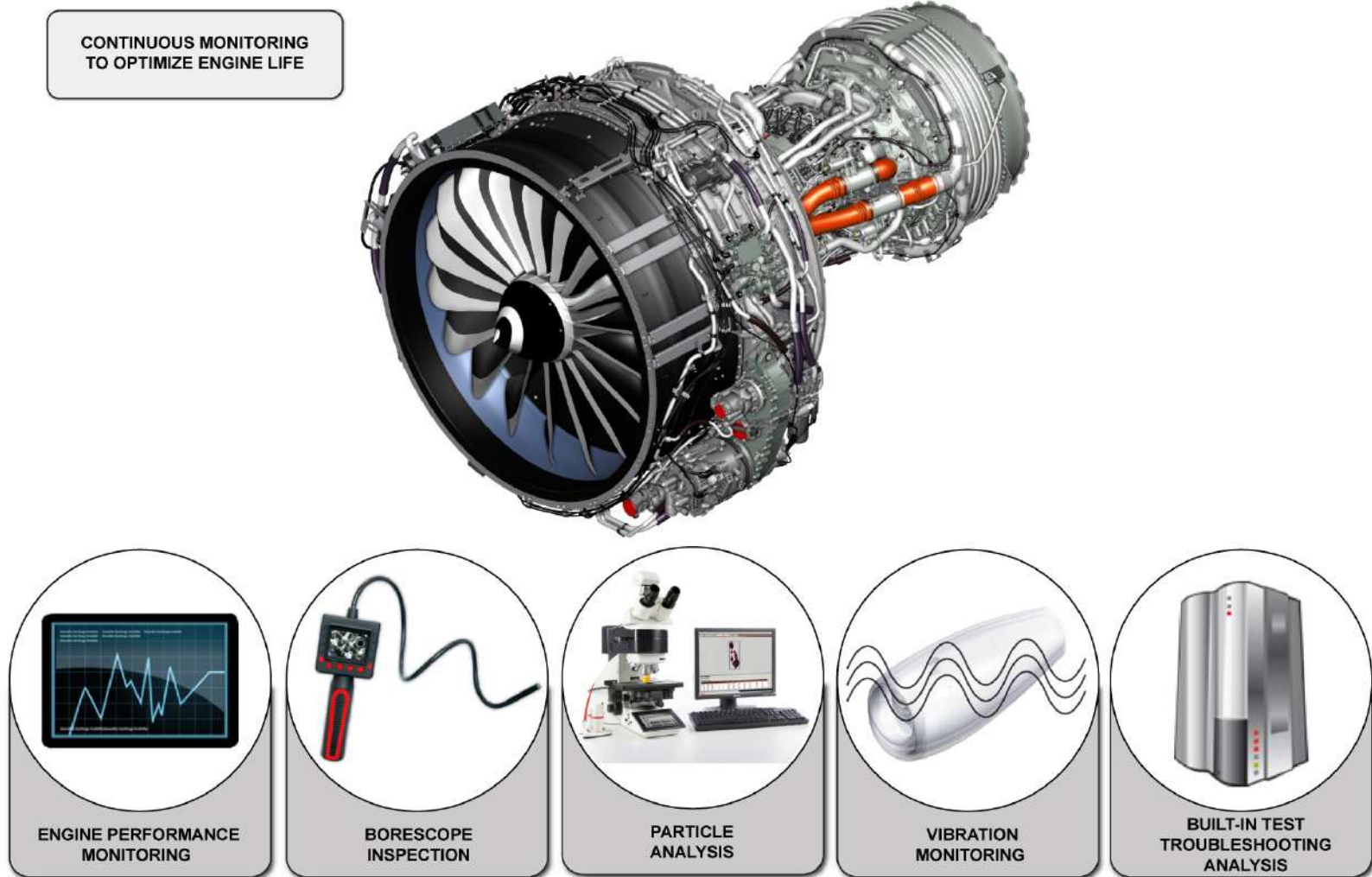
Identify and Locate

The engine uses all modern maintenance techniques to anticipate problems:

- physical parts/engine system inspections,
- borescopic inspection,
- particle analysis,
- engine vibration monitoring,
- engine health condition & performance trend monitoring,
- Built-In Test (BIT) troubleshooting analysis.

Purpose

The engine is designed in accordance with a preventive maintenance concept called "on-condition maintenance". This primary maintenance concept provides inspections or tests to determine the condition of units, systems, or portions of structure with regards to continued serviceability. That enables the engine to have no need for scheduled overhaul on accessories or engine modules. The maintenance on accessories or engine modules will be decided based upon actual condition of such accessories or modules.



On-Condition Maintenance Philosophy

V0

Borescope Ports

Identify and Locate

The borescope ports are on the engine external cases. Borescope Inspection (BSI) port locations are located on the right hand side (around 4 o'clock) of the engine with one left hand side at the combustion chamber port. All ports are fitted with self-locking plugs except for the Low Pressure Compressor (LPC) where there is no plug.

LPC:

The 2 ports (B01 and B02) are located at 3:00 o'clock through the fan flow path fairing.

High Pressure Compressor (HPC):

The 10 ports (B03 to B12) are located between 3:00 and 5:00 o'clock at each HPC stage.

Combustion chamber:

The 2 ports for the combustion chamber are:

- B13 at 3 o'clock,
- B14 at 11 o'clock,

Left and right igniters ports can be used as well.

High Pressure Turbine (HPT):

The 2 ports (B15 and B16) are located at 3:00 o'clock.

Low Pressure Turbine (LPT):

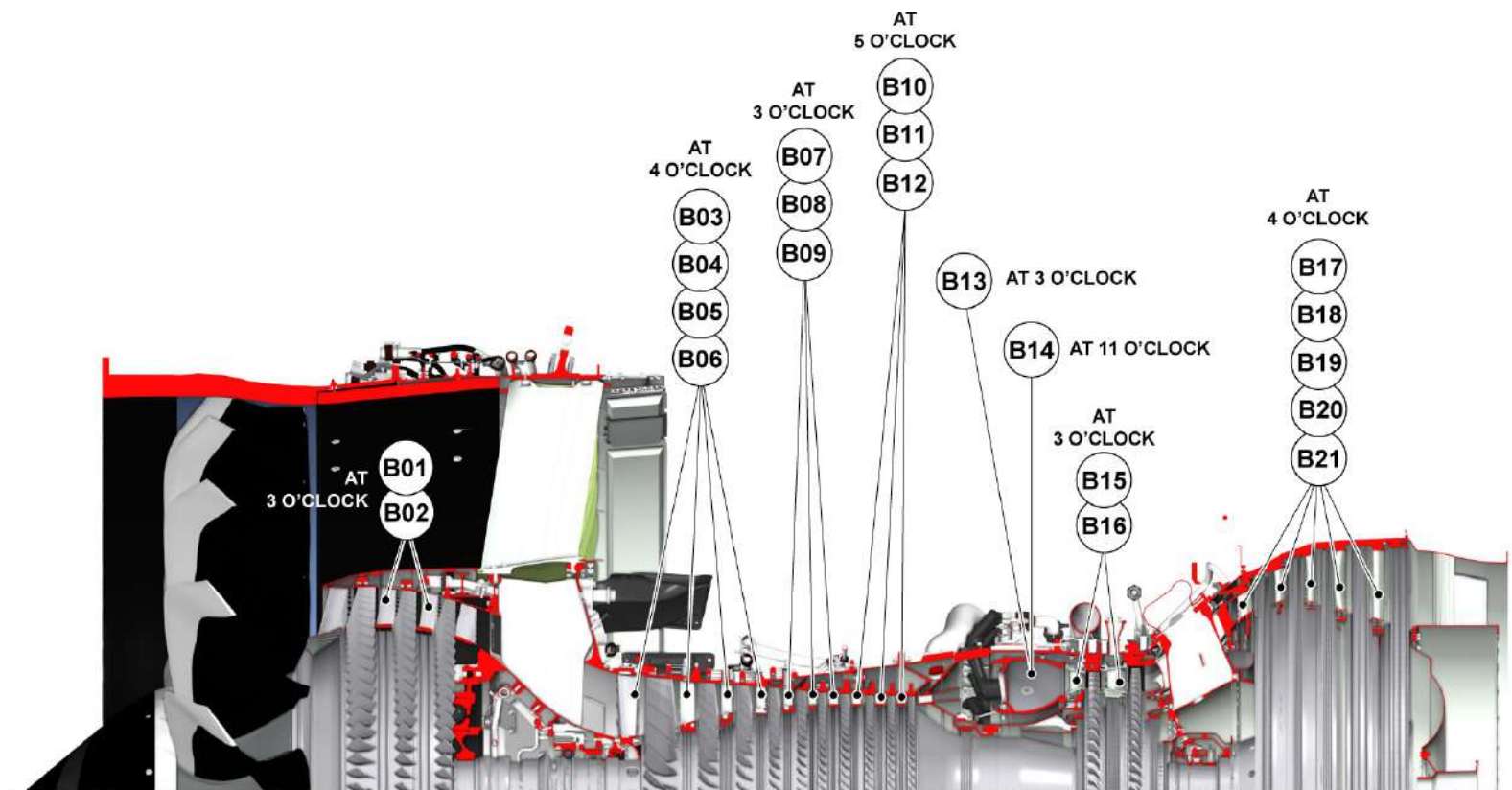
The 5 ports (B17 to B21) are located at 4:00 o'clock.

Purpose

There are 21 borescope ports identified from B01 to B21.

The purpose is to provide access for visual inspection of engine internal parts.

ENABLE VISUAL INSPECTION OF ENGINE INTERNAL PARTS



Borescope Ports

V0

Modular Breakdown

Identify and Locate

The LEAP-1B is a modular concept engine.

There are 3 Major Modules (MM):

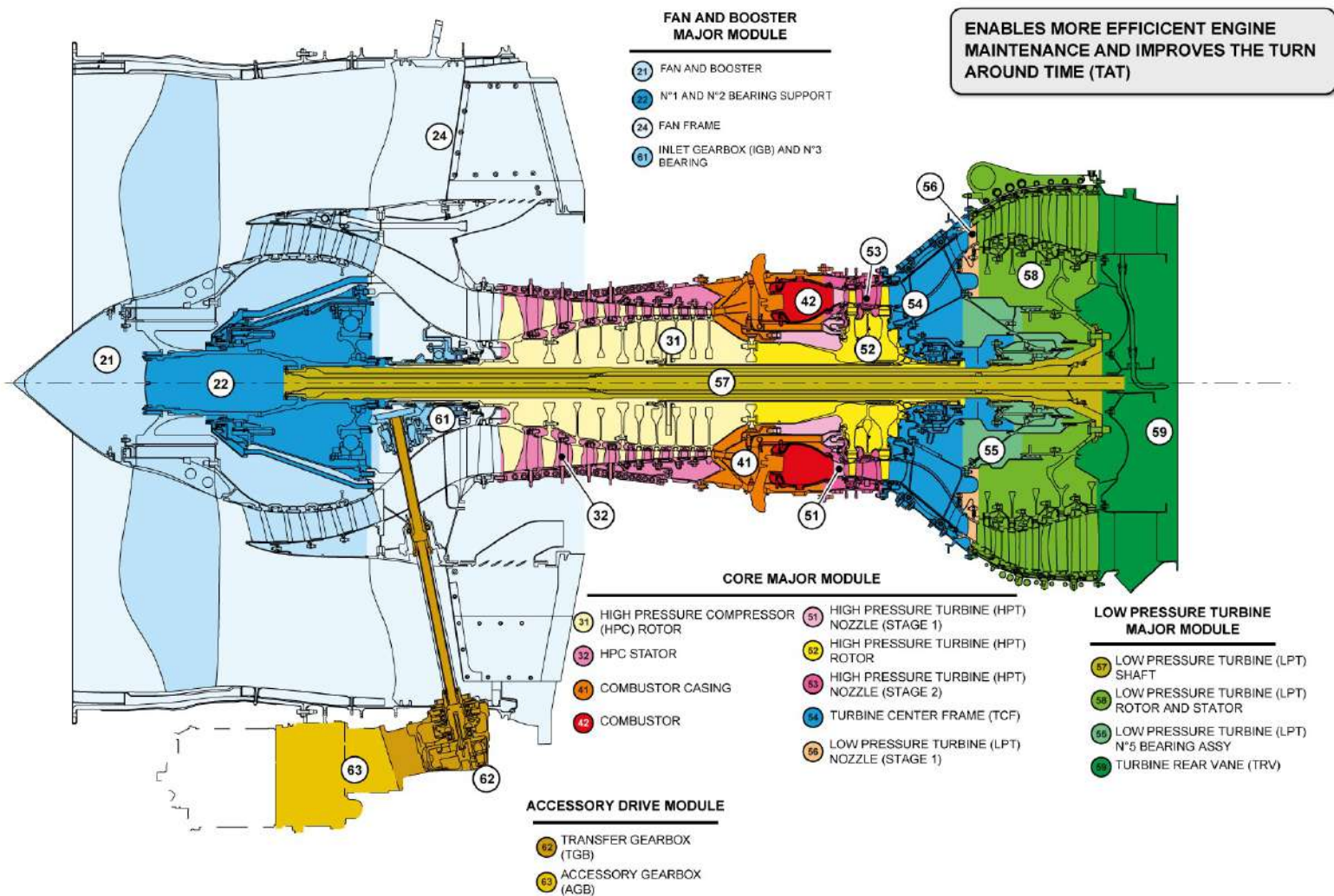
- the fan major module,
 - the core engine major module,
 - the Low Pressure Turbine (LPT) major module,
- and the accessory drive section.

The 3 Major Modules (MM) and the accessory drive section are divided into 20 Minor modules (Mm).

Purpose

With this modular concept, it is possible to replace a damaged minor or major module.

This enables a more efficient maintenance of the engine and improves the Turn Around Time (TAT).



Modular Breakdown

V0



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General Familiarization LEAP-1B



Basic Engine Air Inlet Section

Air Inlet Section

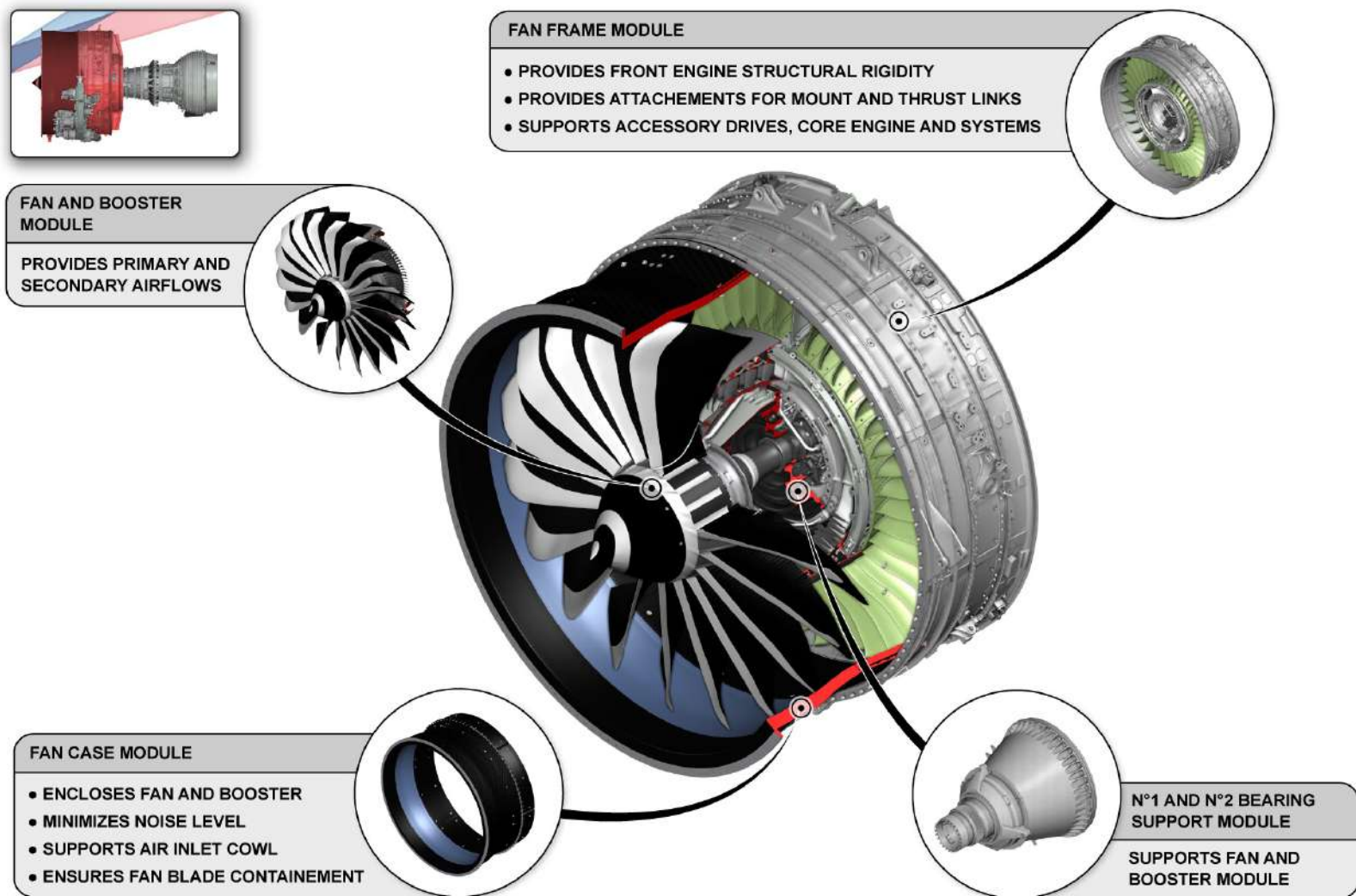
Identify and Locate

The air inlet section is behind the air inlet cowl, at the front of the engine.

Purpose

The main purposes of the air inlet section are to:

- provide the primary and secondary airflows,
- enclose the fan and booster module,
- support the fan and booster module, by the No.1 and No.2 bearing support module,
- support the front of the High Pressure Compressor (HPC) section by the fan frame module,
- support the accessory drives by engine mount fittings and links,
- provide attachments to the forward mount and the thrust links,
- provide structural rigidity at the front of the engine,
- support various powerplant and engine system accessories,
- minimize the noise levels of the fan area,
- support the air inlet cowl,
- ensure fan blade containment.



Air Inlet Section

V0

Air Inlet Section - Details

Identify and Locate

The front spinner is at the front of the engine, forward of the front platform shroud.

The platform front shroud is located between the front spinner and the fan disk and fan blades.

The 18 fan blades are installed behind the front spinner at the front of the engine.

The platform front shroud is between the front spinner cone and the fan blades.

The one-piece acoustic shroud is just aft of the air intake and forward of the abradable liner.

Six acoustic panels are installed aft of the abradable liner.

- to receive the fan blade balance weights that are used to balance the engine rotor.

The fan blades provide kinetic energy to the primary and secondary airflows.

The purposes of the acoustic shroud and the 6 acoustic panels are to:

- provide a smooth aerodynamic surface for the secondary airflow,
- reduce noise.

Purpose

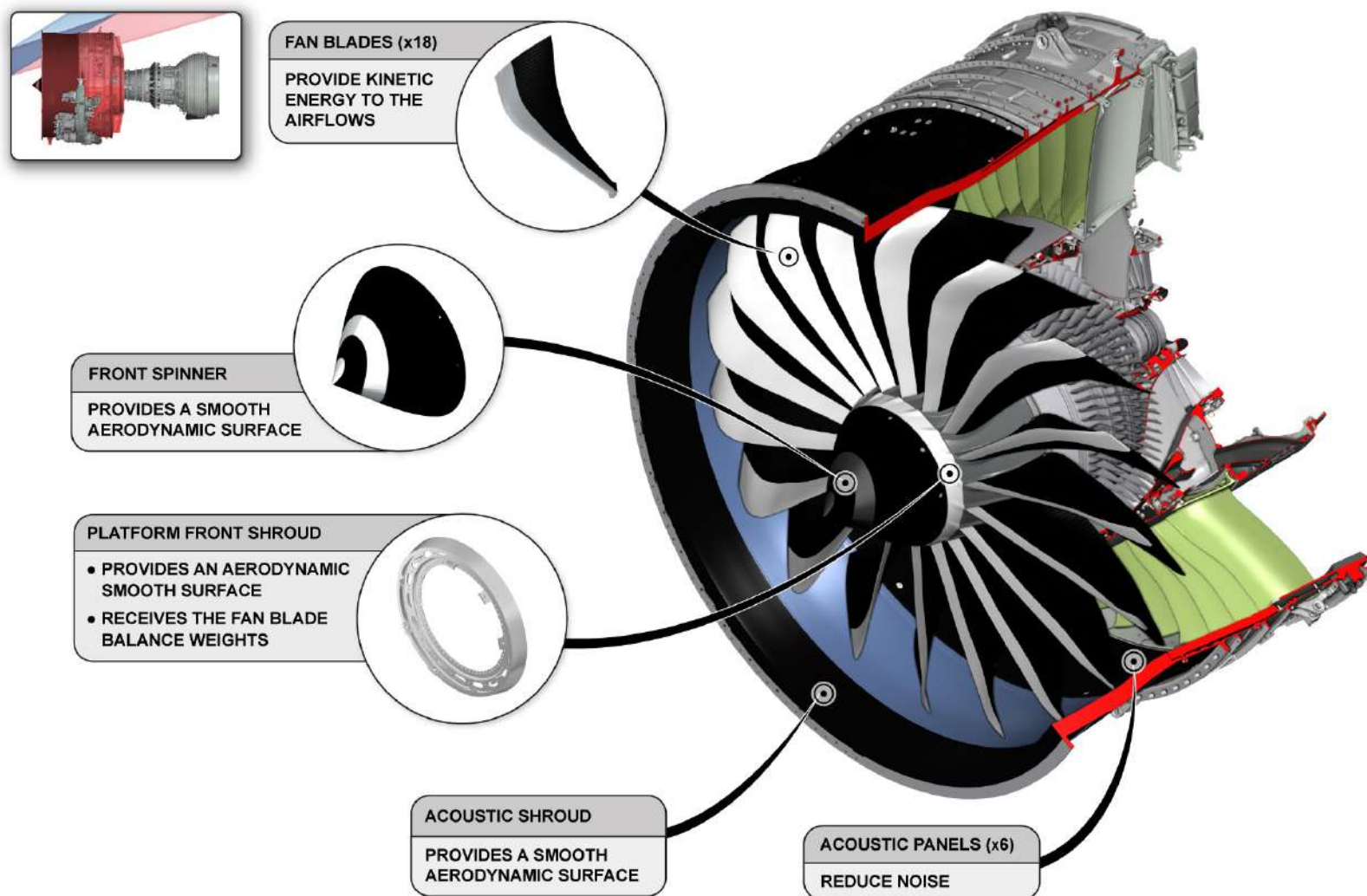
The purpose of the front spinner is to provide a smooth aerodynamic surface to direct the airflow into the fan.

The profile is optimized to:

- prevent Foreign Object Damage (FOD) ingestion in the primary airflow,
- prevent icing phenomenon,
- reduce the noise level.

The main functions of the platform front shroud are:

- to provide an aerodynamic and smooth surface for the airflow,



Air Inlet Section - Details



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General Familiarization LEAP-1B



Basic Engine High Pressure Compressor Section

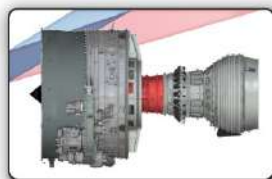
High Pressure Compressor Section

Identify and Locate

The High Pressure Compressor (HPC) section is between the air inlet section and the combustion diffuser section.

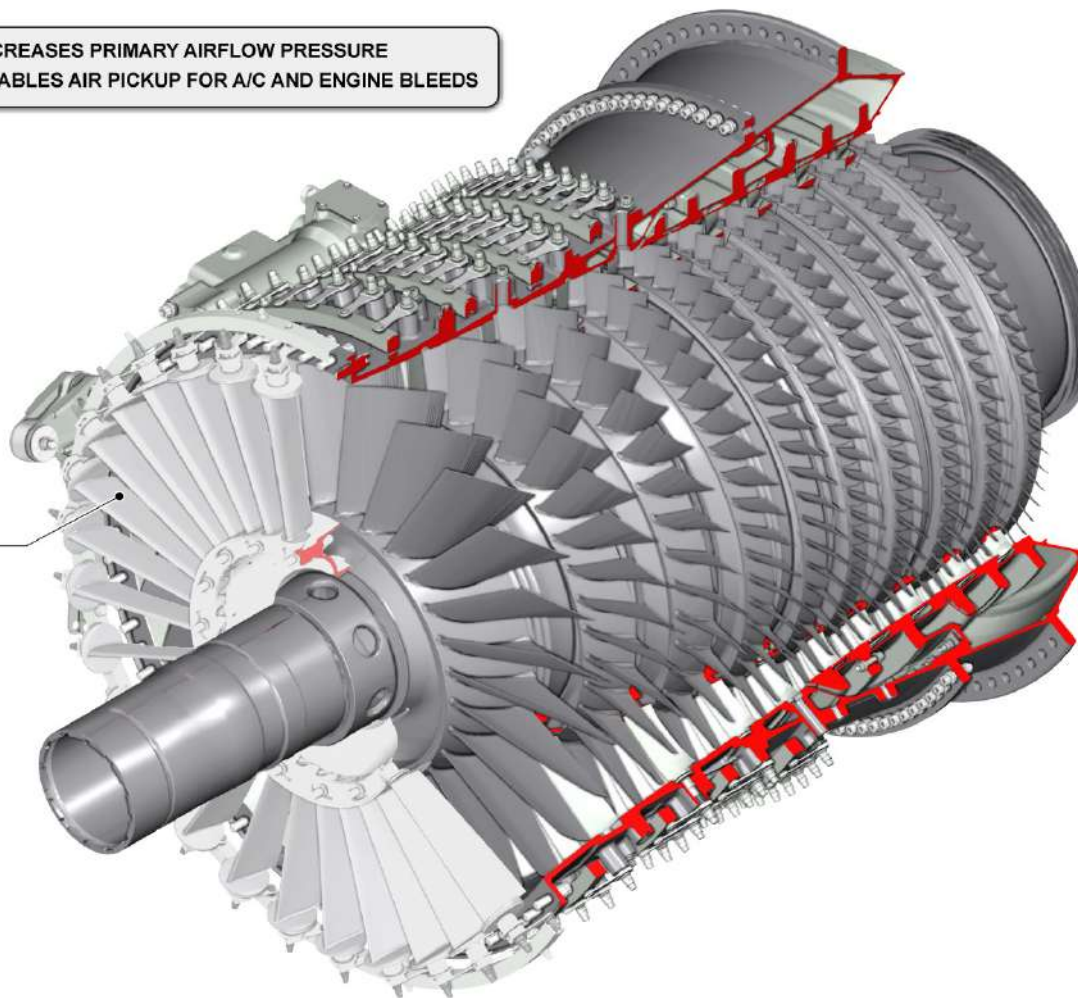
Purpose

The HPC section increases the pressure of the primary airflow for combustion, and provides air for cooling and for customer bleed.



- INCREASES PRIMARY AIRFLOW PRESSURE
- ENABLES AIR PICKUP FOR A/C AND ENGINE BLEEDS

INLET GUIDE VANES (IGVs)
NOT PARTS OF THE HIGH PRESSURE
COMPRESSOR SECTION



High Pressure Compressor Section

V0



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General Familiarization LEAP-1B



Basic Engine Combustion Diffuser Section

JAN 2016

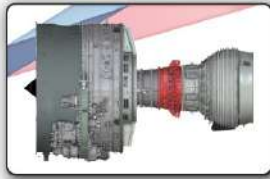
Combustion Diffuser Section

Identify and Locate

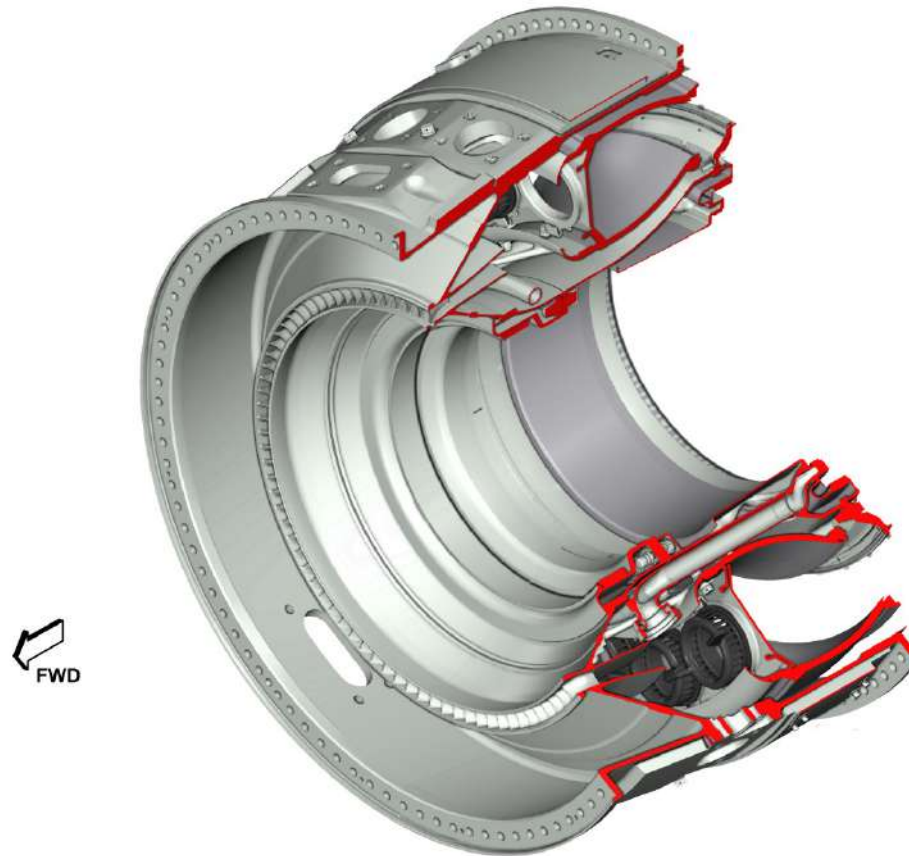
The combustion diffuser section is between the High Pressure Compressor (HPC) section and the High Pressure Turbine (HPT) section.

Purpose

The combustion diffuser section mixes fuel and the air from High Pressure Compressor (HPC) section for efficient and uniform combustion. The combustion diffuser section also provides the structural load path between the rear HPC stator case and the High Pressure Turbine (HPT) stator case, and provides for the 10th stage air.



- BURNS AIR/ FUEL MIXTURE
- DELIVERS GAS ENERGY FROM COMBUSTION TO THE TURBINE SECTION
- ENABLES AIR PICKUP FOR A/C AND ENGINE BLEEDS



Combustion Diffuser Section



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General Familiarization LEAP-1B



Basic Engine Turbine Section

Turbine Section

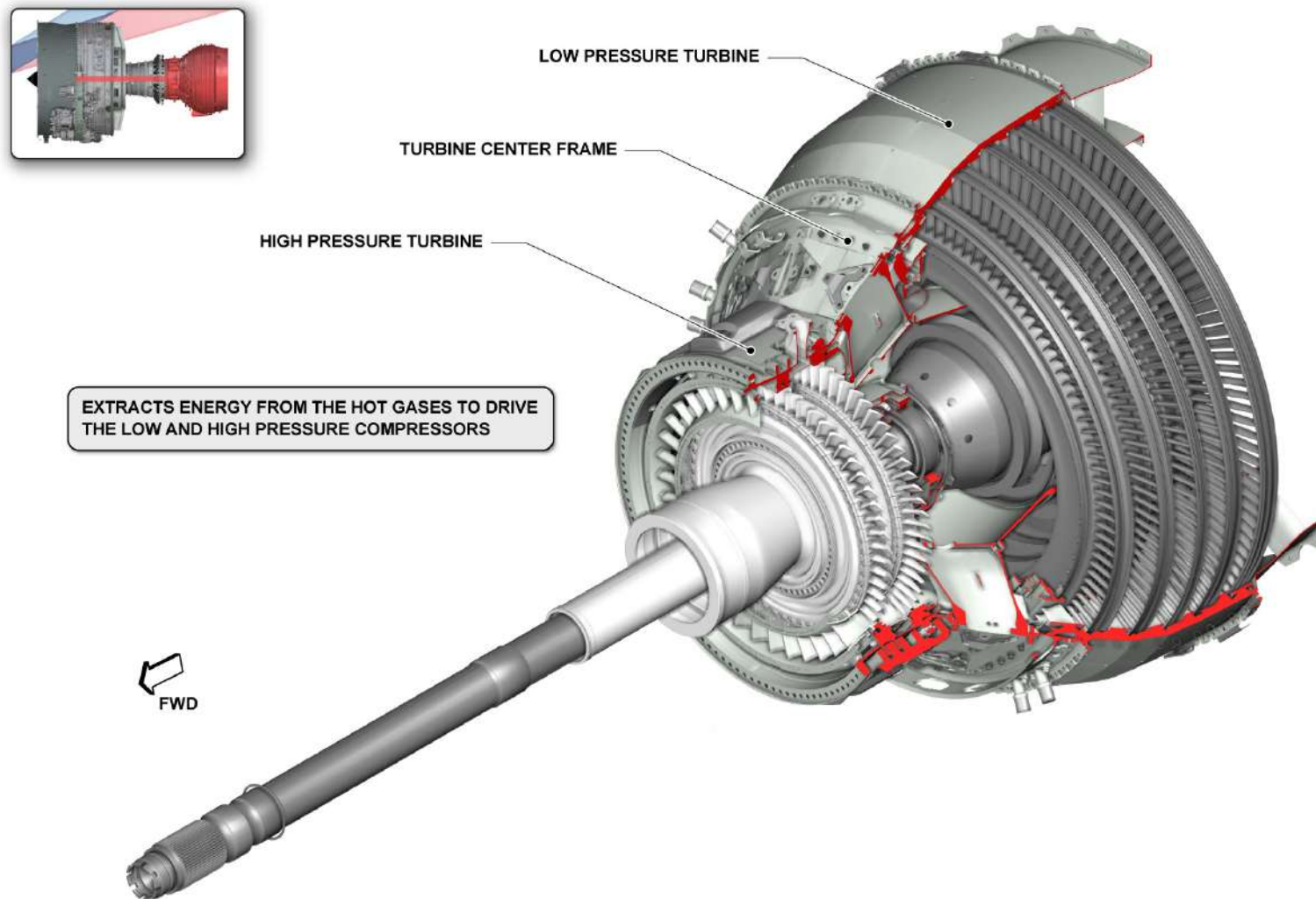
Identify and Locate

The turbine section is at the rear of the engine behind the combustion diffuser section.

Purpose

The turbine section drives the LP and HP compressor rotors:

- the High Pressure Turbine (HPT) extracts the energy from the hot gases to drive the High Pressure Compressor (HPC) rotor,
- the Low Pressure Turbine (LPT) rotor & stator module extracts the energy of gases from the HPT to drive the fan and booster.



Turbine Section



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General Familiarization LEAP-1B



Basic Engine Accessory Drives

Accessory Drives

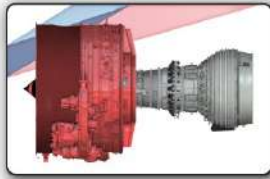
Identify and Locate

The accessory drive section is at the front of the engine, on the fan frame module between 6 o'clock and 8 o'clock.

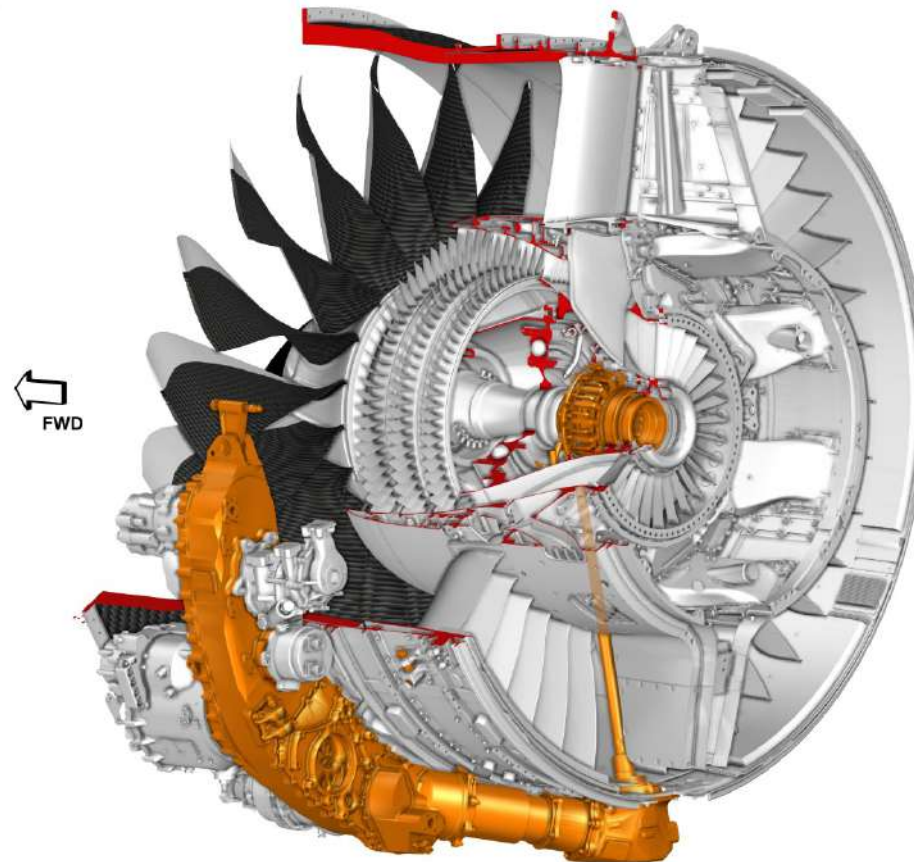
Purpose

The purposes of the accessory drive section are:

- In operating mode: to provide rotation to engine and aircraft accessories,
- In starting mode: to provide rotation to the High Pressure (HP) rotor from the Pneumatic Air Starter (PAS).



- IN OPERATING MODE: TO PROVIDE ROTATION TO ENGINE AND AIRCRAFT ACCESSORIES
- IN STARTING MODE: TO PROVIDE ROTATION TO THE HIGH PRESSURE (HP) SHAFT FROM THE PNEUMATIC AIR STARTER (PAS)



Accessory Drives



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General Familiarization LEAP-1B



Engine Systems

Fuel And Control



Engine Fuel and Control

Identify and Locate

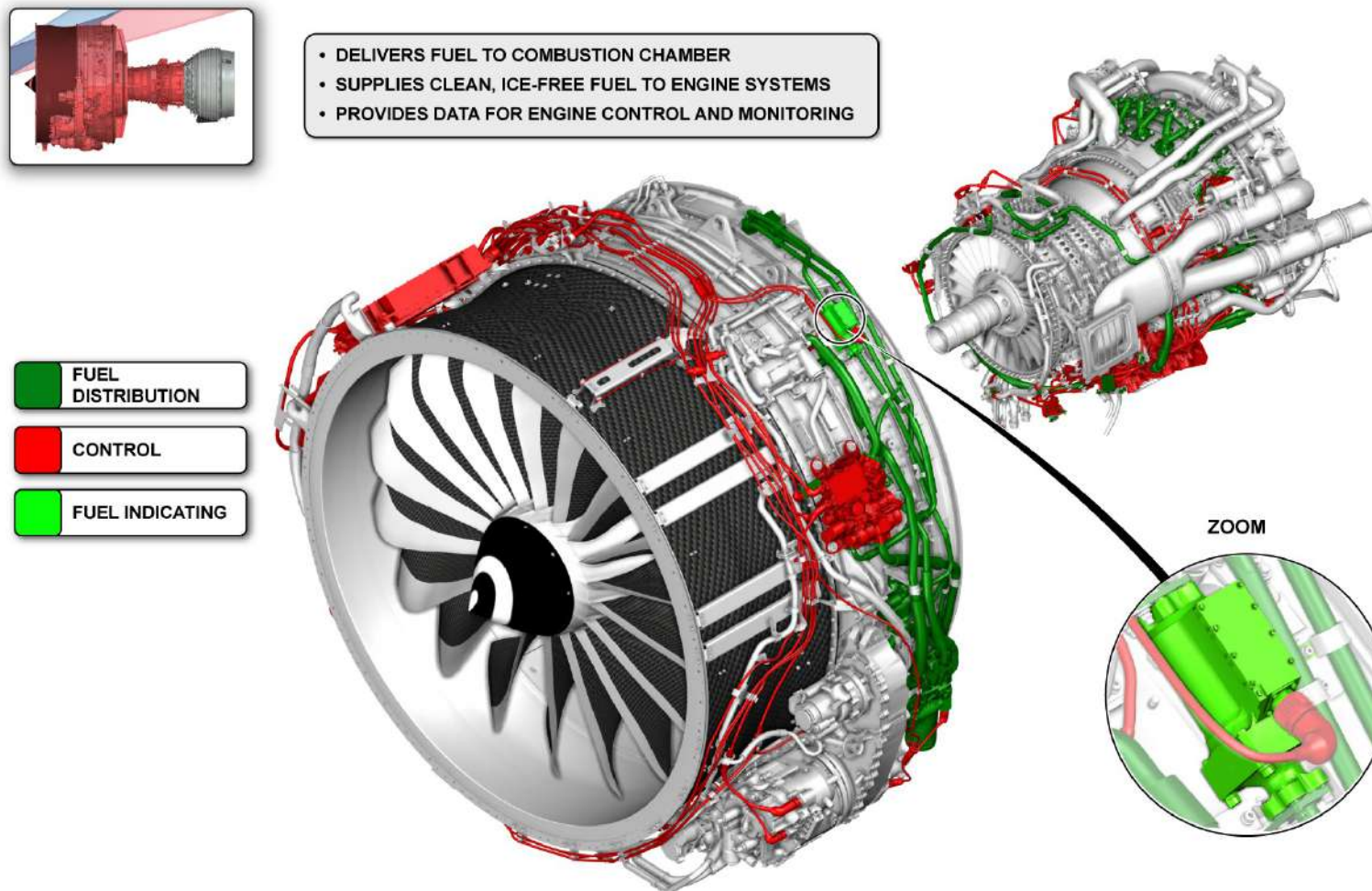
The engine fuel and control system includes the fuel distribution, control and indicating components which are located:

- on the air inlet section between 1 and 4 o'clock and between 8 and 12 o'clock,
- on the engine core assembly at 8 o'clock,
- and all around the combustor section.

Purpose

The engine fuel and control system:

- calculates the required fuel quantity to generate the commanded thrust,
- regulates and supplies the fuel flow to the combustion chamber,
- supplies the fuel flow to the actuated valves and actuators,
- provides data for flight deck indication and engine control, fault monitoring and diagnosis.



Engine Fuel and Control

V0

Fuel - Distribution Components 1/2

Identify and Locate

The Main Fuel Pump (MFP) is located on the aft side of the Accessory Gearbox (AGB) at 8 o'clock and is secured with a V-Band clamp.

The Main Fuel Filter (MFF) assembly is on the air inlet section at 8 o'clock, just behind the Permanent Magnet Alternator (PMA). The MFF comprises the MFF housing and the fuel filter cartridge.

The 18 fuel nozzles are equally distributed all around the combustion section.

The Servo Fuel Heater (SFH) is installed on the air inlet section at 9 o'clock, just aft of the AGB.

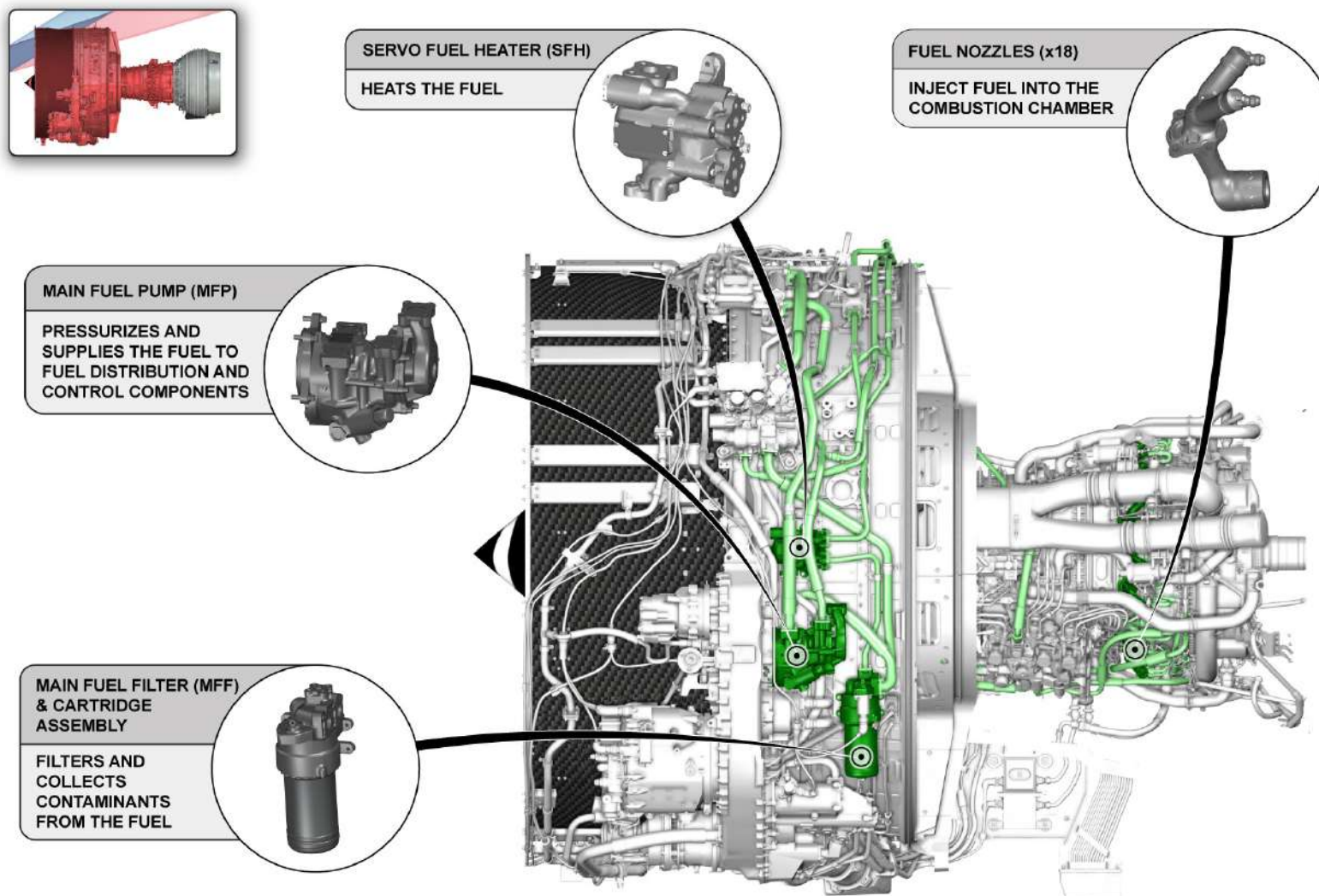
Purpose

The MFP pressurizes the fuel coming from the aircraft fuel tanks and supplies it to distribution and control components.

In the MFF assembly, the fuel filter cartridge filters and collects any contaminant from the fuel to protect downstream components.

The SFH heats the fuel for servo valves and actuators.

The 18 identical fuel nozzles distribute and spray the fuel into the combustion chamber.
Four of the fuel nozzles are enriched via a specific pressure line.



Fuel - Distribution Components 1/2

V0

Fuel - Distribution Components 2/2

Identify and Locate

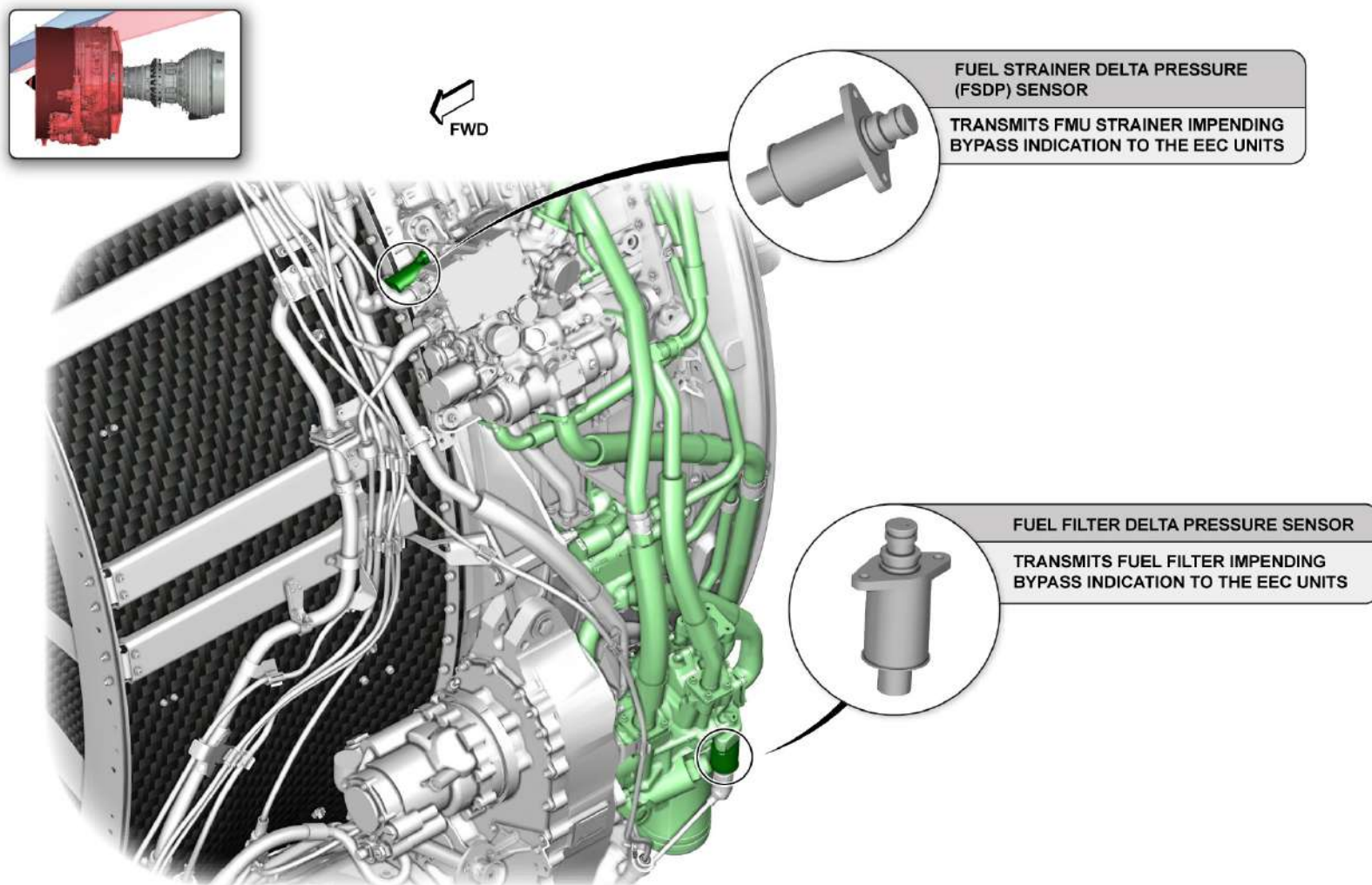
The fuel filter delta pressure sensor is mounted on the cap of the Main Fuel Filter (MFF) which is installed on the air inlet section at 8 o'clock.

The Fuel Strainer Delta Pressure (FSDP) sensor is mounted on the Fuel Metering Unit (FMU) at 10 o'clock.

Purpose

The fuel filter delta pressure sensor transmits the pressure drop across the fuel filter and is used to detect fuel filter clogging and impending bypass for aircraft and maintenance annunciation.

The FSDP transmits the pressure drop across the fuel strainer to the EEC units.
Both EEC units (channel A and channel B) use this information to sense impending bypass of strainer due to an increase in differential pressure.



Fuel - Distribution Components 2/2

Fuel - Control Components 1 of 3

Identify and Locate

The Electronic Engine Control (EEC) units are mounted on the air inlet section between the 1 and 3 o'clock positions. The EEC units are mounted on shock absorbers for vibration damping. The EEC units are equipped with electrical harness connection points, and an electrical bonding strap.

The Pressure Sub-System (PSS) box is mounted on the air inlet section at 4 o'clock. The PSS box is mounted on shock absorbers for vibration damping.

The data entry plug is attached to the engine via a lanyard and mounted on the PSS box.

The High Pressure Compressor (HPC) section inlet temperature (T25) sensor is mounted between the 3 and 4 o'clock positions on the fan frame hub with the sensing element installed in the primary flow path.

The engine wiring harnesses are installed all around the engine, connecting the EEC units to engine accessories, sensors and to the aircraft through the pylon.

Purpose

The first main function of the EEC is to process system input information and perform calculations and comparisons with stored memory information. The second main function of the EEC is to adjust the fuel and air systems in accordance with thrust command, in order to produce the requested thrust and the necessary engine airflow for cooling. The EEC

transmits information to the aircraft for flight deck, maintenance reporting and engine condition monitoring.

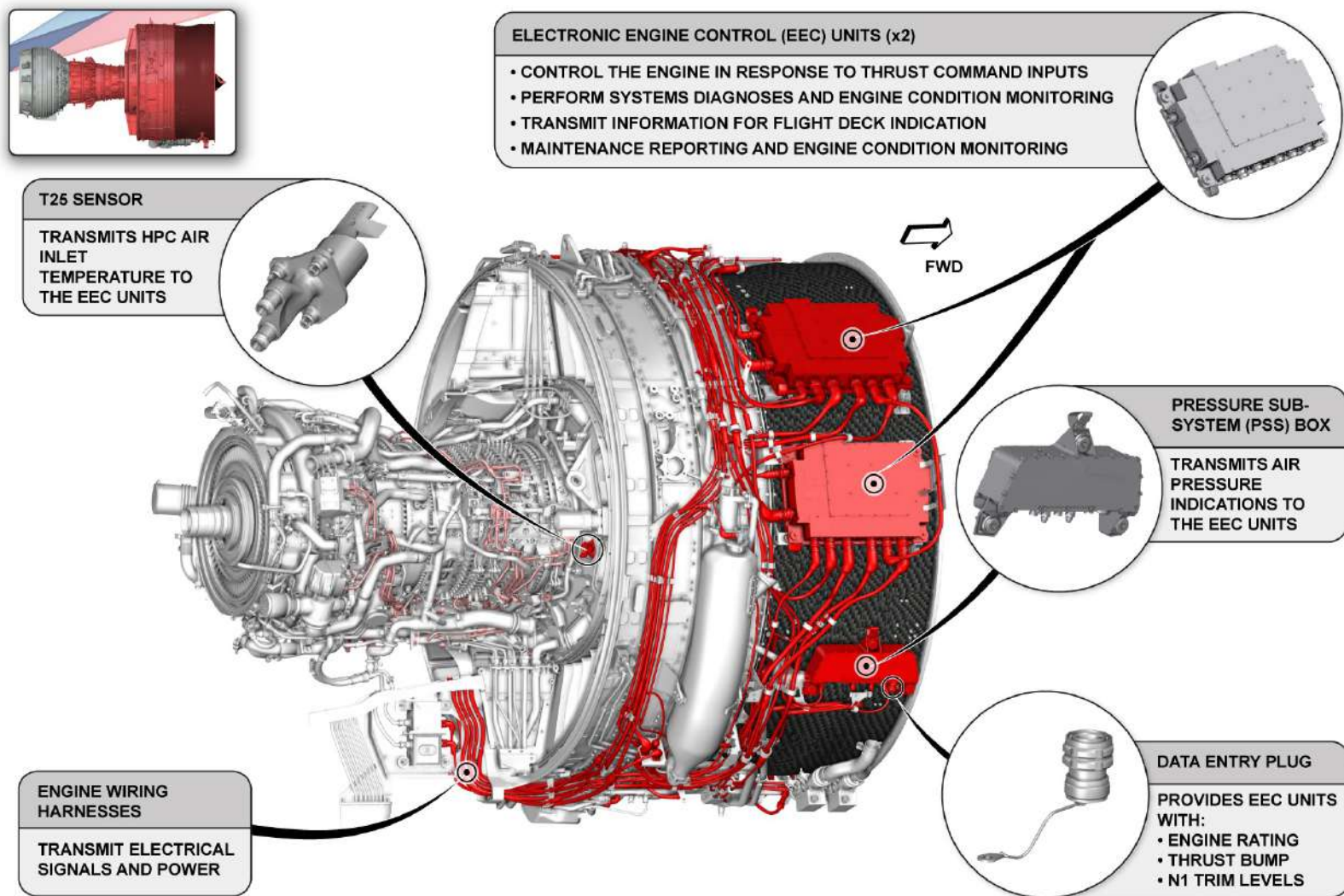
The first main function of the PSS box is to perform signal conversions of the pressure system inputs and communicate with each EEC unit (channel A and channel B) for engine control, fault monitoring and diagnosis/prognosis monitoring.

The second main function of the PSS box is to provide Cross-Channel Data Link (CCDL) backup.

The T25 sensor transmits the total temperature at the inlet of the High Pressure Compressor (HPC) to the EEC units, which is used for engine control.

The purpose of the data entry plug is to provide both channels of the EEC system with the engine rating, thrust bump and N1 trim levels.

The engine wiring harnesses transmit electrical signals between the engine sensors, actuators and the EEC units.



Fuel - Control Components 1 of 3

V0

Fuel - Control Components 2 of 3

Identify and Locate

The air inlet section total air temperature (T12) sensor is on the inlet cowl at 12 o'clock, just before the engine fan case.

The High Pressure Compressor (HPC) section discharge temperature (T3) sensor is mounted aft of the HPC section at 10 o'clock.

The fuel manifold pressure sensor is attached to the Split Control Unit/Servo Valve Assembly (SCU/SVA) housing.

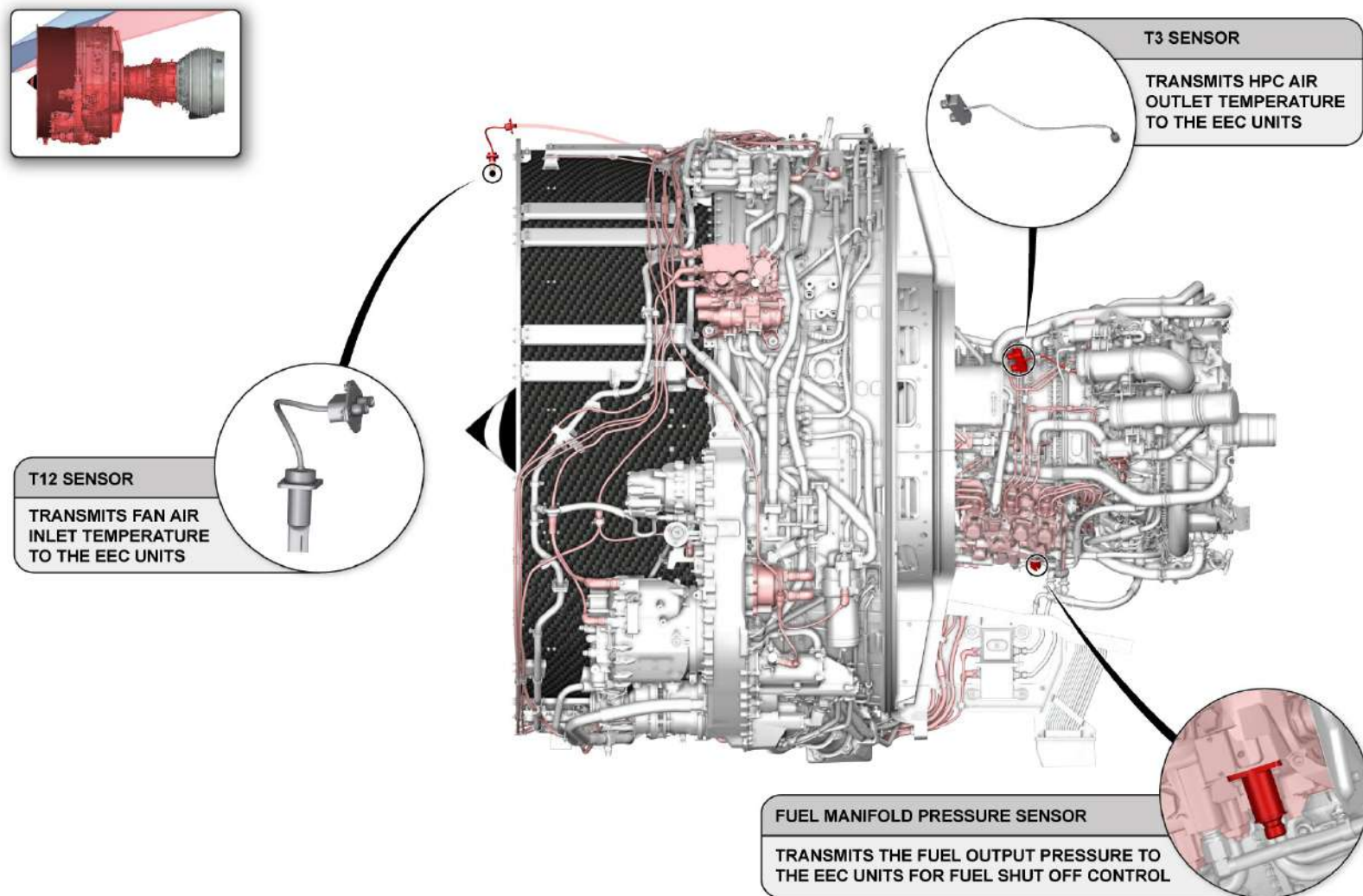
Note: The SCU/SVA is mounted on the core engine in the 8 o'clock position.

Purpose

The T12 sensor transmits the air temperature of the fan inlet to the Electronic Engine Control (EEC) units, which is used for engine control.

The T3 sensor transmits the temperature of the HPC section discharge airflow to the EEC units which is used for engine control.

The fuel manifold pressure sensor transmits the SCU/SVA fuel output pressure to the EEC units, to confirm the engine's fuel shut off valve is functioning properly.



Fuel - Control Components 2 of 3

V0

Fuel - Control Components 3 of 3

Identify and Locate

The Fuel Metering Unit (FMU) is mounted on the air inlet section at 10 o'clock, just above the Accessory Gearbox (AGB).

The Permanent Magnet Alternator (PMA) is mounted on the aft side of the AGB at 8 o'clock.

The Split Control Unit/Servo Valve Assembly (SCU/SVA) is mounted on the engine core at 8 o'clock.

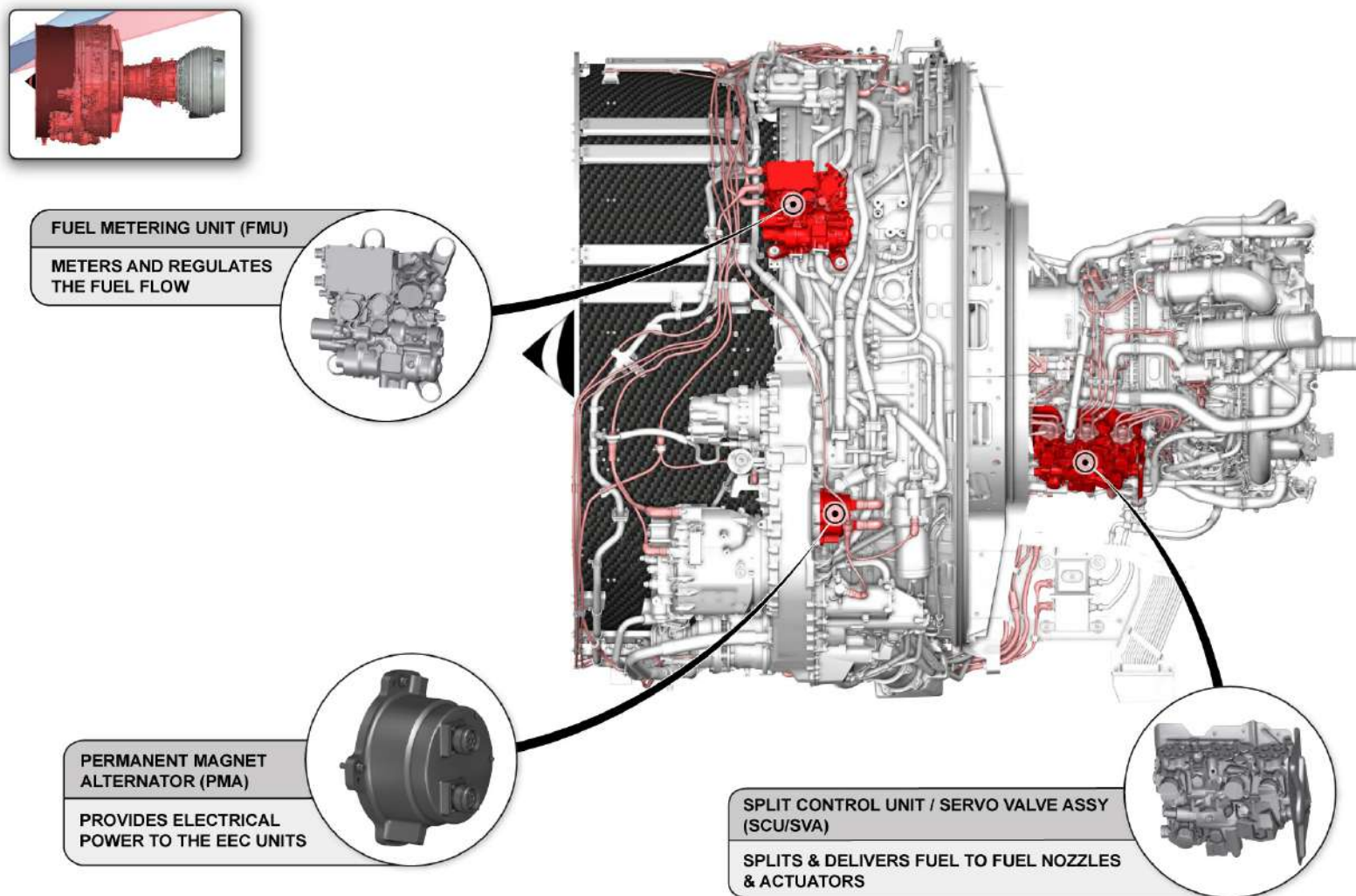
Purpose

The purpose of the FMU is to meter and regulate the fuel flow that is supplied to the combustor.

The SCU/SVA splits the fuel flow and supplies fuel to:

- the fuel nozzles,
- the various servo valves and actuators.

To operate independently from aircraft power sources, the PMA provides dedicated electrical power to the EEC units.



Fuel - Control Components 3 of 3

V0

Fuel - Indicating

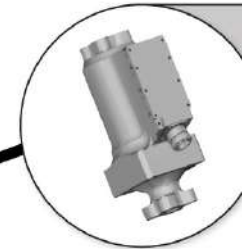
Identify and Locate

The fuel flow transmitter is mounted on the aft side of the air inlet section at 11 o'clock.

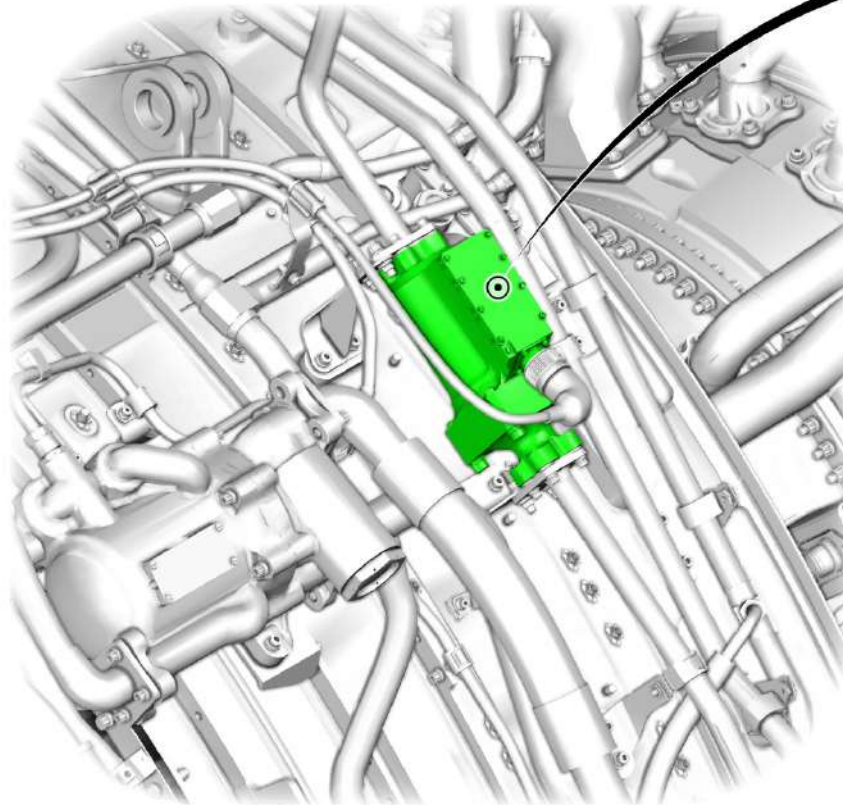
Purpose

The fuel flow transmitter provides the Electronic Engine Control (EEC) units with a measurement of the fuel flow used for combustion.

The EEC units transmit fuel flow rate and used fuel information for flight deck indicating purposes.



**FUEL FLOW
TRANSMITTER**
TRANSMITS FUEL FLOW
INFORMATION TO THE
ELECTRONIC ENGINE
CONTROL (EEC) UNITS



Fuel - Indicating

V0



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General Familiarization LEAP-1B



Engine Systems

Engine Starting
and Ignition



Ignition Components

Identify and Locate

The engine ignition system is located on the core engine between 5 and 6 o'clock.

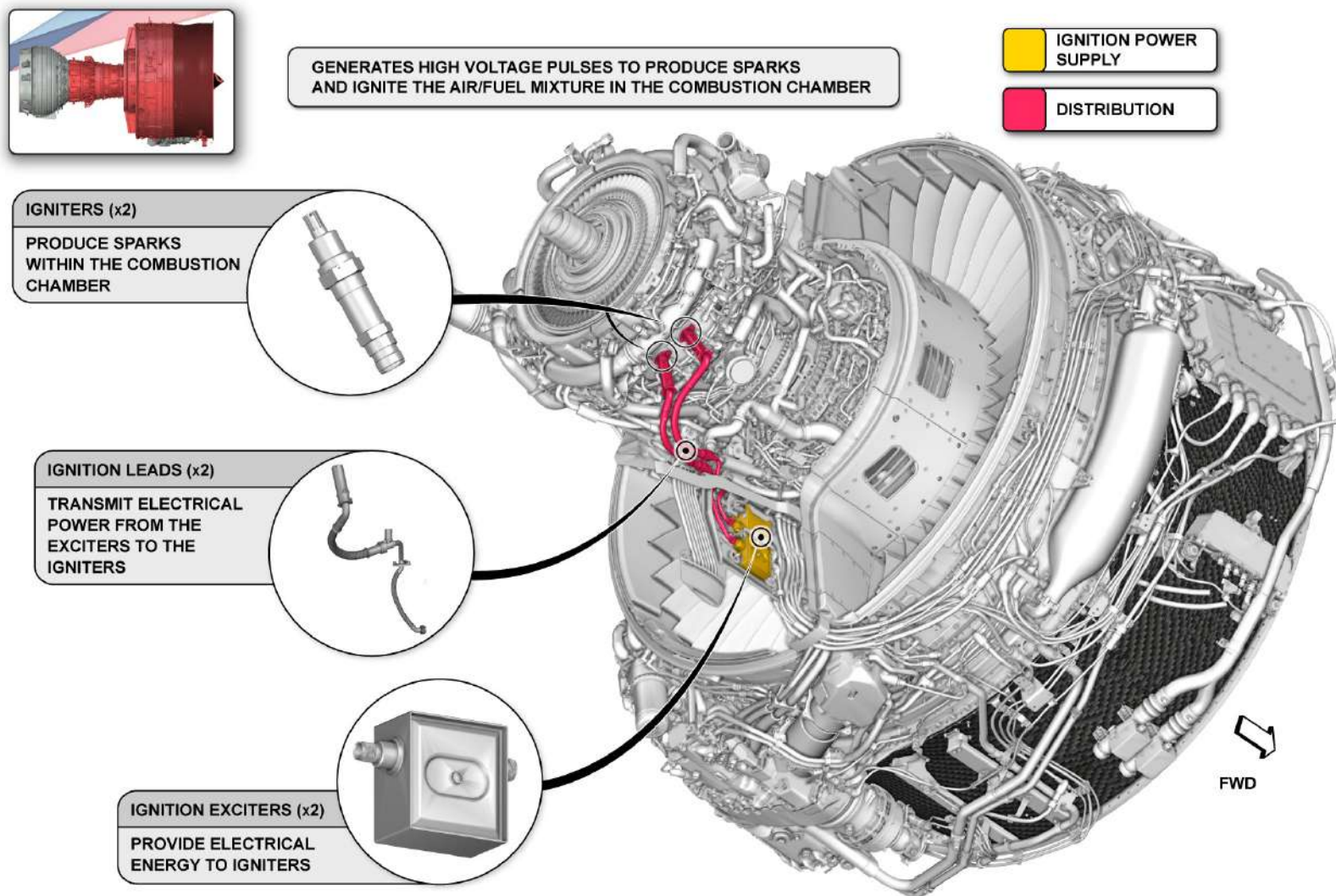
The ignition distribution system includes:

- two ignition exciters located at 6 o'clock, mounted on the A/C Z-Deck in the bifurcation area,
- two ignition leads which run from the ignition exciters to the igniters,
- two igniters (left & right) installed on the combustion chamber at the 5:30 and 6:00 ALF positions.

The electrical power supply comes from the aircraft (115 VAC).

Purpose

The ignition system generates high voltage pulses to produce sparks and ignite the air/fuel mixture in the combustion chamber.



Ignition Components

V0

Starting Components

Identify and Locate

The starting system is located between the 7 o'clock and 8 o'clock positions and mounted on the forward side of the Accessory Gearbox (AGB).

The starting system includes:

- one Pneumatic Air Starter (PAS) mounted on the forward side of the AGB,
- one PAS magnetic plug, installed on the outboard side of the PAS,
- one Starter Air Valve (SAV) mounted on the PAS,
- one SAV cable mounted on the SAV.

Purpose

The purposes of the engine starting system are:

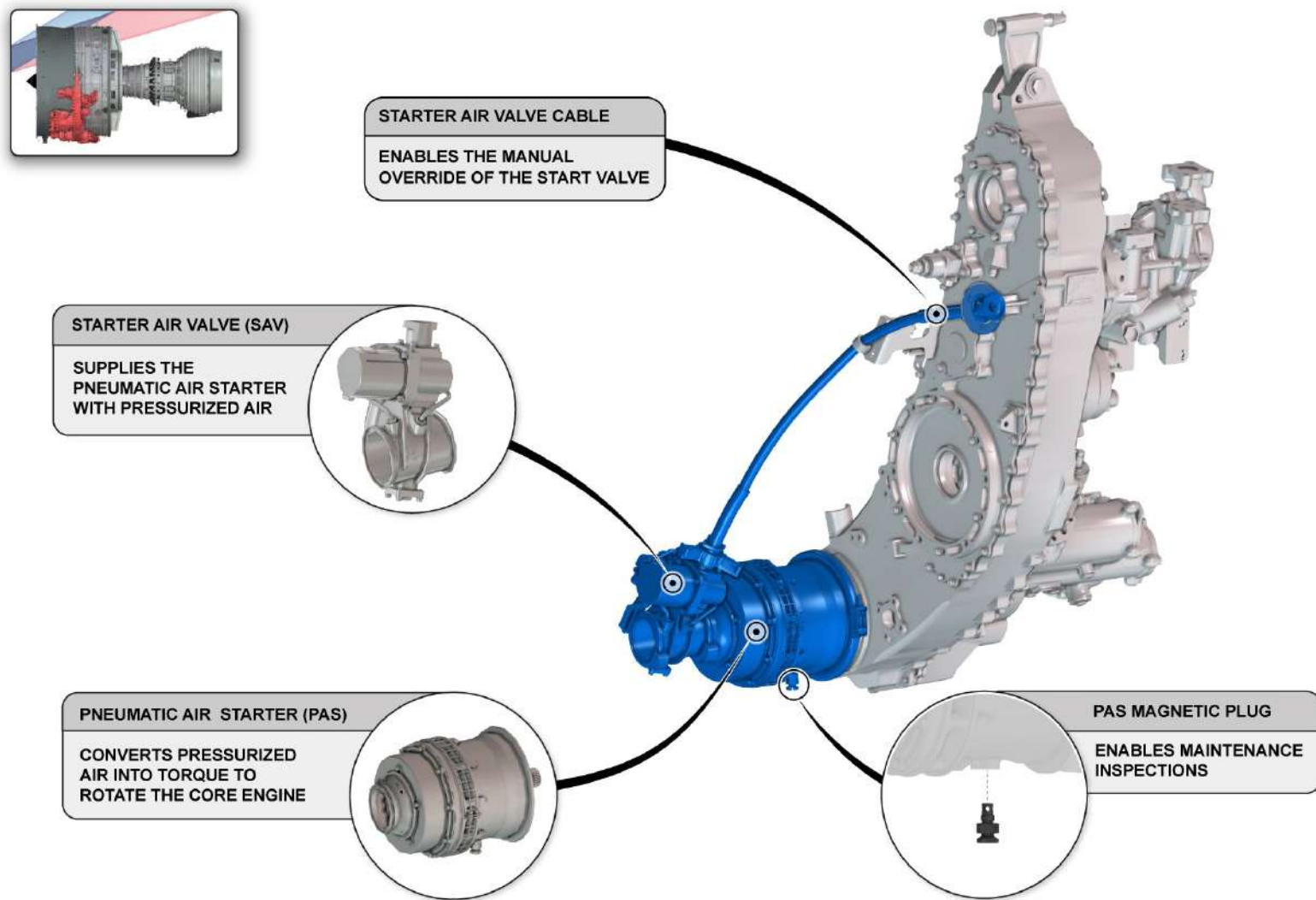
- to rotate the engine up to self-sustaining operation during the starting sequence,
- to rotate the engine for leak-check,
- to conduct compressor wash cycles during maintenance action.

The purpose of the PAS is to convert pressurized air supply into a driving torque to rotate the engine.

The purpose of the SAV is to control the airflow supplied to the PAS.

In case of failure, the Starter Air Valve (SAV) cable enables a manual opening or closure of the SAV.

The PAS magnetic plug collects magnetic particles for inspection.



Starting Components

V0



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General Familiarization LEAP-1B



Engine Systems



Air System

Identify and Locate

operatibility during steady-state and transient operations.

The engine air system includes the following sub-systems:

- engine anti-icing,
- engine cooling,
- compressor control.

The engine anti-icing system is located on the right-hand side of the engine with a pneumatic supply manifold attached to the booster split fairing.

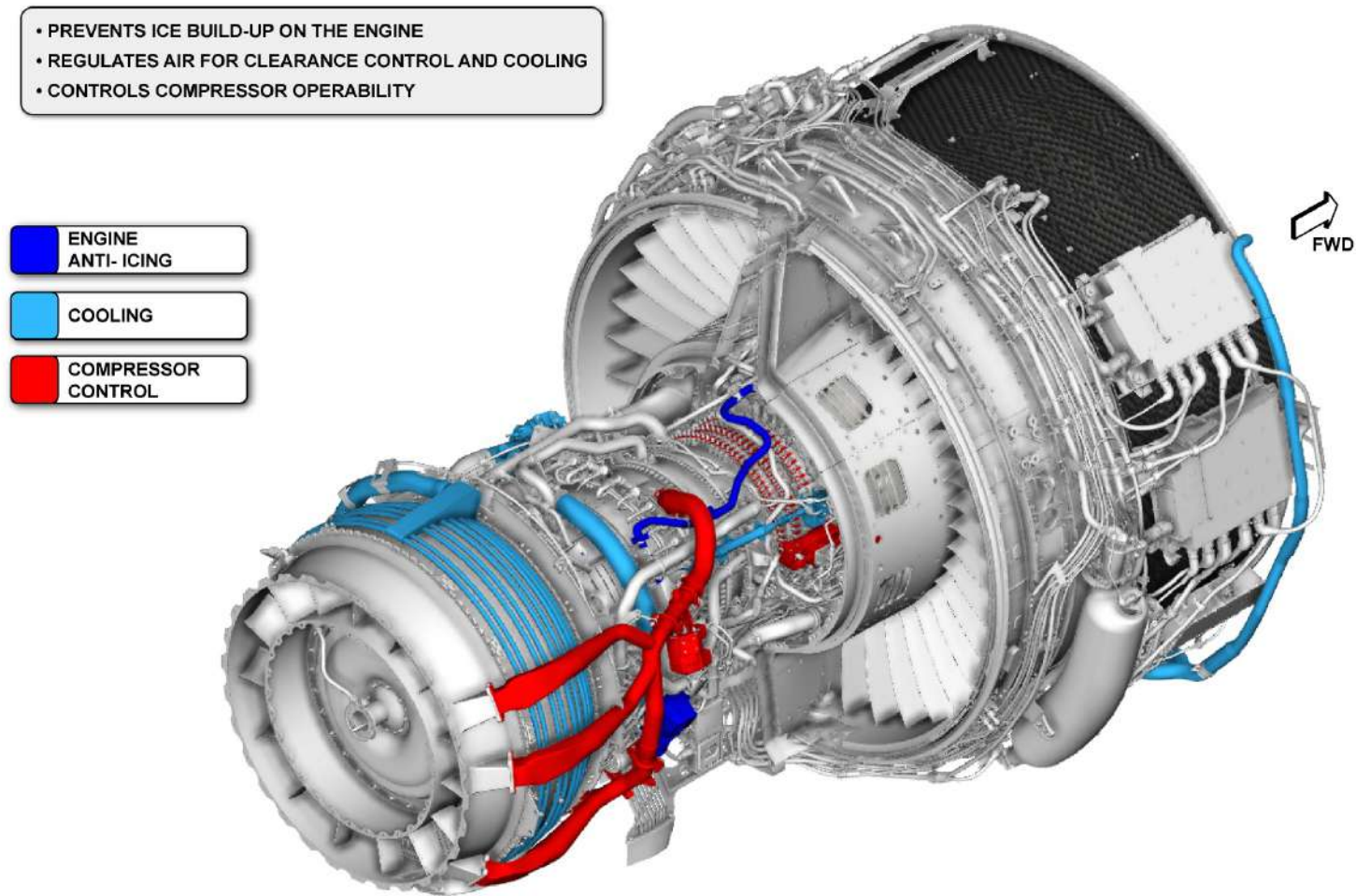
The engine cooling system is located on the air inlet section, the High Pressure Compressor (HPC) section and the Low Pressure Turbine (LPT) section.

The compressor control system is located aft of the air inlet section and on the engine core.

Purpose

The engine air system regulates air circuits throughout the engine:

- the engine anti-icing system directs hot air to prevent ice build-up on the engine flow splitter,
- the engine cooling system regulates air throughout the engine core components, engine control system and nacelle cooling, turbine clearance and compressor control and High Pressure Turbine (HPT) blade efficiency purposes,
- the compressor control actuates the Variable Stator Vanes (VSVs), Variable Bleed Valves (VBVs) and Transient Bleed Valve (TBV) to maintain compressor



Air System

V0

Start Bleed / Booster Anti-Ice Valve

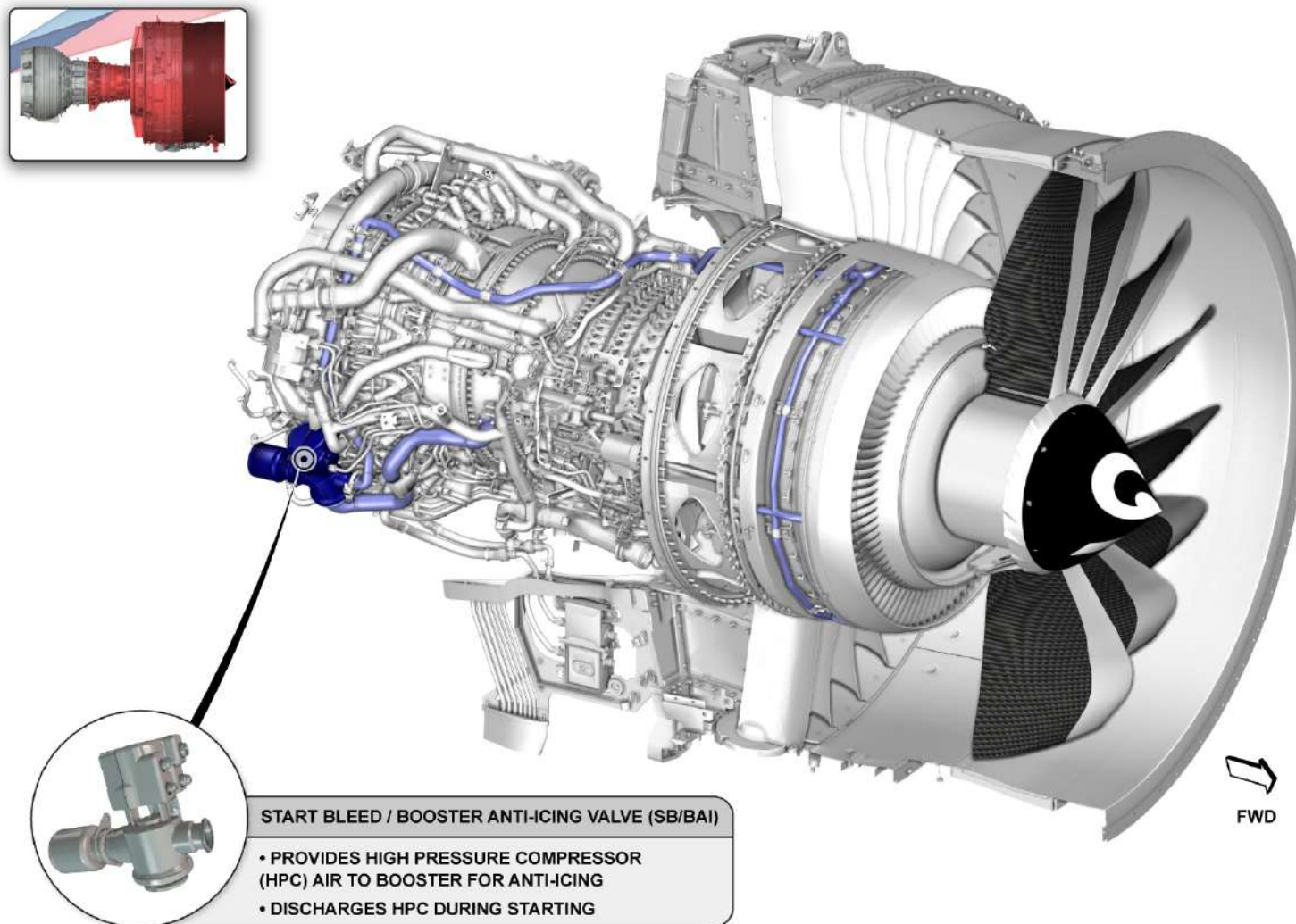
Identify and Locate

The booster anti-icing system consists of the Start Bleed/Booster Anti-Ice (SB/BAI) valve and actuator. It is located on the engine core at 3 o'clock with manifold supply tubing routed at 1 o'clock.

Purpose

The SB/BAI valve serves to control 7th stage High Pressure Compressor (HPC) bleed for dual functionality.

The SB/BAI valve is used to unload the HPC during engine starting for increased stall margin. The SB/BAI valve also provides hot HPC 7th stage air to the Low Pressure Compressor (Booster) flow splitter forward end to prevent ice accumulation.



Start Bleed / Booster Anti-Ice Valve

V0

HPTACC and LPTACC valves

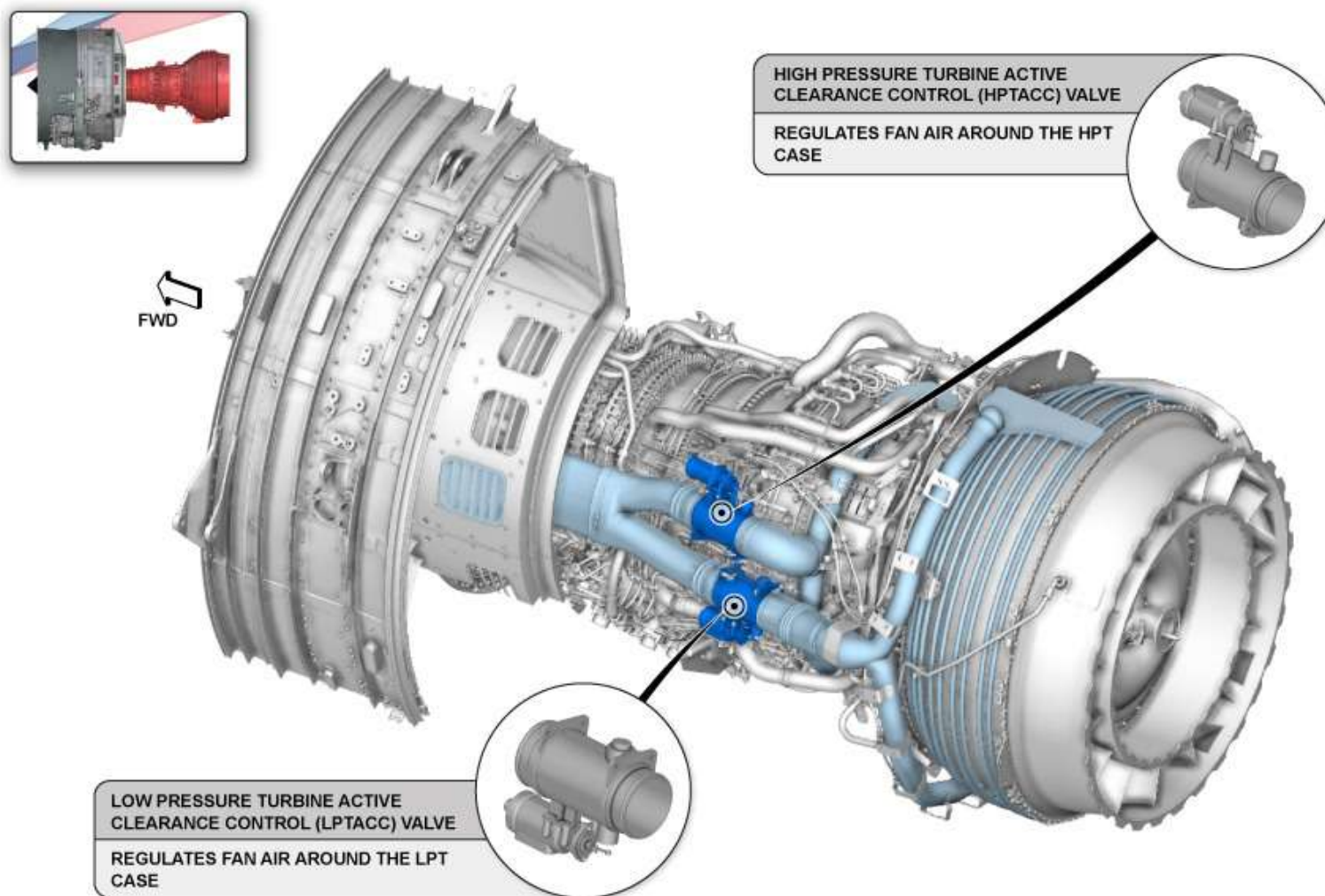
Identify and Locate

The High Pressure Turbine Active Clearance Control (HPTACC) and the Low Pressure Turbine Active Clearance Control (LPTACC) valves are located on the engine core at 9 o'clock.

Purpose

The HPTACC valve regulates the fan airflow to cool the outer case of the High Pressure Turbine (HPT) in order to enhance the efficiency of the HPT.

The LPTACC valve regulates the fan airflow to cool the outer case of the Low Pressure Turbine (LPT) in order to enhance the efficiency of the LPT.



HPTACC and LPTACC valves

Modulated Turbine Cooling System

Identify and Locate

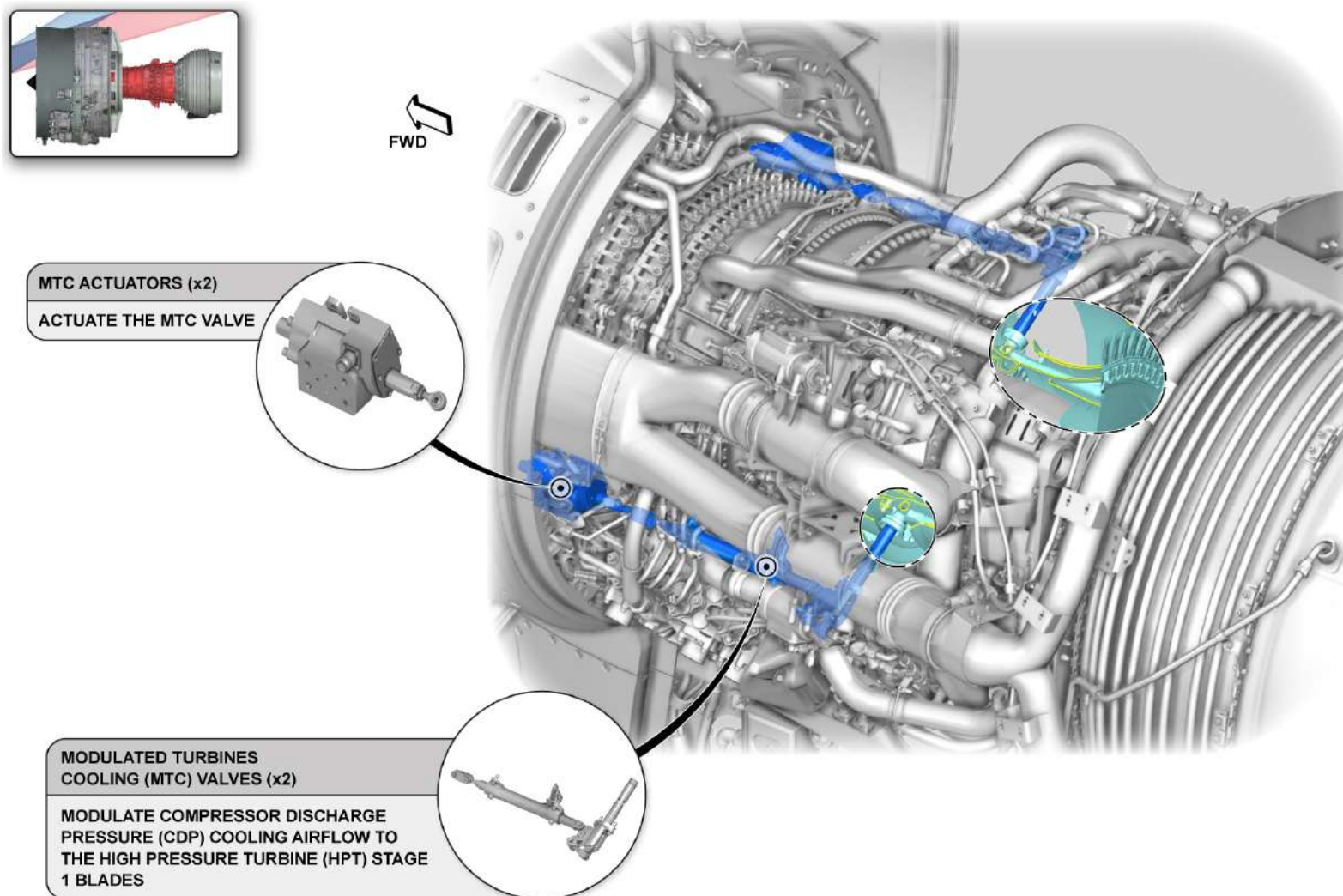
The Modulated Turbine Cooling (MTC) system consists of the following components:

- 2 MTC actuators located on the front of the High Pressure Compressor (HPC) at 2 and 8 o'clock,
- 2 MTC valves located on the Combustion Diffuser Nozzle (CDN) case at 2 and 8 o'clock.

The MTC linkages connect the MTC actuators to the MTC valves.

Purpose

The MTC system modulates 10th stage HPC, Compressor Discharge Pressure (CDP) cooling airflow to the stage 1 High Pressure Turbine (HPT) blade roots and blade internal cooling passages to improve Specific Fuel Consumption (SFC) at cruise and low temperature operating conditions.



Modulated Turbine Cooling System

V0

Electronic Engine Control Cooling Blower

Identify and Locate

Two Electronic Engine Control (EEC) cooling blowers are installed on the fan case at 5 o'clock.

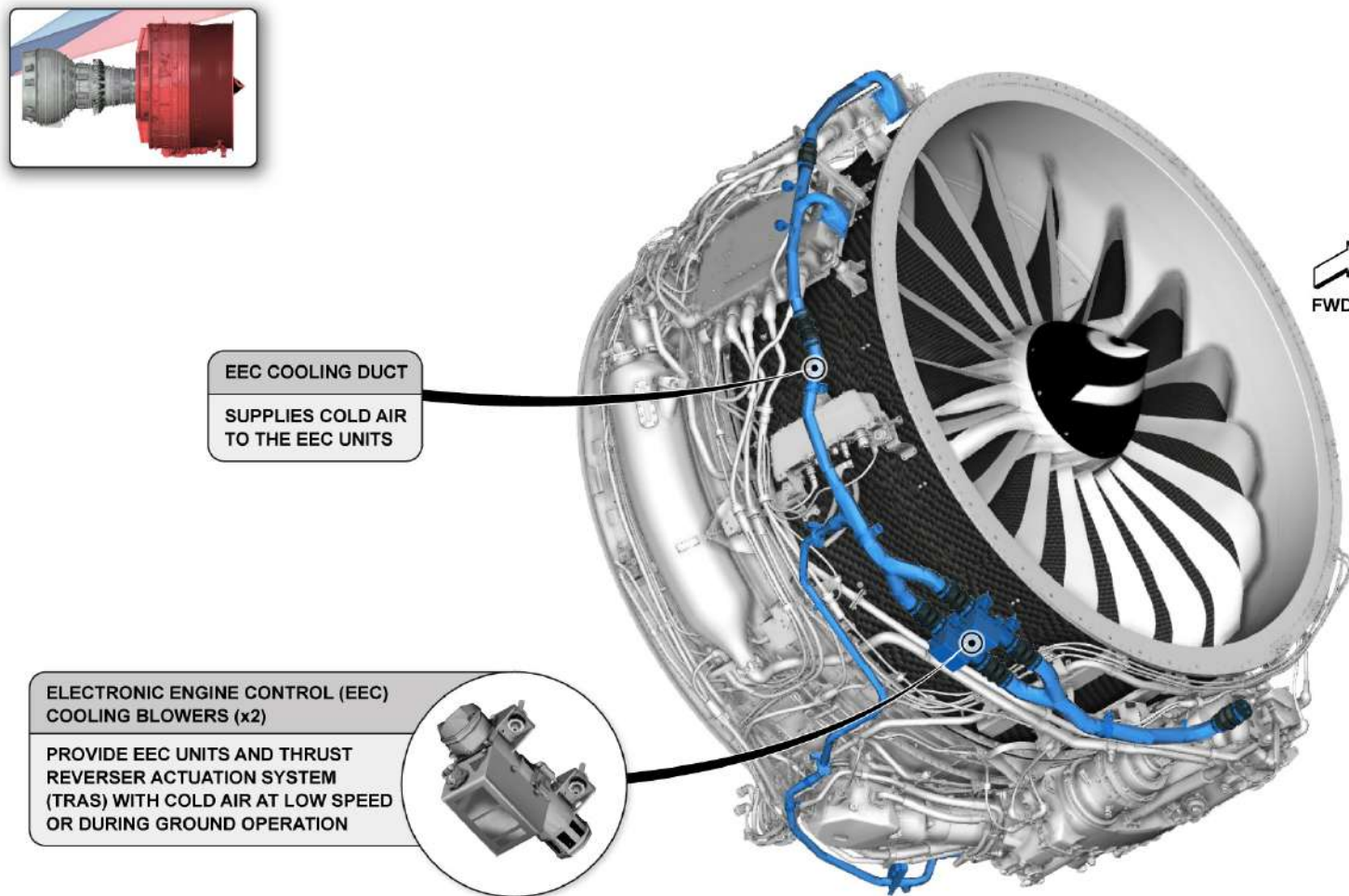
The EEC cooling duct is located between the 1 o'clock and 6 o'clock positions on the fan case. An air pickup is located downstream of the blowers to the Thrust Reversers Actuator (TRA) system.

Purpose

The EEC cooling blowers provide external fresh air to cool both EEC units when:

- the aircraft speed is low,
- the internal EEC units temperature is above a specific limit,
- the aircraft is on the ground.

The EEC cooling blowers also provide air to cool the Thrust Reverser Actuation System (TRAS).



Electronic Engine Control Cooling Blower

V0

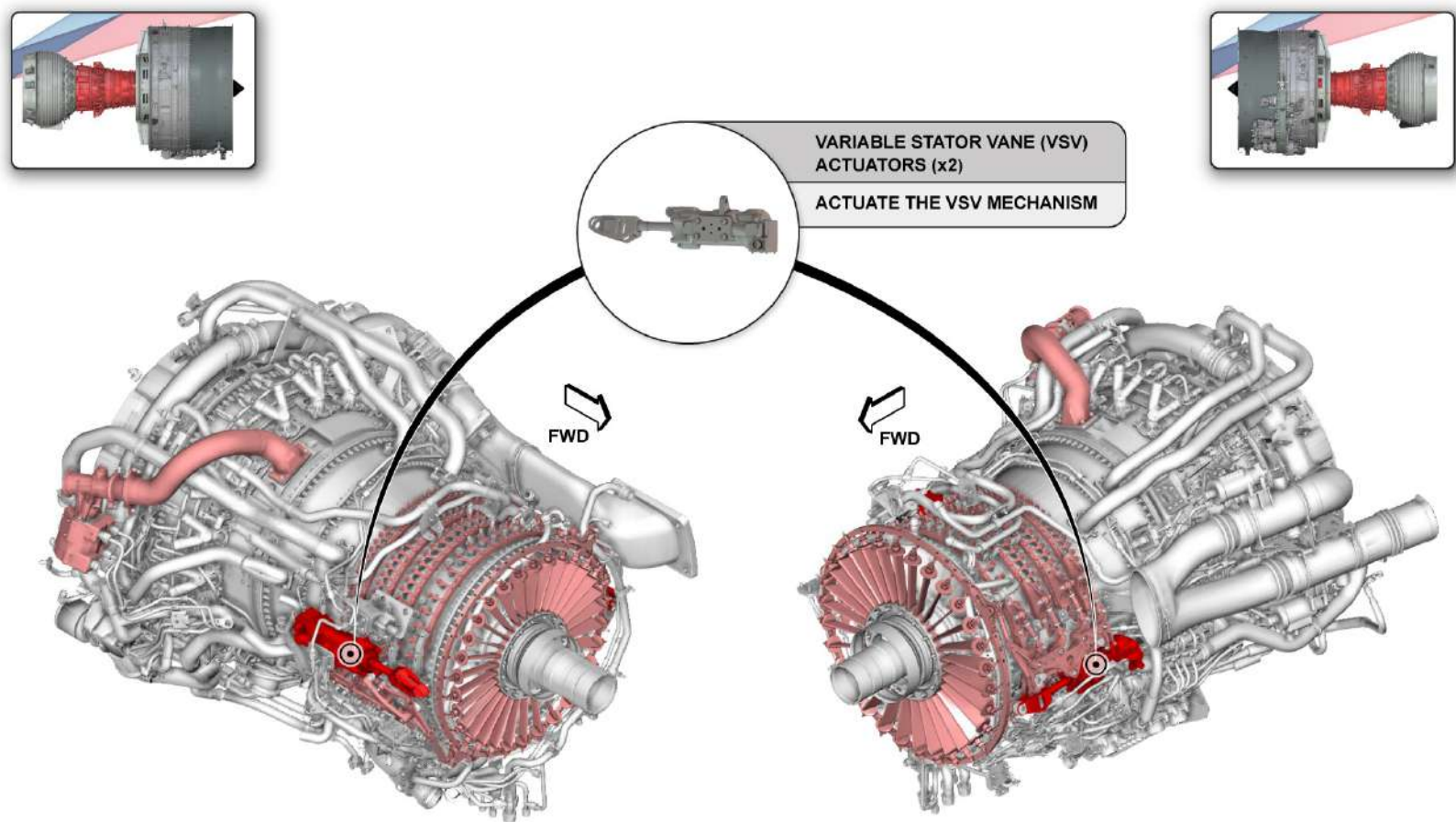
Variable Stator Vane Actuators

Identify and Locate

The two Variable Stator Vane (VSV) actuators are located on each side and at the front of the High Pressure Compressor (HPC) at 3 and 9 o'clock.

Purpose

The VSV actuators control the position of the VSVs and the Inlet Guide Vanes (IGVs) to regulate the airflow through the HPC, in order to provide optimum compressor performance.



Variable Stator Vane Actuators

V0

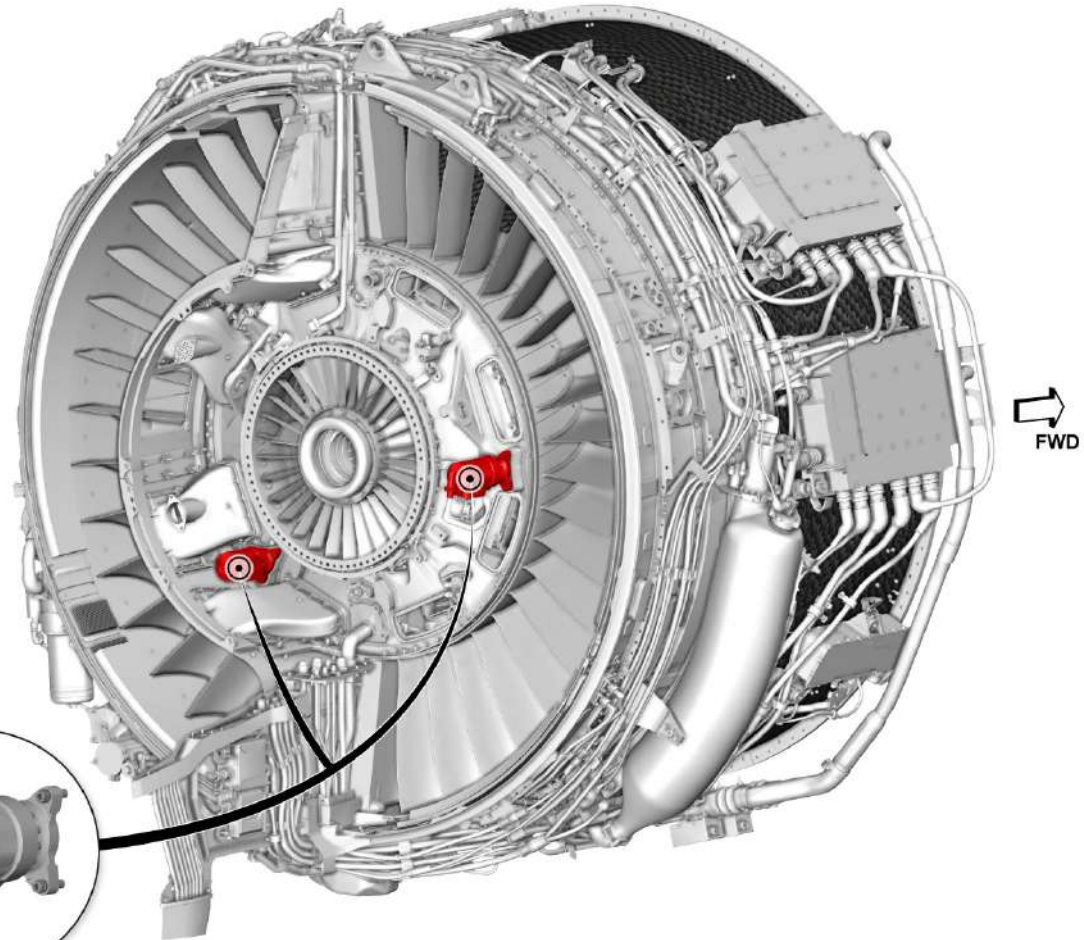
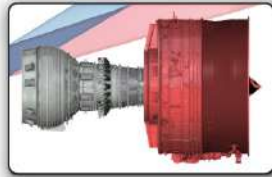
Variable Bleed Valve Actuators

Identify and Locate

Two Variable Bleed Valve (VBV) actuators are located on the fan frame rear flange at 3 and 7 o'clock.

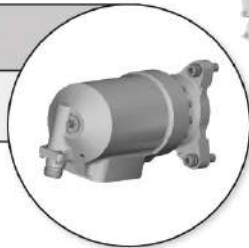
Purpose

The two VBV actuators work in pair to control the positions of the 8 variable bypass scoops in order to optimize the operation of low and high pressure compressors at low speeds and during transient operations.



VARIABLE BLEED VALVE (VBV)
ACTUATORS (X2)

ACTUATE 8 VBV SCOOPS
MECHANISMS



Variable Bleed Valve Actuators

V0

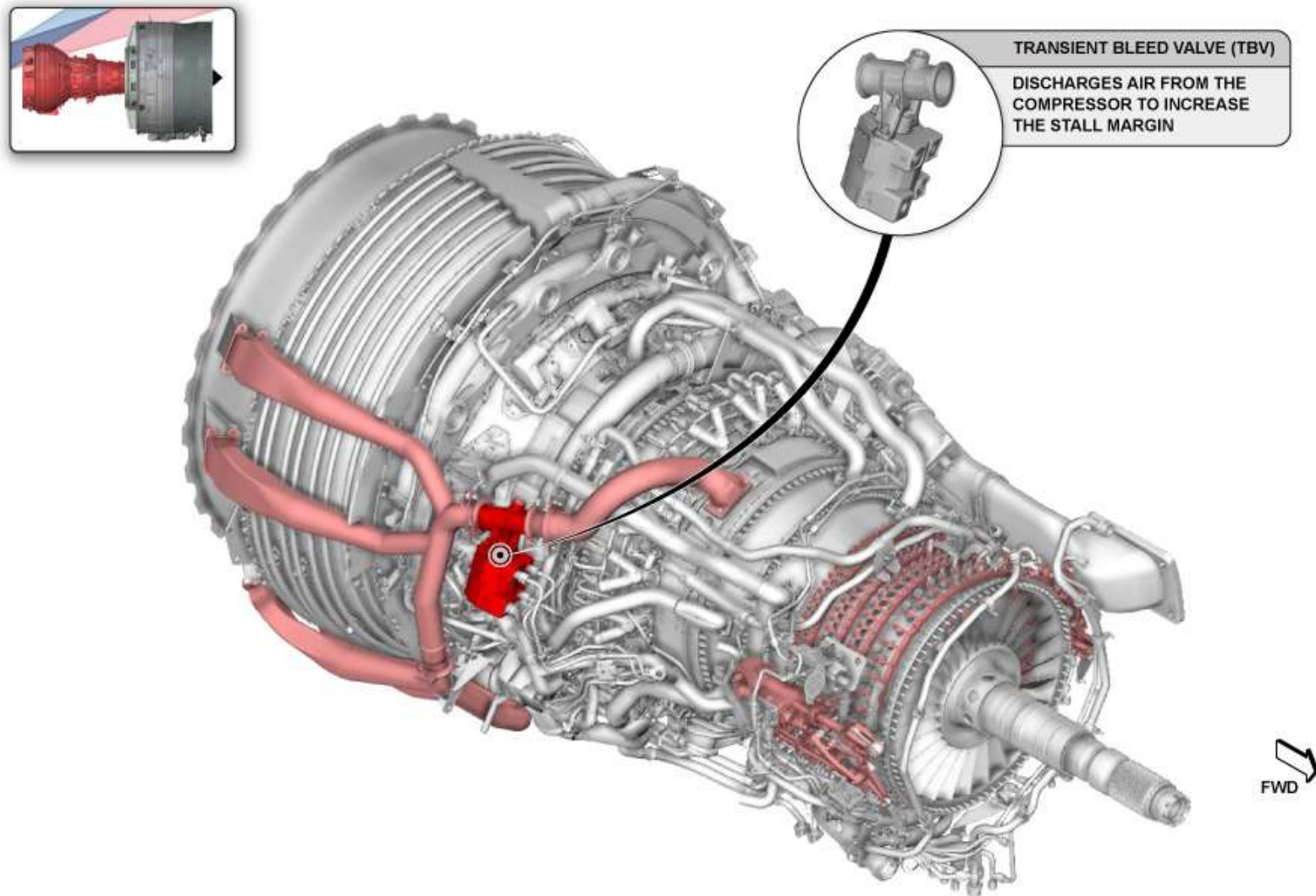
Transient Bleed Valve

Identify and Locate

The Transient Bleed Valve (TBV) is located on the combustion section between 2 and 3 o'clock.

Purpose

The purpose of the TBV is to bleed compressor discharge (10th stage) air to unload the compressor during engine transient phases and increase the stall margin.



Transient Bleed Valve

V0



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General Familiarization LEAP-1B



Engine Systems

Engine Indicating 

Indicating Systems

Identify and Locate

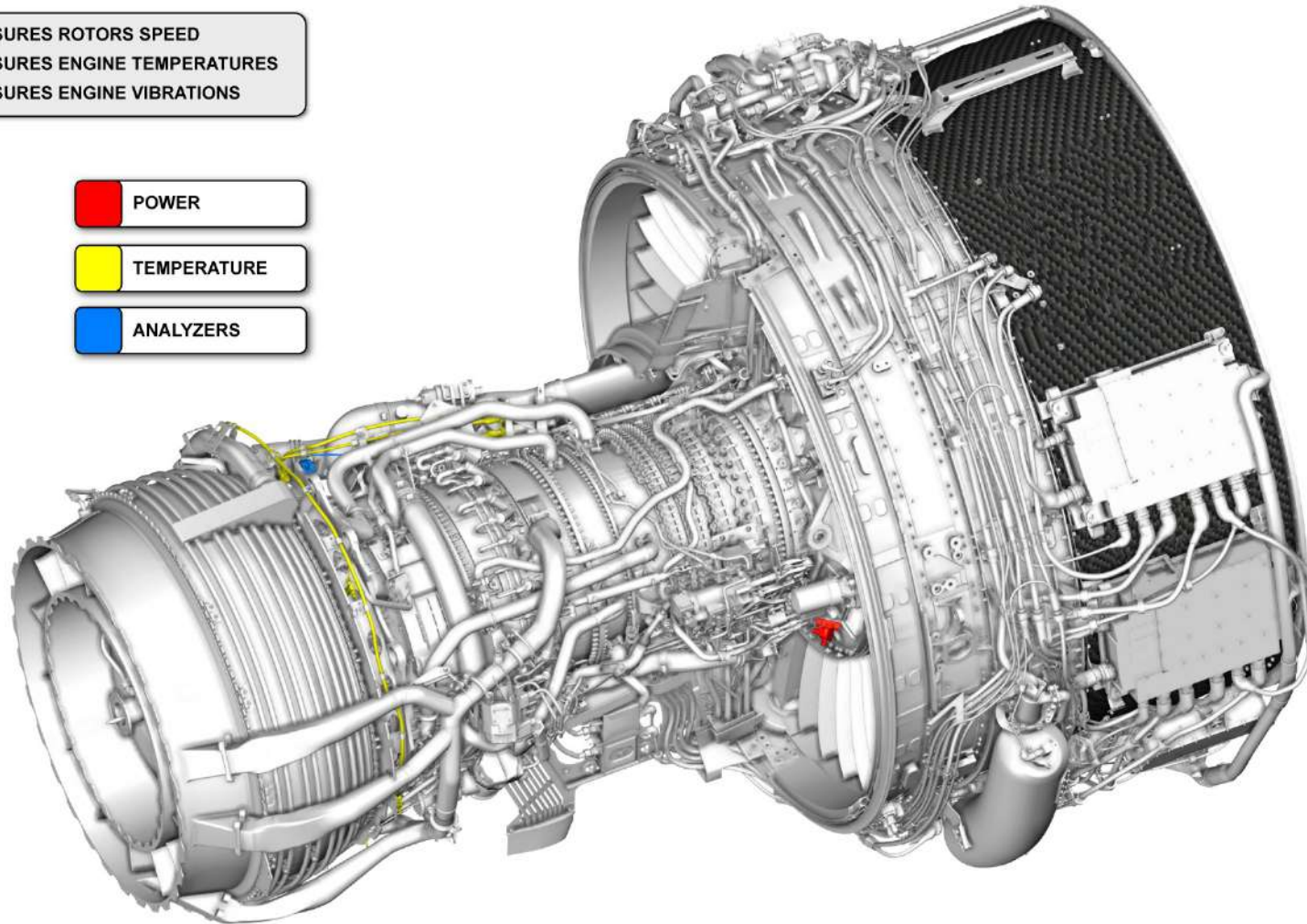
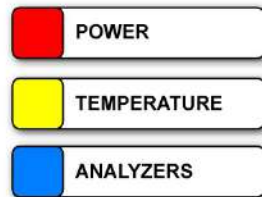
The engine indicating system consists of the:

- power indicating system (N1 and N2 sensors),
- Exhaust Gas Temperature system (EGT) sensors,
- vibration analyzers (No.1 bearing and Turbine Center Frame).

Purpose

The engine indicating system provides engine rotor speed, temperature and vibration parameter signals to Electronic Engine Control (EEC) units for regulation and maintenance purposes.

- MEASURES ROTORS SPEED
- MEASURES ENGINE TEMPERATURES
- MEASURES ENGINE VIBRATIONS



Indicating Systems

V0

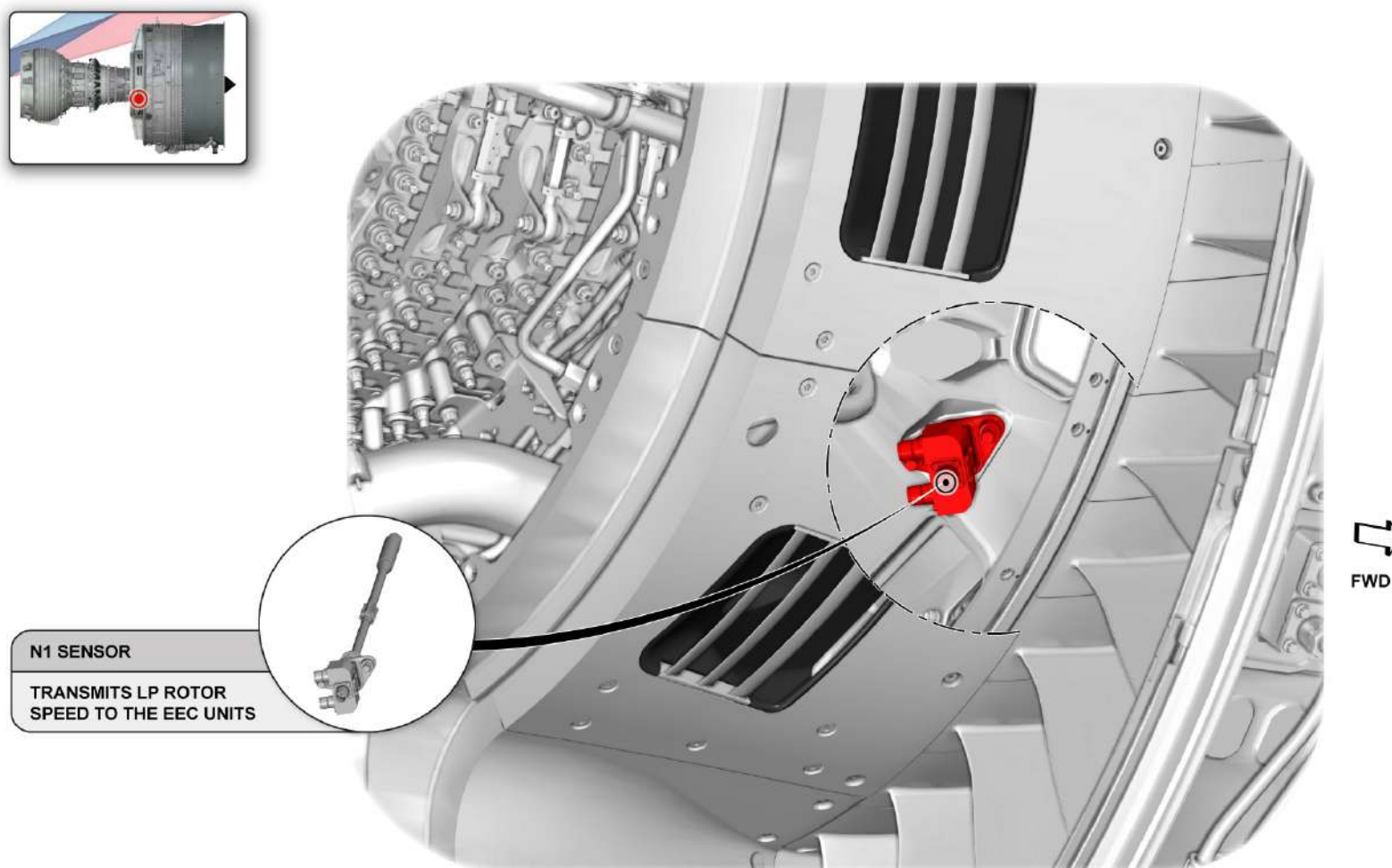
N1 Speed Sensor

Identify and Locate

The fan speed (N1) sensor is mounted in the N1 sensor guide, inside the 5 o'clock strut of the fan hub.

Purpose

The purpose of the N1 speed sensor is to measure the rotational speed of the Low Pressure (LP) rotor for engine control, vibration monitoring and flight deck indication.



N1 Speed Sensor

V0

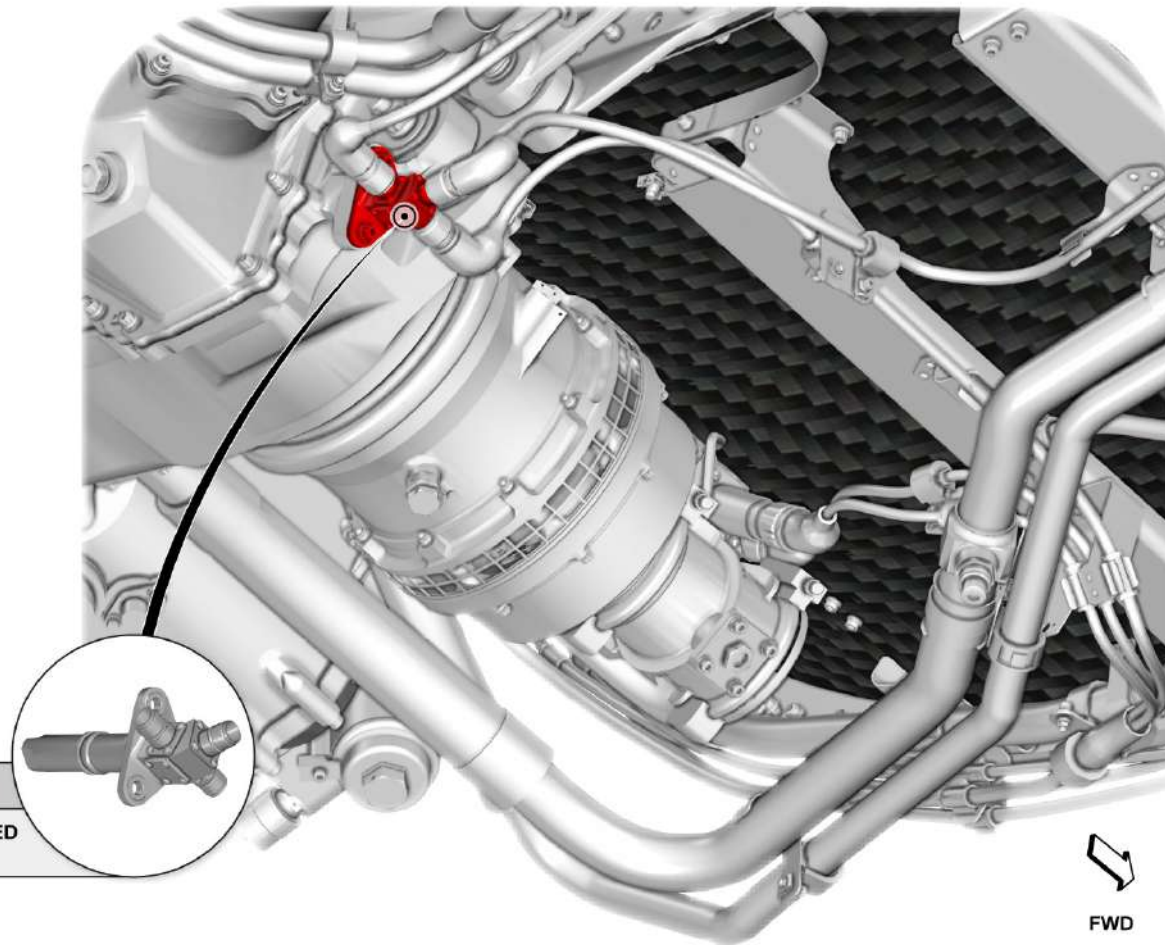
N2 Speed Sensor

Identify and Locate

The core speed (N2) sensor is installed on the Accessory Gearbox (AGB) at the 7 o'clock position.

Purpose

The purpose of the N2 speed sensor is to measure the rotational speed of the High Pressure (HP) rotor for engine control, vibration monitoring, over-speed detection and flight deck indication.



N2 SENSOR

**TRANSMITS HP ROTOR SPEED
TO THE EEC UNITS**

N2 Speed Sensor

V0

Exhaust Gas Temperature Sensors

Identify and Locate

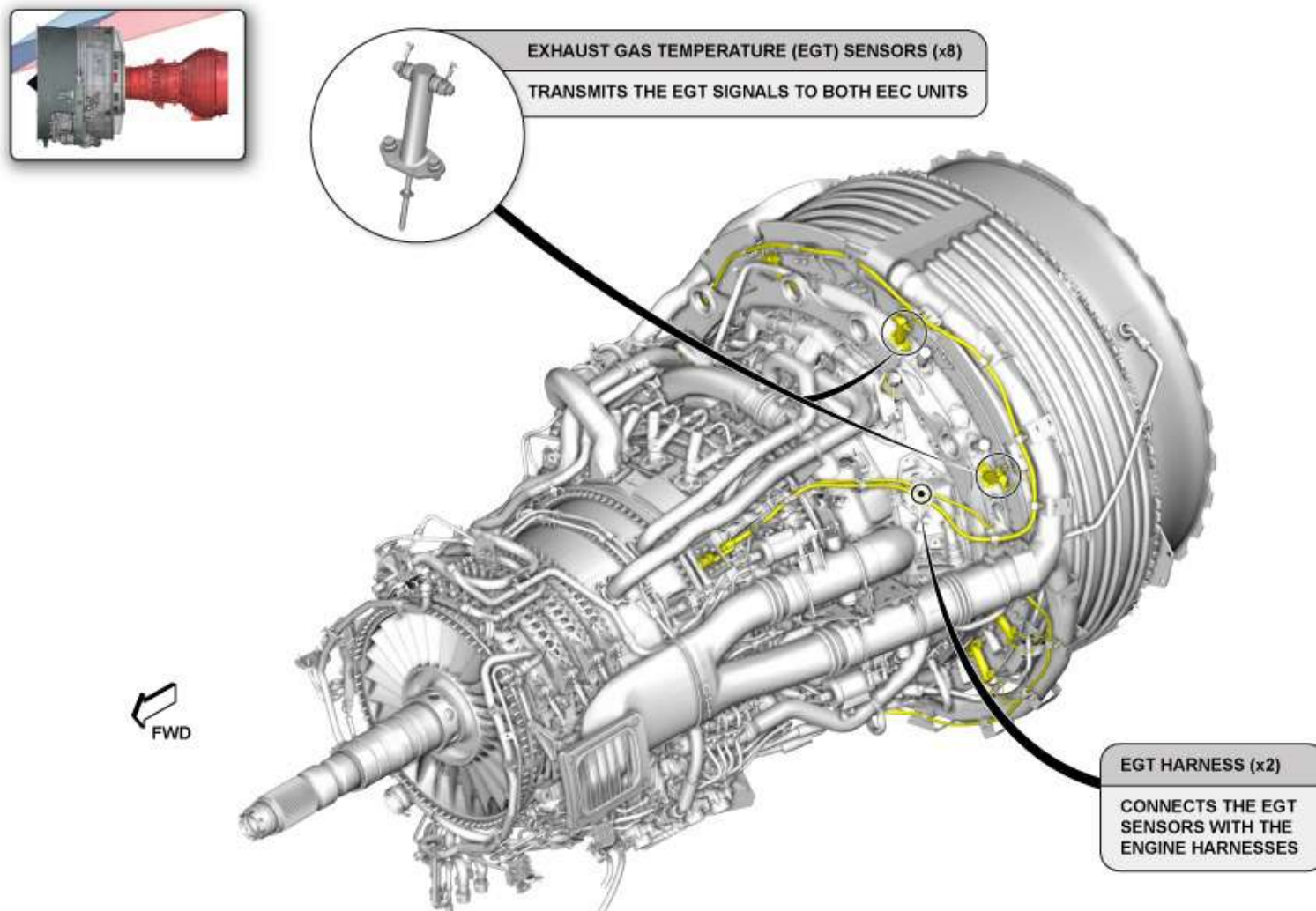
The 8 Exhaust Gas Temperature (EGT) sensors are installed equally spaced around the Turbine Center Frame (TCF) at the inlet of the Low Pressure Turbine (LPT) section.

T48 harness No.1 is located on the right side of the TCF.
T48 harness No.2 is located on the left side of the TCF.

Purpose

The EGT sensors measure the temperature of the exhaust air for flight deck indicating and engine condition health monitoring.

The T48 harnesses connect EGT sensors to Electronic Engine Control (EEC) units.



Exhaust Gas Temperature Sensors

Vibration Analysers

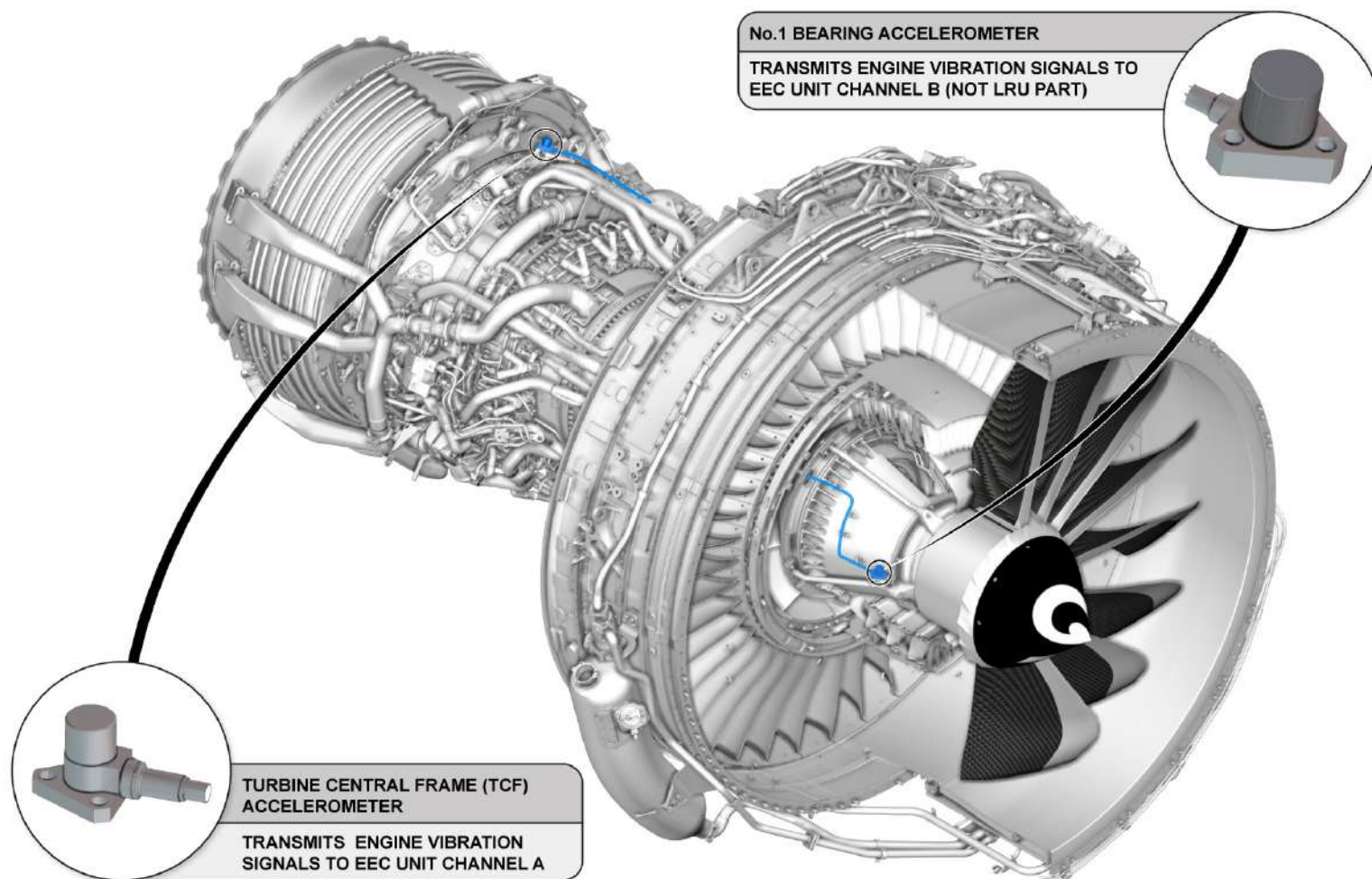
Identify and Locate

There are two vibration sensors:

- the N°1 bearing accelerometer is installed on the N°1 and N°2 bearing support at 4 o'clock and goes through the fan hub strut,
- the Turbine Center Frame (TCF) accelerometer is installed on the TCF at 12 o'clock.

Purpose

The No.1 bearing accelerometer transmits engine vibration signals to the Electronic Engine Control (EEC) unit channel B, and the TCF accelerometer to the EEC unit channel A.



Vibration Analysers

V0



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General Familiarization LEAP-1B



Engine Systems

Oil



Oil System

Identify and Locate

units for aircraft flight deck indicating and monitoring.

Oil system components are mainly located on the right and left sides of the fan frame.

The oil system is composed of 3 sub-systems:

- storage,
- distribution,
- indicating.

The oil storage system consists of the oil tank, located on the right side of the fan frame.

The oil distribution system is mainly located on the left side of the engine.

The oil indicating system is located mainly on the fan frame, and the Accessory Gear Box (AGB).

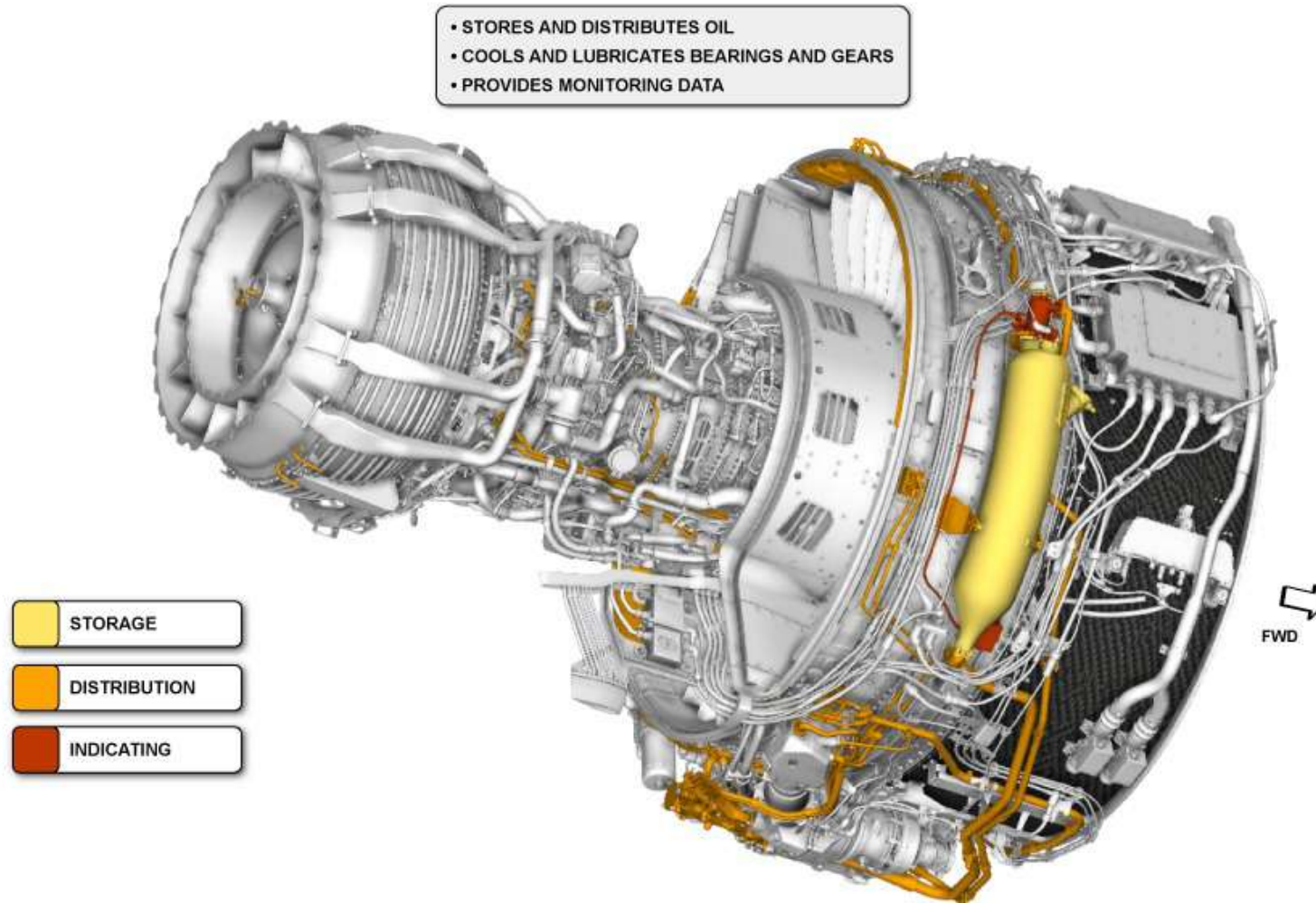
Purpose

The purpose of the oil system is to store, distribute, cool and lubricate engine bearings and gears.

The purpose of the oil storage system is to keep a sufficient quantity of oil to continuously supply the oil distribution system.

The oil distribution system supplies and scavenges oil to and from the engine and accessory drive gears and bearings.

The purpose of the oil indicating system is to provide oil system data via the Electronic Engine Control (EEC)



Oil System

V0

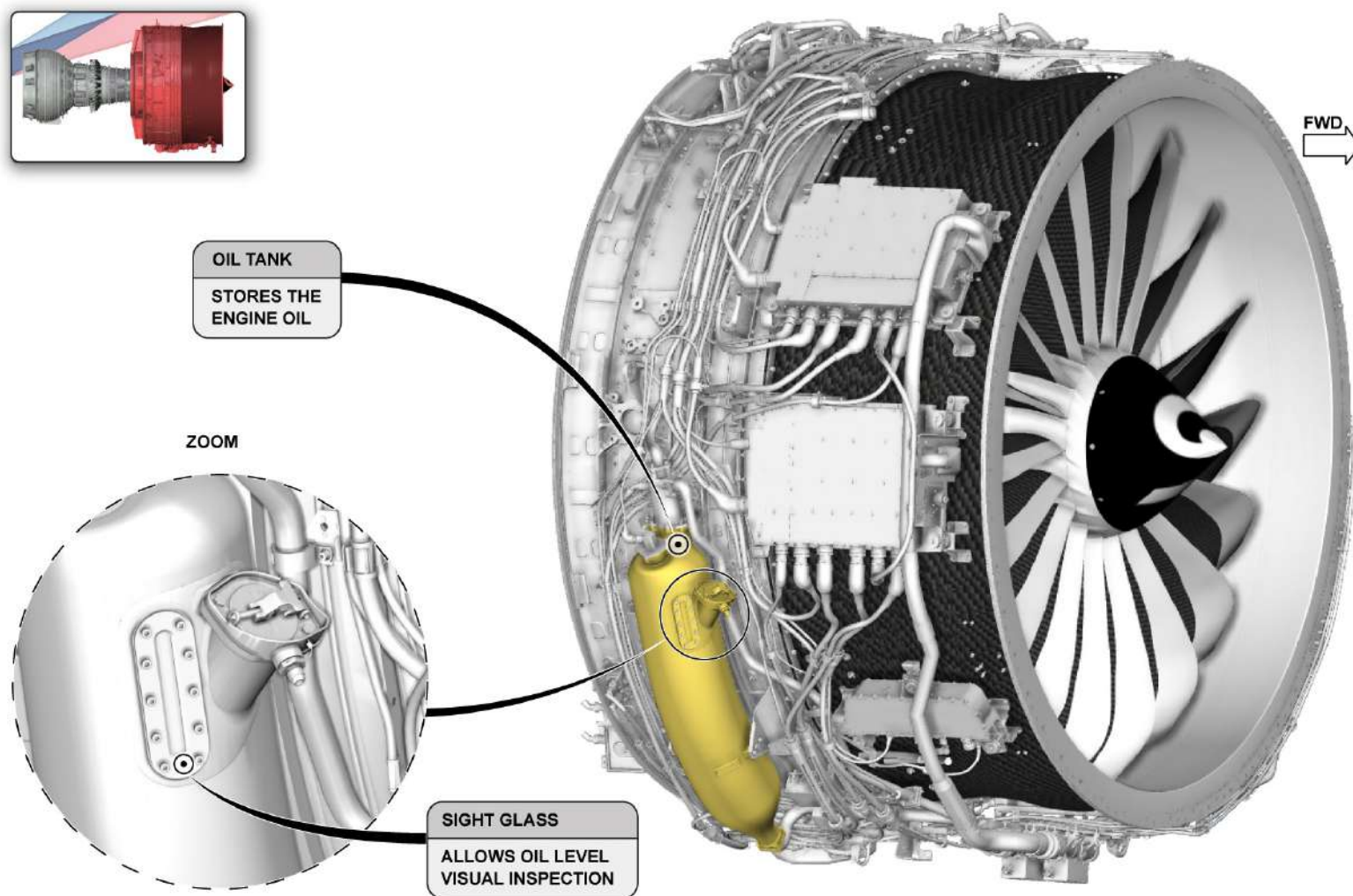
Oil Tank

Identify and Locate

The oil tank is installed on the right hand side of the air inlet section at 4 o'clock.

Purpose

The purpose of the oil tank is to store the engine oil.



Oil Tank

V0

Oil - Distribution Components 1 of 2

Identify and Locate

The oil lubrication unit is mounted on the rear face of the Accessory Gearbox (AGB), at 7 o'clock.

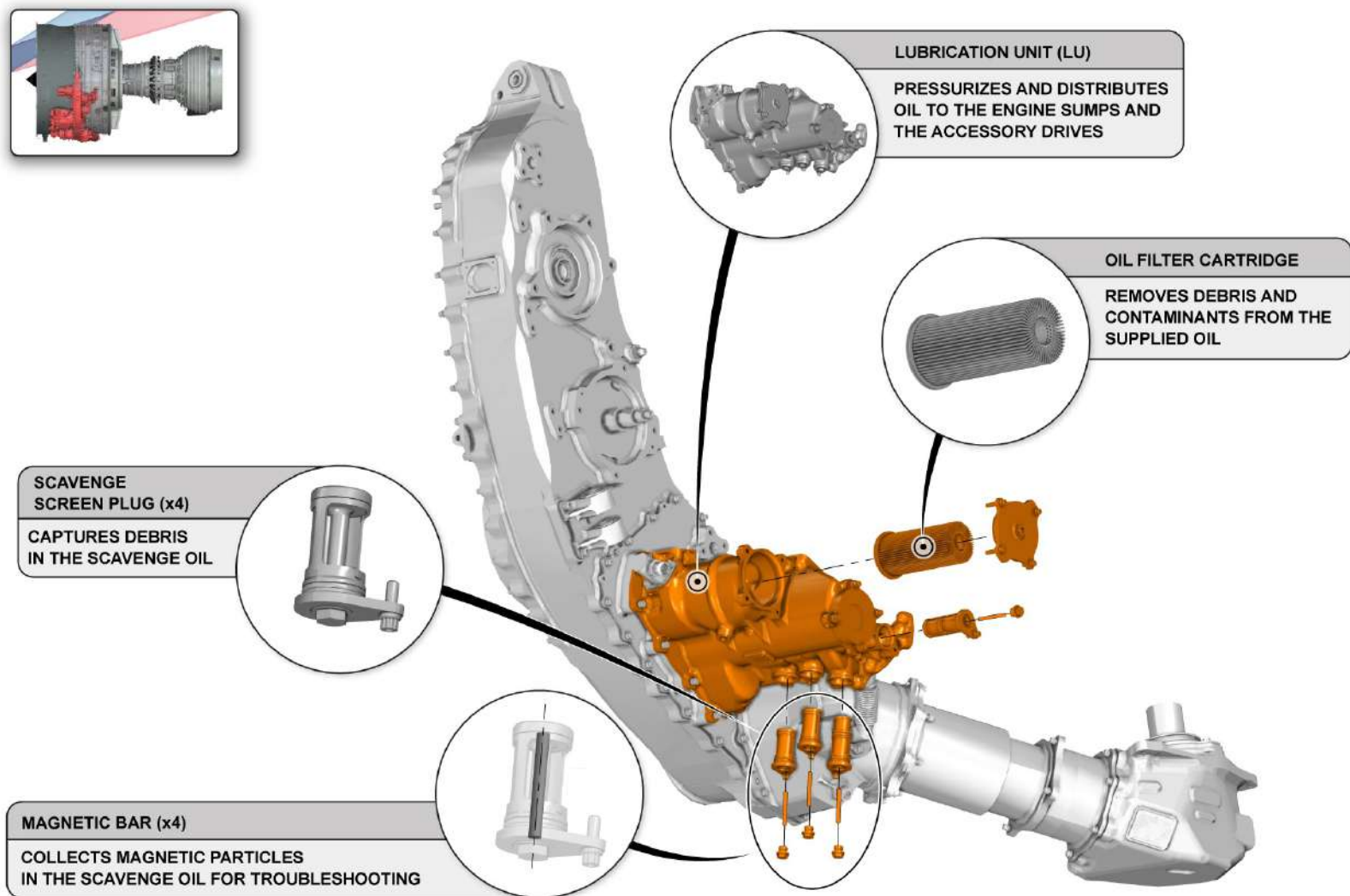
The oil filter cartridge and the four scavenge screen plugs with magnetic bars are located in the lubrication unit.

Purpose

The oil lubrication unit pressurizes oil and delivers it to the 3 sumps and the accessory drives.

The oil is filtered through the oil filter cartridge.

The 4 scavenge screen plugs collect big contaminants from oil before those contaminants enter the oil lubrication unit (scavenge area). The magnetic bars collect the metallic particles.



Oil - Distribution Components 1 of 2

V0

Oil - Distribution Components 2 of 2

Identify and Locate

The Main Fuel/Oil Heat Exchanger (MHX) is located on the air inlet section at 10 o'clock.

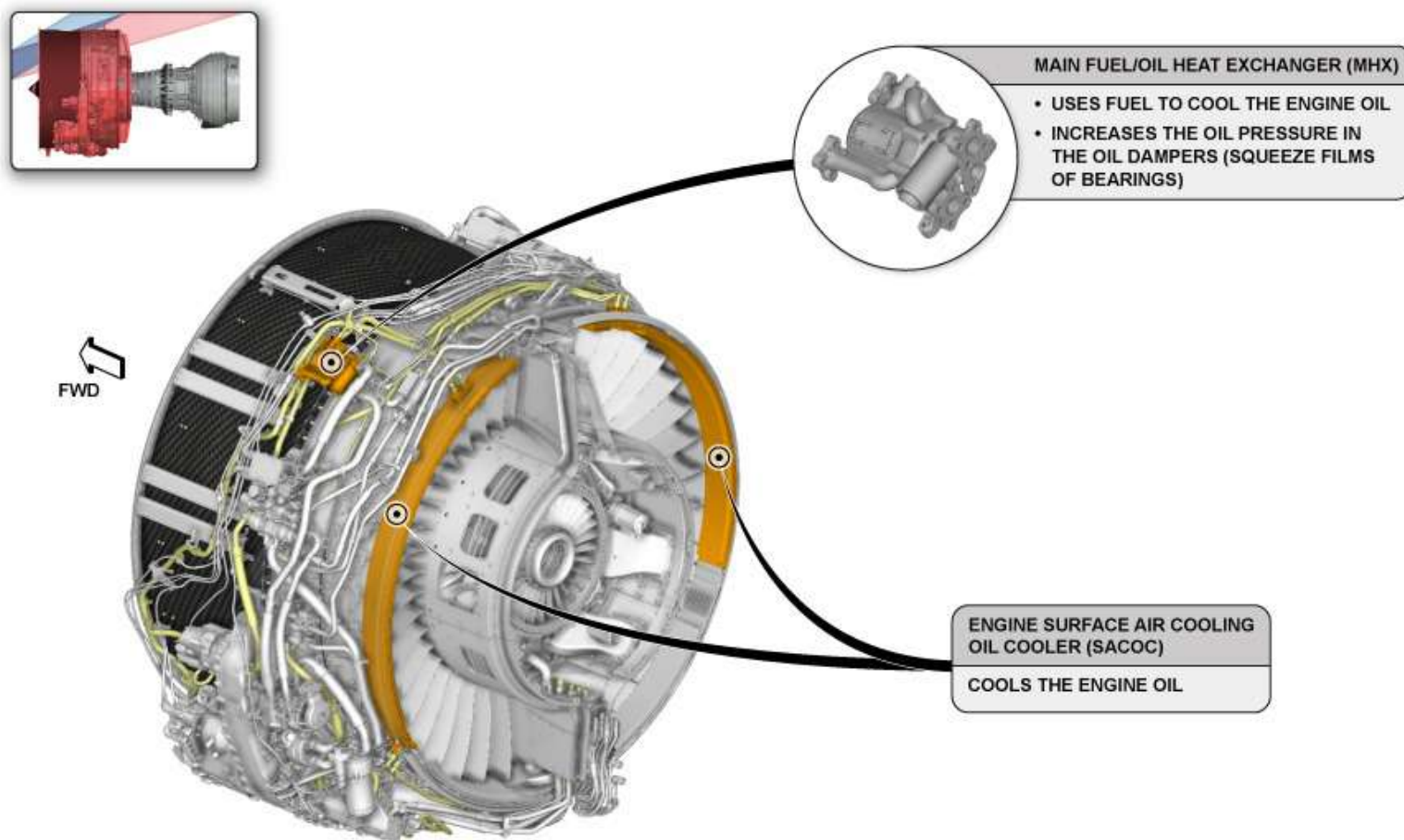
The engine Surface Air Cooling Oil Cooler (SACOC) is composed of two segments fixed to the inner surface of the fan frame, between 8 and 11 o'clock for segment 1 and between 1 to 4 o'clock for segment 2.

Purpose

The purpose of the MHX is to cool the engine oil with the fuel, and increases the oil pressure for the damper bearings

(the damper bearings provide squeeze film that is a thin oil stream between the bearing housing and its support, in order to damp the shaft vibrations).

The purpose of the SACOC is to cool the engine oil with the secondary airflow.



Oil - Distribution Components 2 of 2

Oil Debris Monitoring System

Identify and Locate

The Oil Debris Monitoring System (ODMS) is composed of:

- an air/oil separator located on the fan frame at 3 o'clock, on the scavenge line just above the oil tank,
- an ODMS sensor mounted on the air/oil separator,
- an ODMS unit mounted on the fan case, on the right hand bottom side of the oil tank, with its harness.

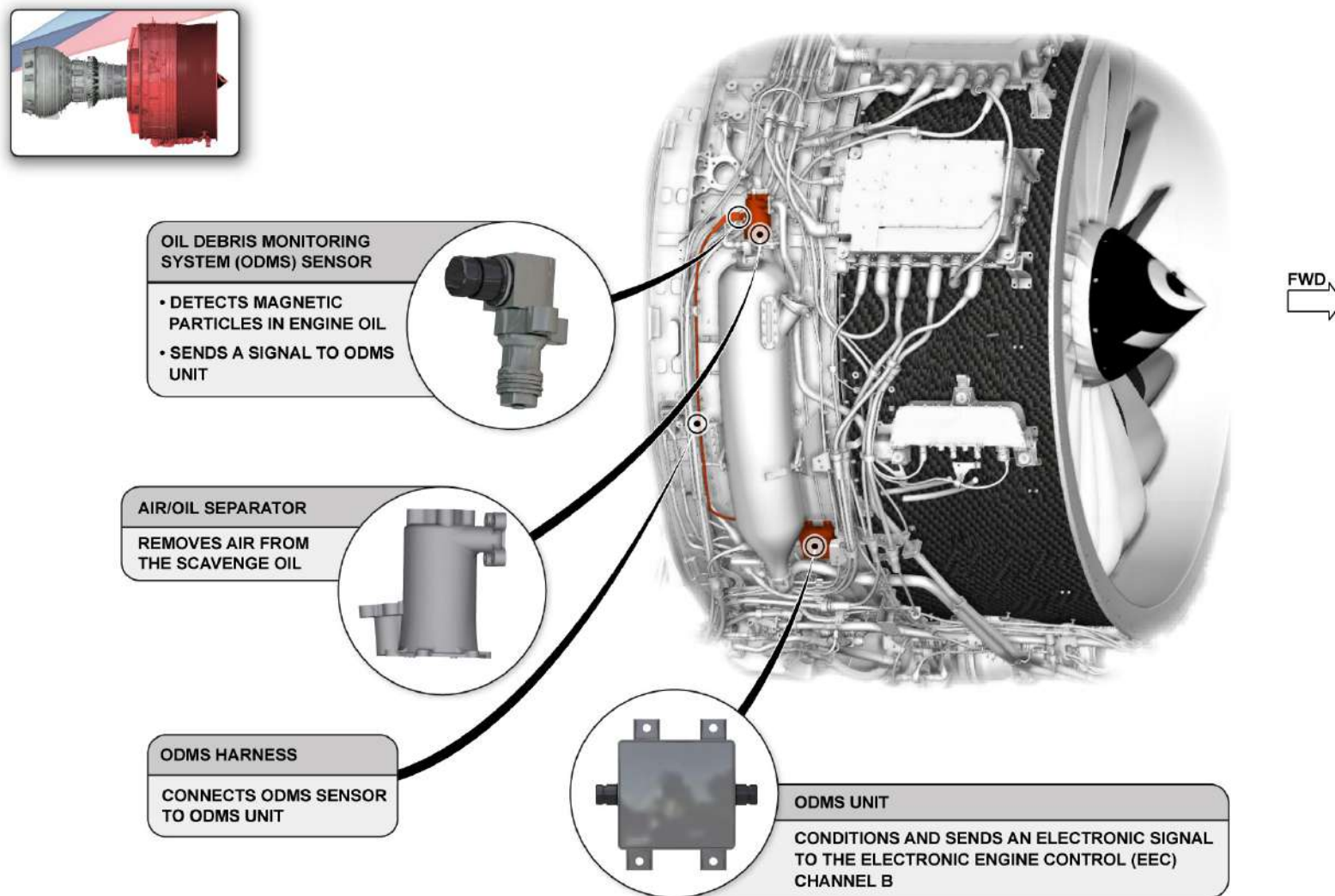
Purpose

The ODMS detects magnetic debris in the engine oil and sends a signal to Electronic Engine Control (EEC) channel B in case of magnetic debris detection.

The air/oil separator removes air from the oil in the scavenge line and redistributes the oil to the oil supply circuit.

The ODMS sensor sends an electrical signal to the ODMS unit in case of oil contamination with magnetic debris.

Above a defined threshold, the ODMS unit sends an electrical signal to EEC channel B.



Oil Debris Monitoring System

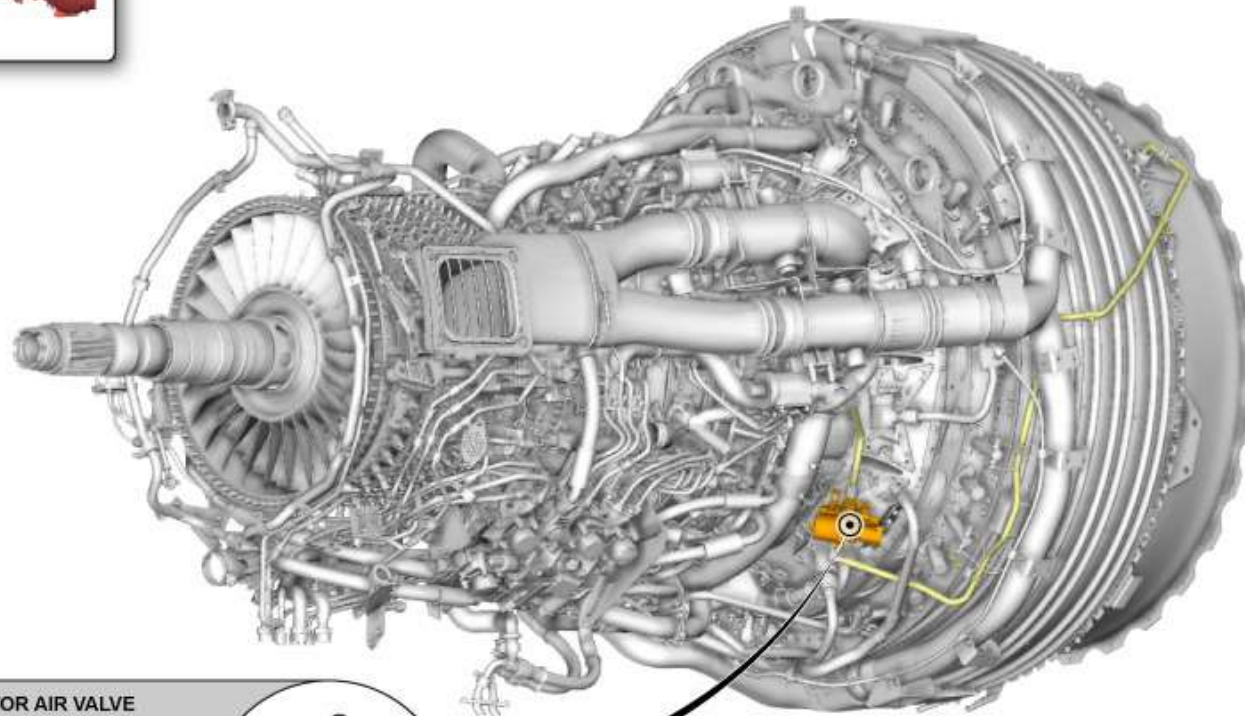
Oil Eductor Air Valve

Identify and Locate

The oil eductor air valve is mounted at 8 o'clock, on a bracket fixed to the combustion case and the High Pressure Turbine (HPT) case.

Purpose

The oil eductor air valve increases the differential pressure across the forward sump (A) seals at low engine speed to prevent oil leaks.



OIL EDUCTOR AIR VALVE

INCREASES DELTA
PRESSURE ACROSS THE
FORWARD SUMP (A) SEALS
TO PREVENT OIL LEAKS



Oil Eductor Air Valve

V0

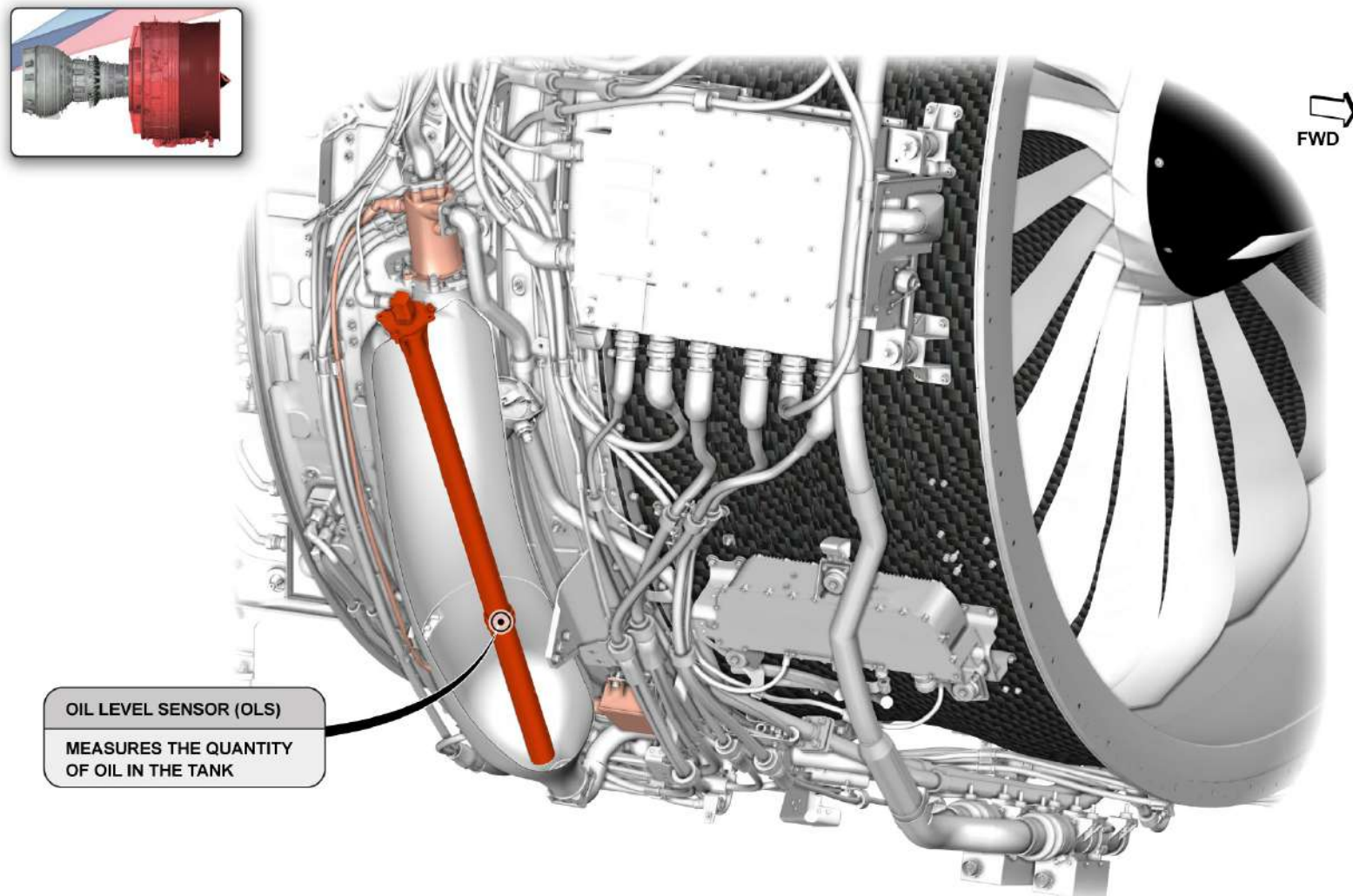
Oil Level Sensor

Identify and Locate

The Oil Level Sensor (OLS) mounted on the top flange of the oil tank.

Purpose

The Oil Level Sensor (OLS) measures the oil level in the oil tank for flight deck display.



Oil Level Sensor

V0

OPTS And OFDPS

Identify and Locate

The Oil Pressure and Temperature Sensor (OPTS) is mounted at the Accessory Gearbox (AGB) inlet interface with the oil supply,

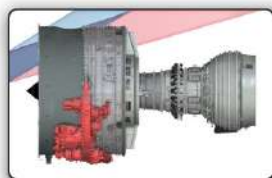
The Oil Filter Differential Pressure Sensor (OFDPS) mounted on the Lube Unit (LU), located on the rear face of AGB at 7 o'clock.

Purpose

The OPTS provides the Electronic Engine Control (EEC) units with:

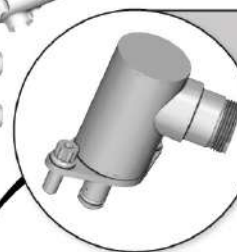
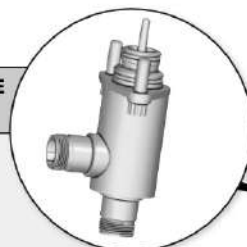
- the oil pressure,
- the oil temperature.

The OFDPS provides both EEC units with the value of the differential pressure between the inlet and the outlet of the oil filter to monitor the clogging condition.



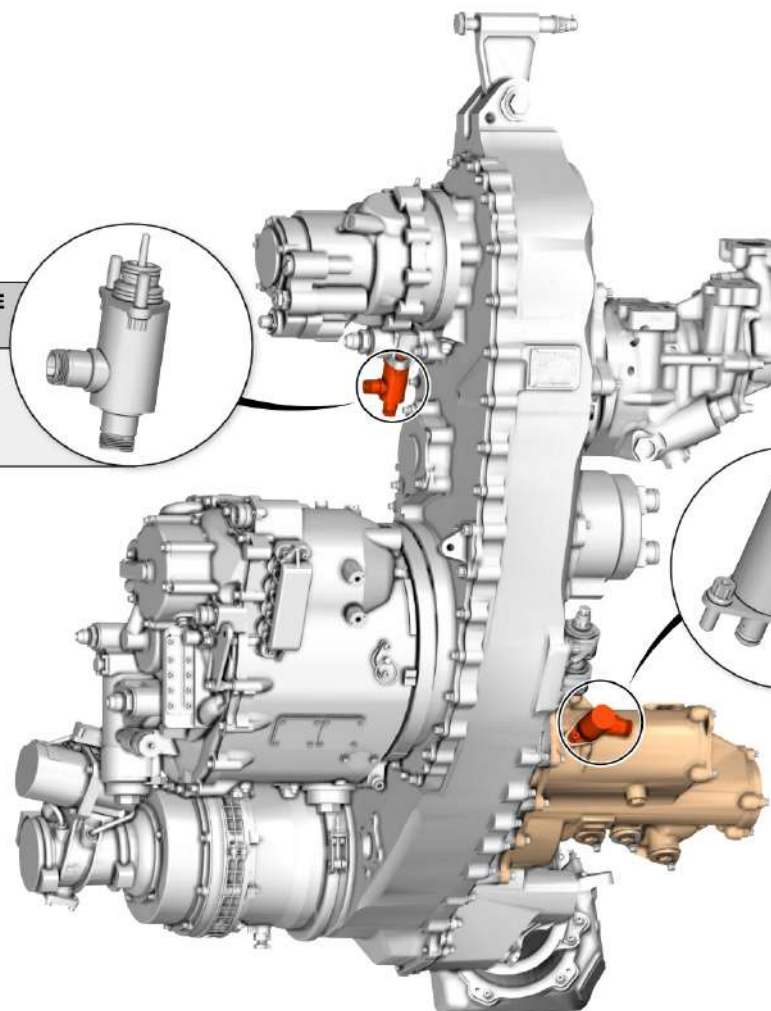
**OIL PRESSURE AND TEMPERATURE
SENSOR (OPTS)**

TRANSMITS ENGINE OIL
TEMPERATURE AND PRESSURE
TO ELECTRONIC ENGINE
CONTROL (EEC) UNITS



**OIL FILTER DIFFERENTIAL
PRESSURE SENSOR (OFDPS)**

TRANSMITS OIL FILTER
IMPENDING BYPASS
INDICATION TO EEC UNITS



OPTS And OFDPS

V0



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